CompTIA Server+

Configuring Server High Availability

Introduction

Server+
Load Balancing
NIC

Welcome to the **Configuring Server High Availability** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

To ensure that resources are highly available on the network, additional configurations can be done to ensure high availability and disaster recovery in the event of a device outage or failure. These configurations can include configuring Network Interface Card (NIC) teaming and Network Load Balancing.

Network Load Balancing can be configured by adding an additional server with the same role installed to ensure that if the primary server fails, a failover can occur for high availability. For example, the Internet Information Services role can be installed on both servers to create a failover for an internal website.

Network Interface Card teaming is configured by adding an additional network card to the device. The additional NIC can then be configured to share the network traffic for the device, ensuring high availability.

A further configuration that can be made to ensure resources are highly available in the event of a disaster is to configure multipathing. In essence, different network routes to the same resource are configured, ensuring the resource is available if a network device becomes unresponsive.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Configure Network Load Balancing
- Exercise 2 - Configuring NIC Teaming on a Server

After completing this module, you should be able to:

- Install and Configure a Web Server
- Install the Network Load Balancing Feature
- Configure a Network Load Balancing Cluster
- Configure a Forward Lookup Zone for Network Load Balancing Cluster
- Verify the Created Network Load Balancing Cluster
- Configure a Secondary Network Interface Card (NIC)
- Configure NIC Teaming

Exam Objectives

The following exam objectives are covered in this module:

2.4 Explain the key concepts of high availability for servers

- Clustering
- Redundant server network infrastructure

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PFSENSE -** (Virtual Firewall)
- **PLABDC01 -** (Windows Server 2019 - Domain Controller)
- **PLABDM01 -** (Windows Server 2019 - Domain Member)
- **PLABKALI -** (Kali 2021.1 - Standalone Server)
- **PLABWIN10 -** (Windows 10 - Domain Member)
- **PLABWEB01 -** (Windows Server 2019 - Standalone Server)

Click **Next** to proceed to the first exercise.

Exercise 1 - Configure Network Load Balancing

Network load balancing is configured to ensure the high availability of resources in the network, for example, an internally hosted intranet site.

In this exercise, network load balancing will be configured on two servers on the network.

Learning Outcomes

After completing this exercise, you should be able to:

- Install and Configure a Web Server
- Install the Network Load Balancing Feature
- Configure a Network Load Balancing Cluster
- Configure a Forward Lookup Zone for Network Load Balancing Cluster
- Verify the Created Network Load Balancing Cluster

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01 -** (Windows Server 2019 - Domain Controller)
- **PLABDM01 -** (Windows Server 2019 - Domain Member)
- **PLABWIN10 -** (Windows 10 - Domain Member)

Task 1 - Install and Configure a Web Server

Web servers can be used to host a company's internal or external websites. These websites must be highly available.

In this task, a web server will be installed to be configured to utilize network load balancing.

Step 1

Connect to **PLABDC01**.

Right-click **Start** and select **Windows Powershell (Admin)**.

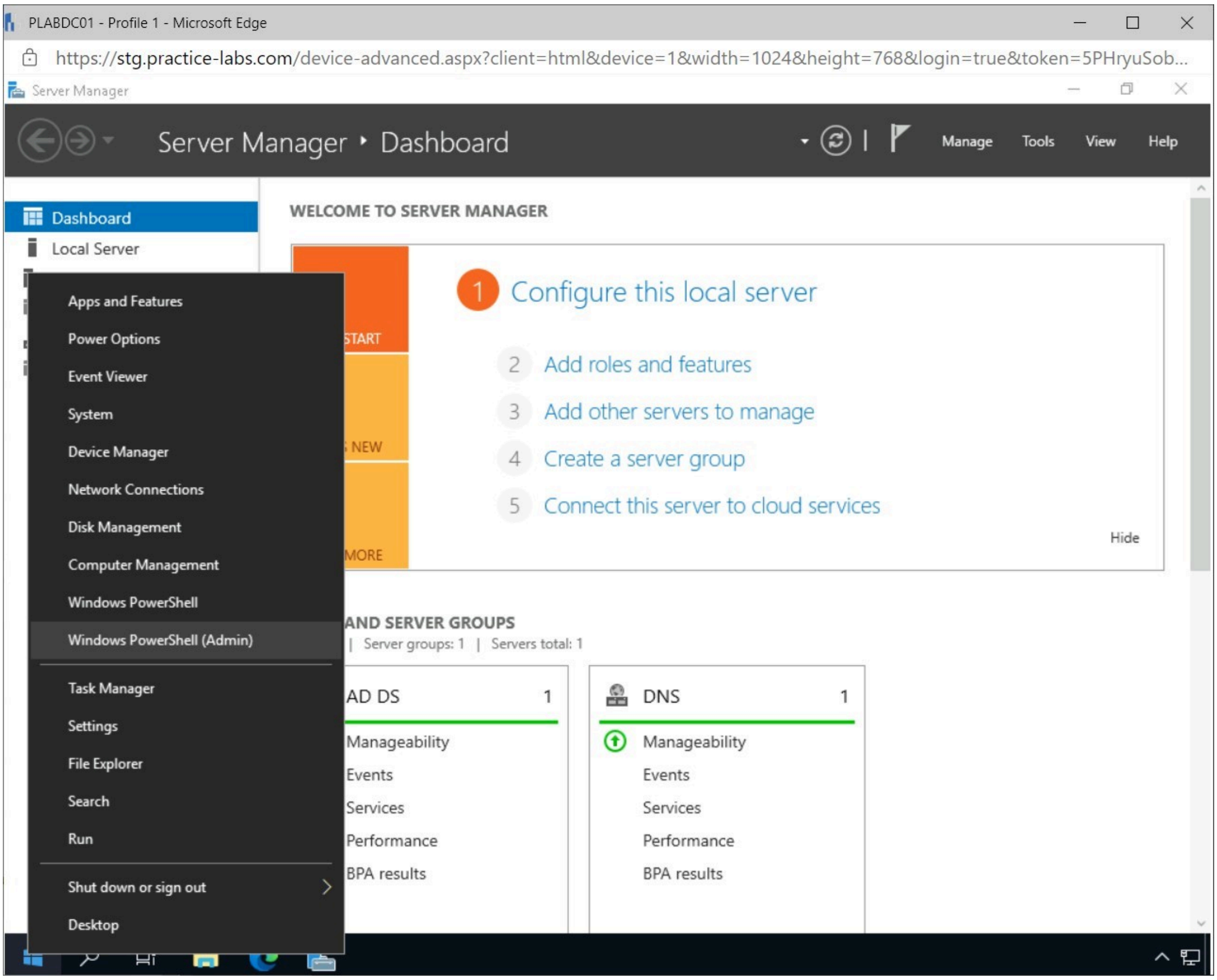

Figure 1.1 Screenshot of PLABDC01: Displaying right-clicking Start and selecting Windows PowerShell (Admin).

Step 2

If the **User Account Control** pop-up window appears, click **Yes**.

Type the following command in the **Administrator: Windows PowerShell** window and press **Enter**.

```
Install-WindowsFeature -Name Web-Server -
IncludeManagementTools
```

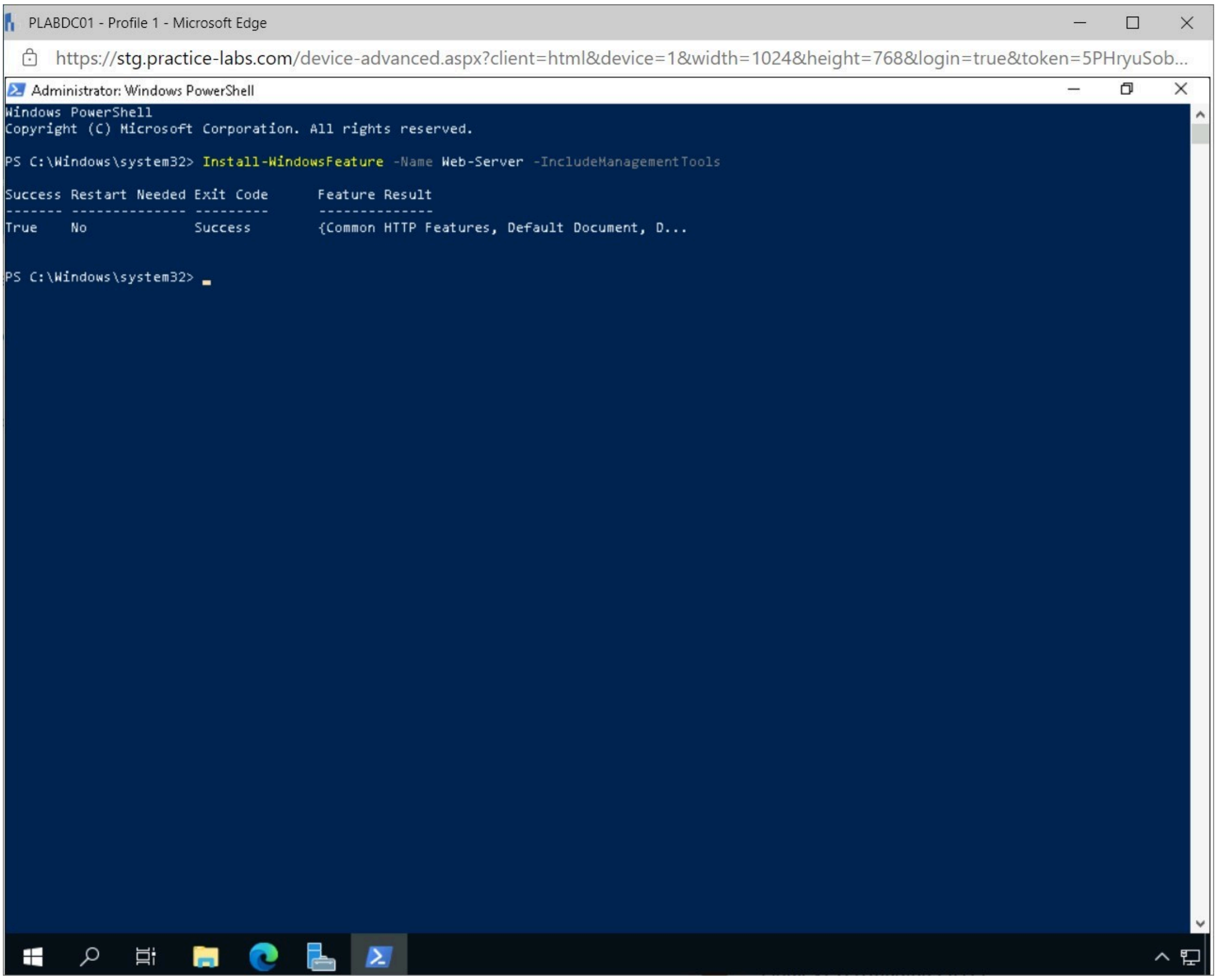

Figure 1.2 Screenshot of PLABDC01: Displaying executing the command in Windows Powershell.

Note: *By executing the command, the IIS server role will be installed on the PLABDC01 device. The IIS Server role can be used to host a company's internal or external website.*

Step 3

Right-click **Start** and select **Settings**.

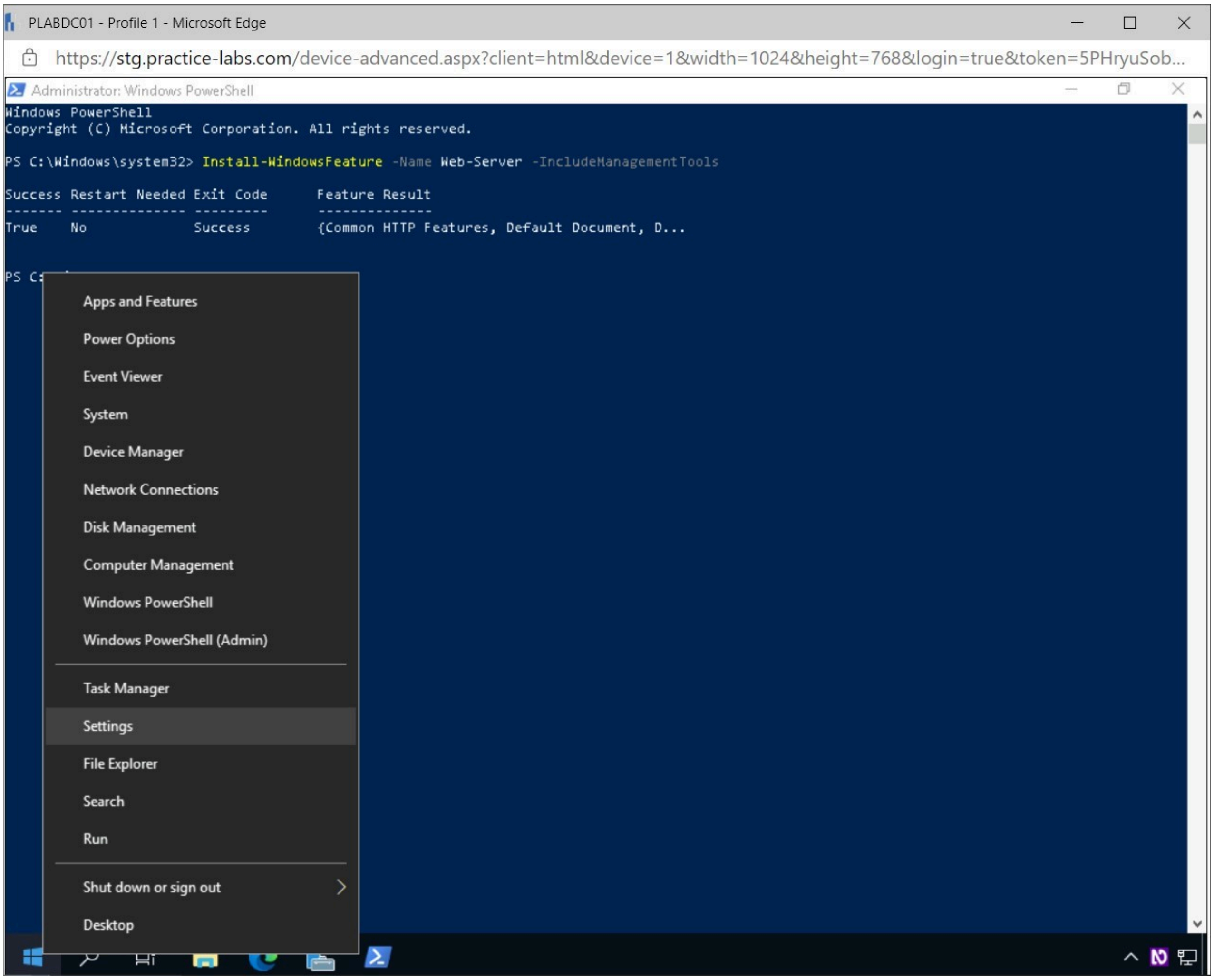

Figure 1.3 Screenshot of PLABDC01: Displaying right-clicking Start and selecting Settings.

Step 4

In the **Windows Settings** window, select **Network & Internet**.

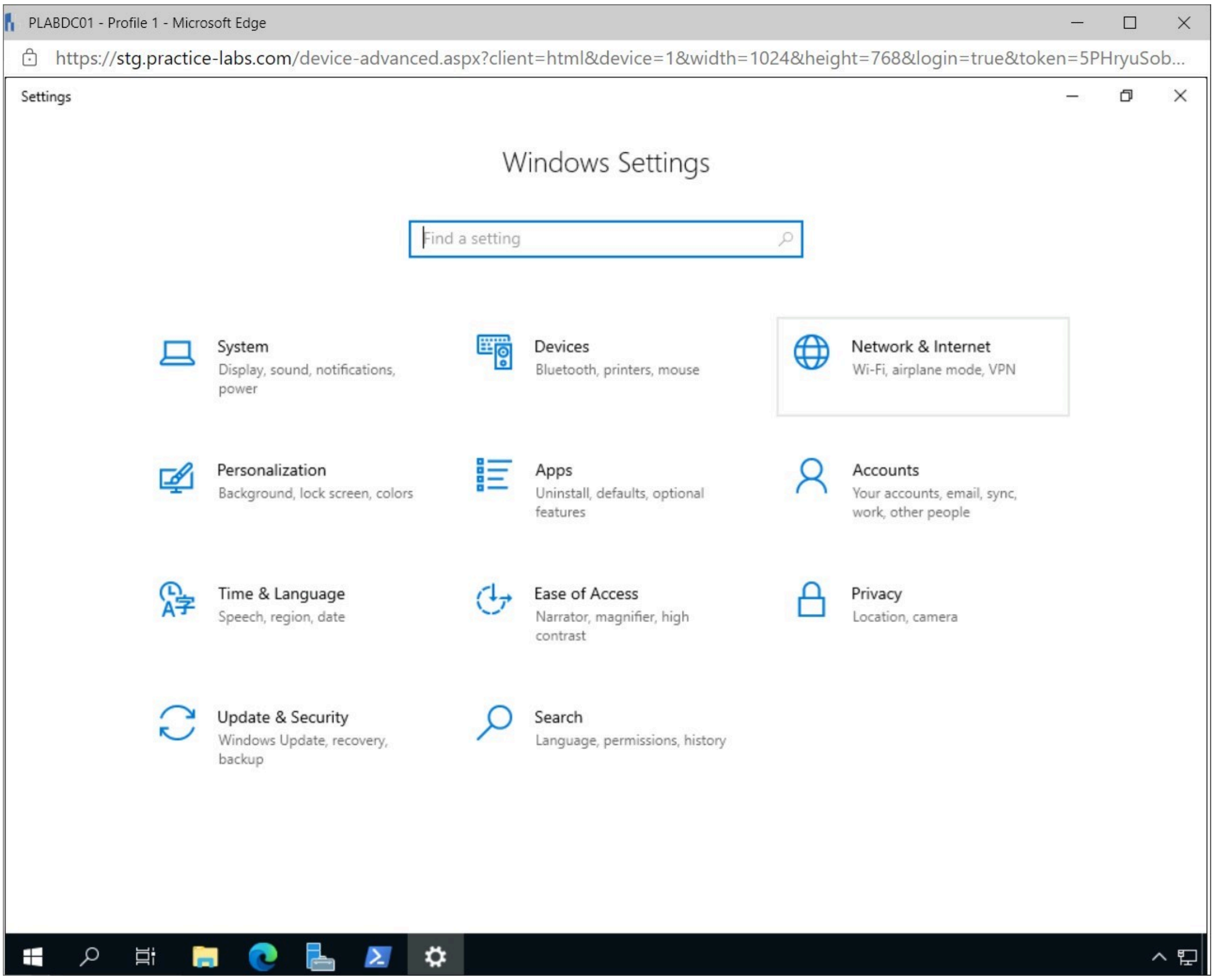

Figure 1.4 Screenshot of PLABDC01: Displaying selecting Network & Internet in the Windows Settings window.

Step 5

In the left pane of the **Settings - Network & Internet** window, select **Proxy**.

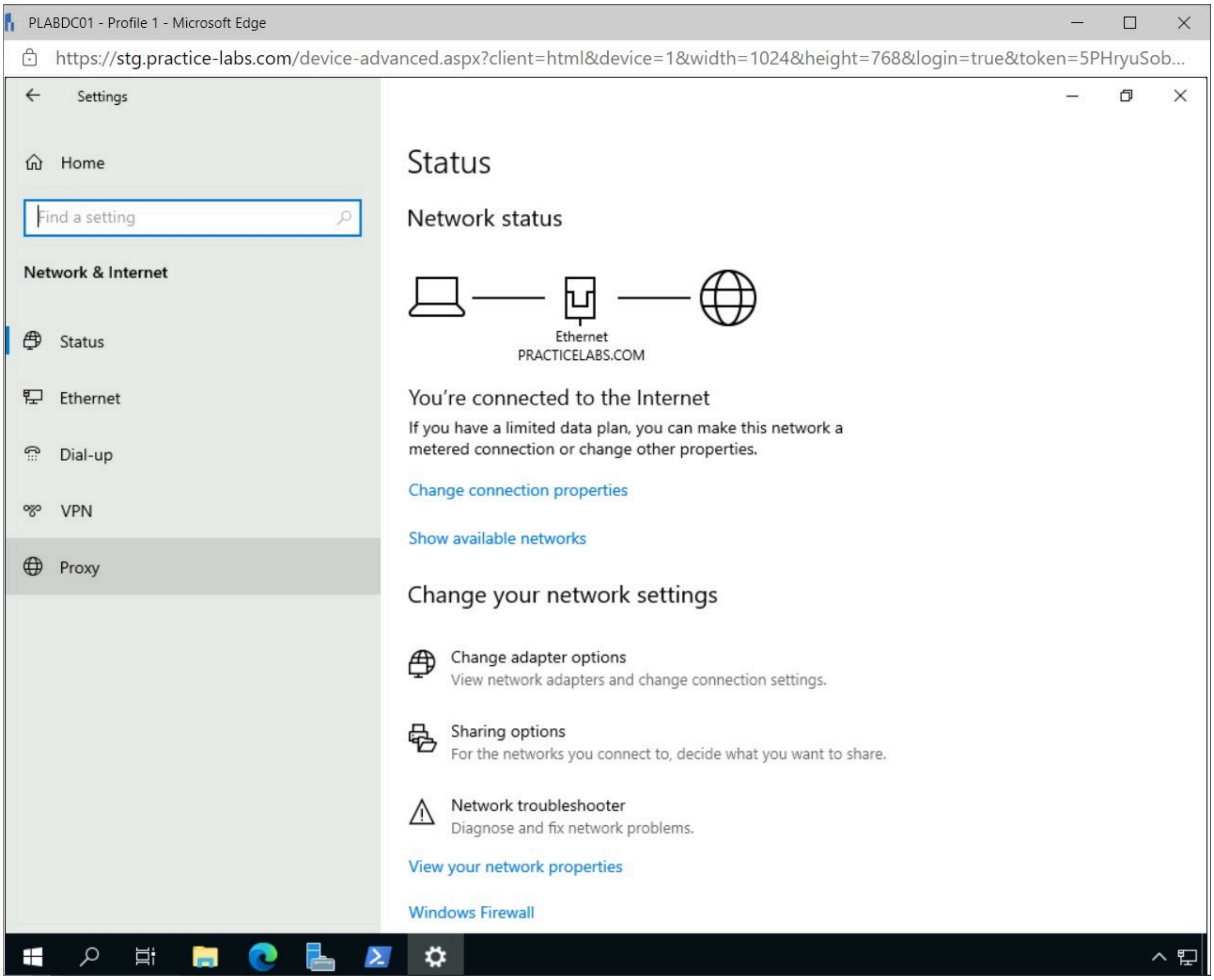

Figure 1.5 Screenshot of PLABDC01: Displaying selecting Proxy in the Settings - Network & Internet window.

Step 6

In the **Manual proxy setup** pane, add the following exception and click **Save**.

```
;plab*
```

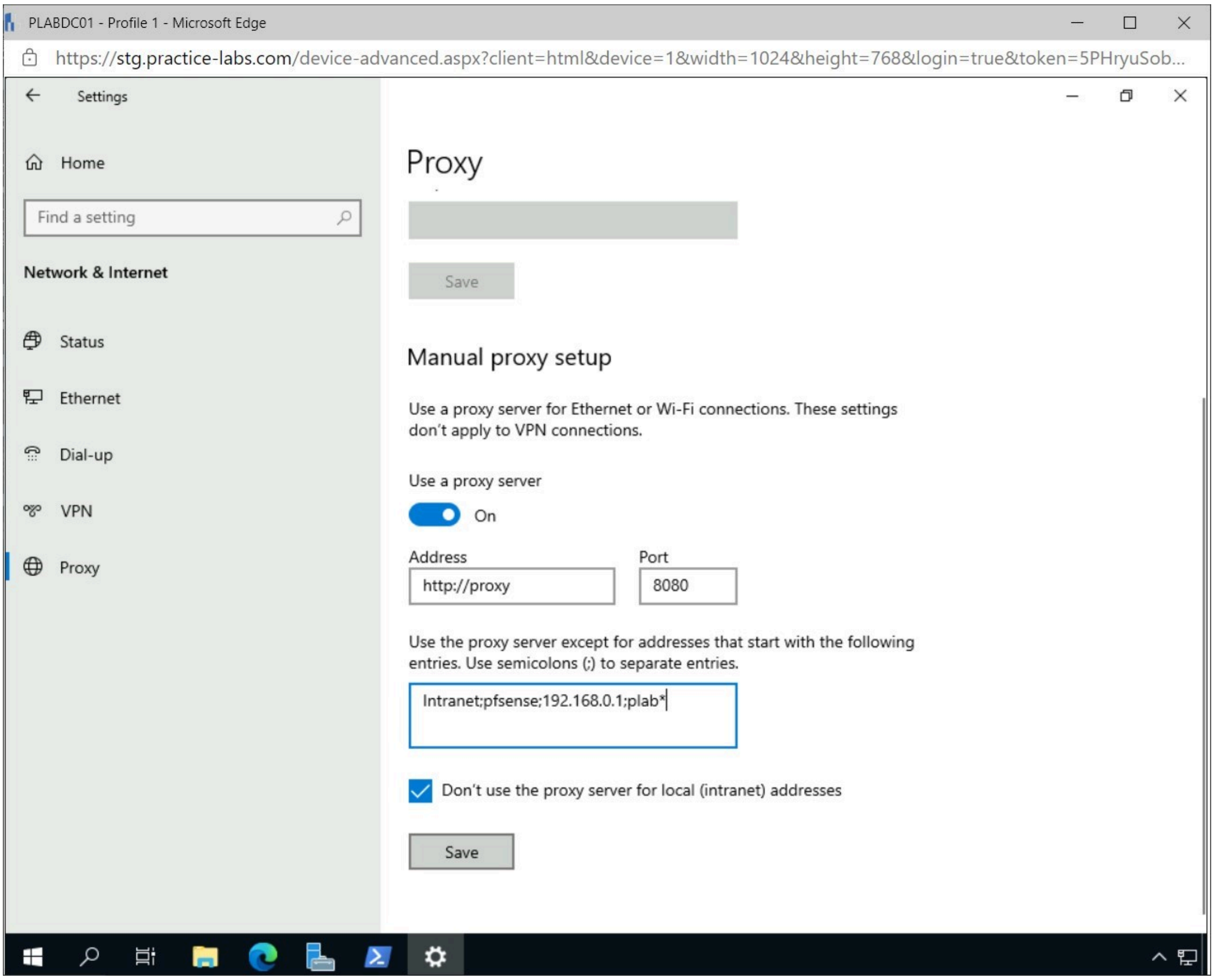

Figure 1.6 Screenshot of PLABDC01: Displaying adding a proxy exception to the Manual proxy setup.

> ***Note:*** *A proxy exception must be added to ensure that the recently created web server's website will be accessible in the lab environment.*

Step 7

Close the **Settings** window.

Close the **Windows PowerShell** window.

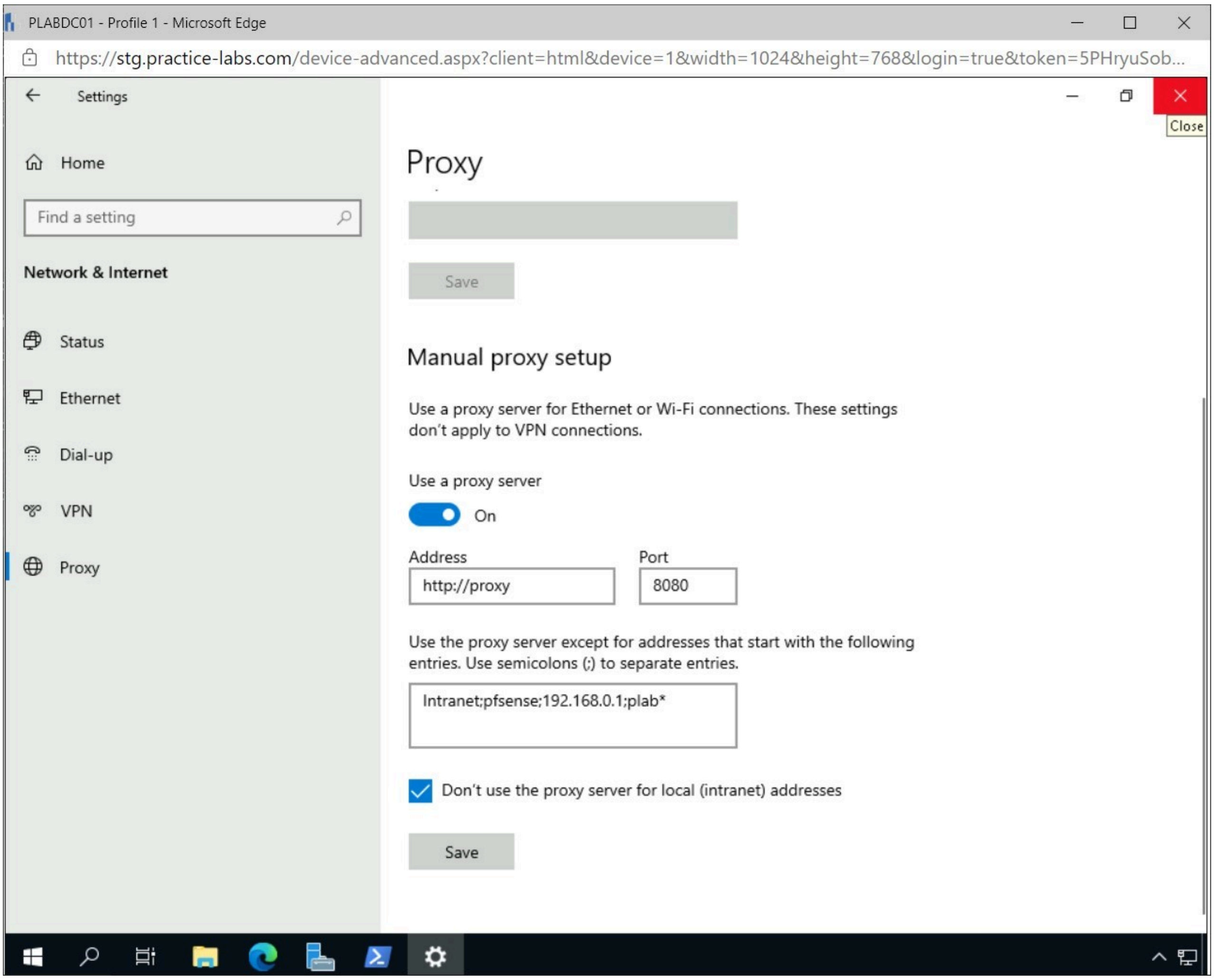

Figure 1.7 Screenshot of PLABDC01: Displaying closing the Settings window.

Step 8

Open **Microsoft Edge** from the **Taskbar** and browse to the following URL:

```
http://plabdc01
```

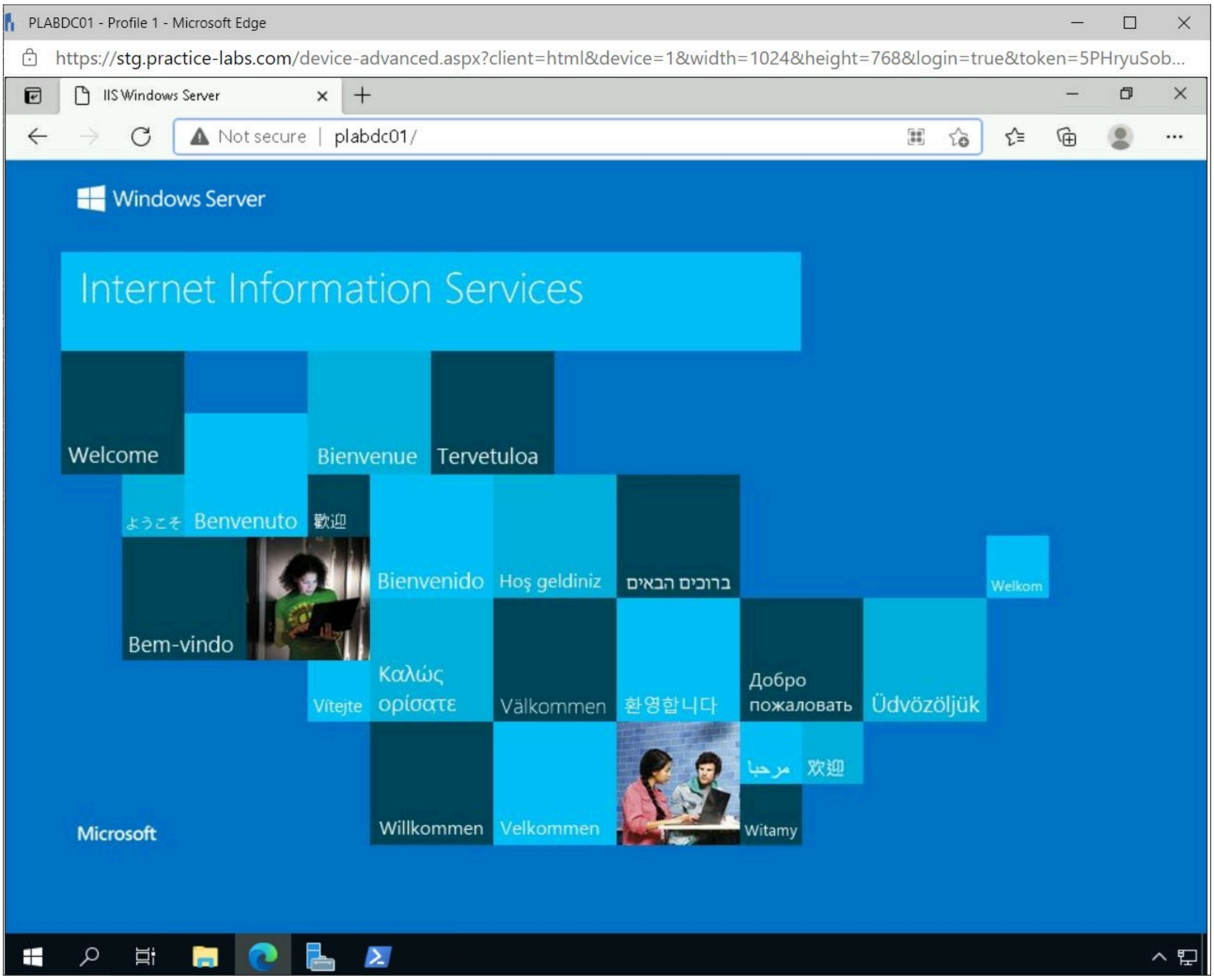

Figure 1.8 Screenshot of PLABDC01: Displaying browsing to the internal website using Microsoft Edge.

> ***Note:*** *After the Web Server was installed on the PLABDC01, it needs to be verified if the web server is functioning correctly. The above screenshot shows that it is functioning correctly.*

Step 9

Close **Microsoft Edge**.

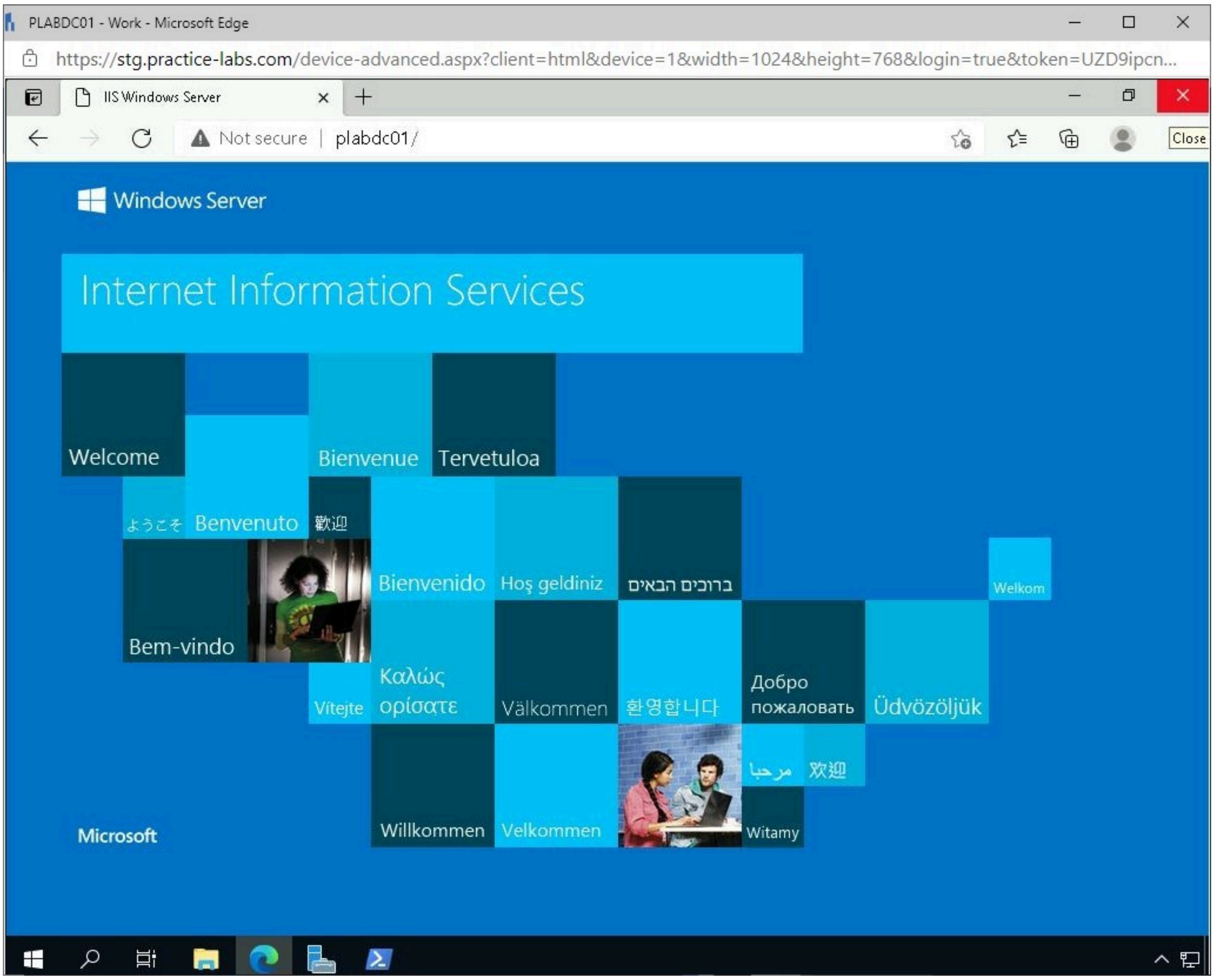

Figure 1.9 Screenshot of PLABDC01: Displaying closing the Microsoft Edge browser.

Step 10

Connect to **PLABDM01**.

Right-click **Start** and select **Windows Powershell (Admin)**.

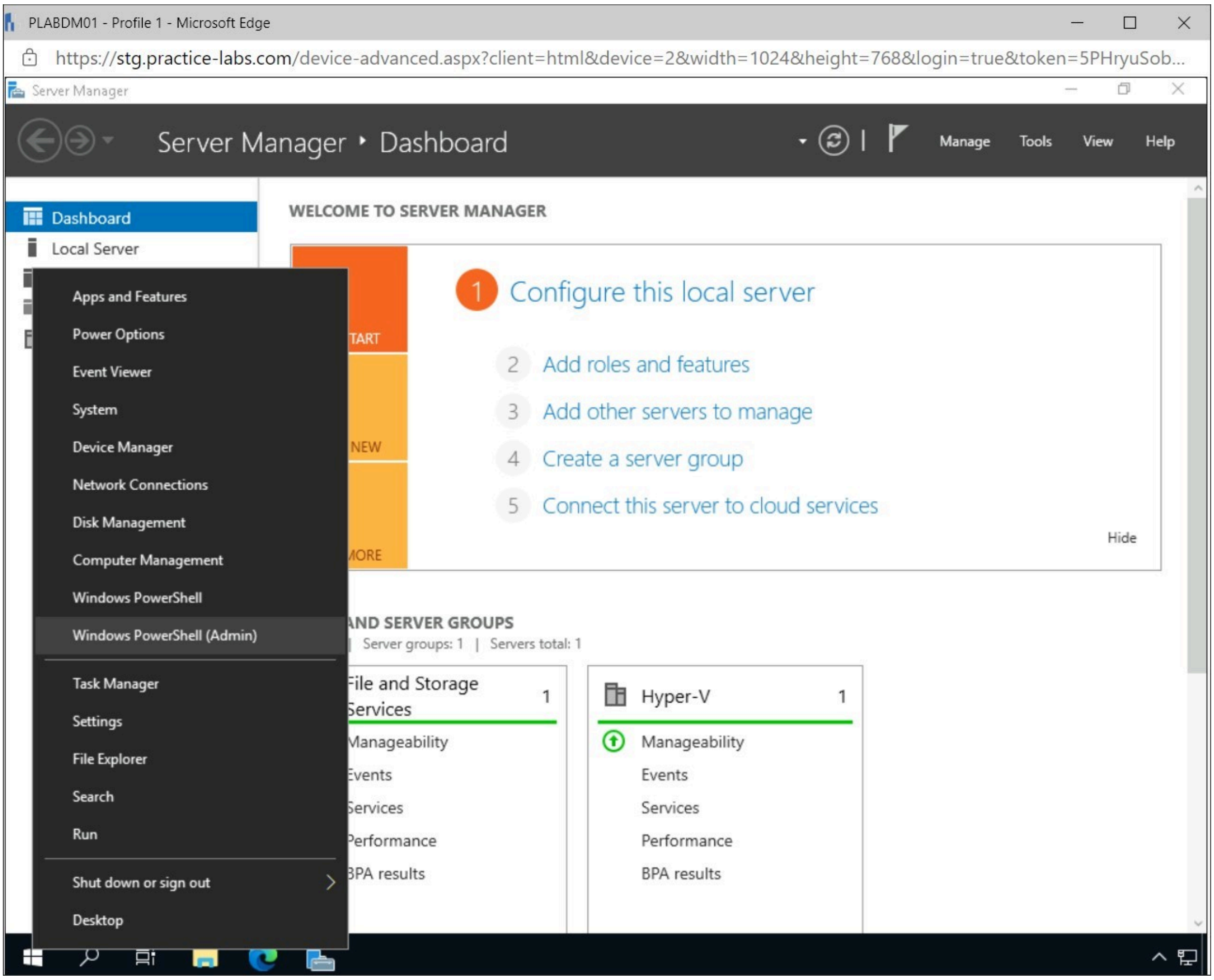

Figure 1.10 Screenshot of PLABDM01: Displaying right-clicking Start and selecting Windows PowerShell (Admin).

Step 11

In the **User Account Control** pop-up window, click **Yes**.

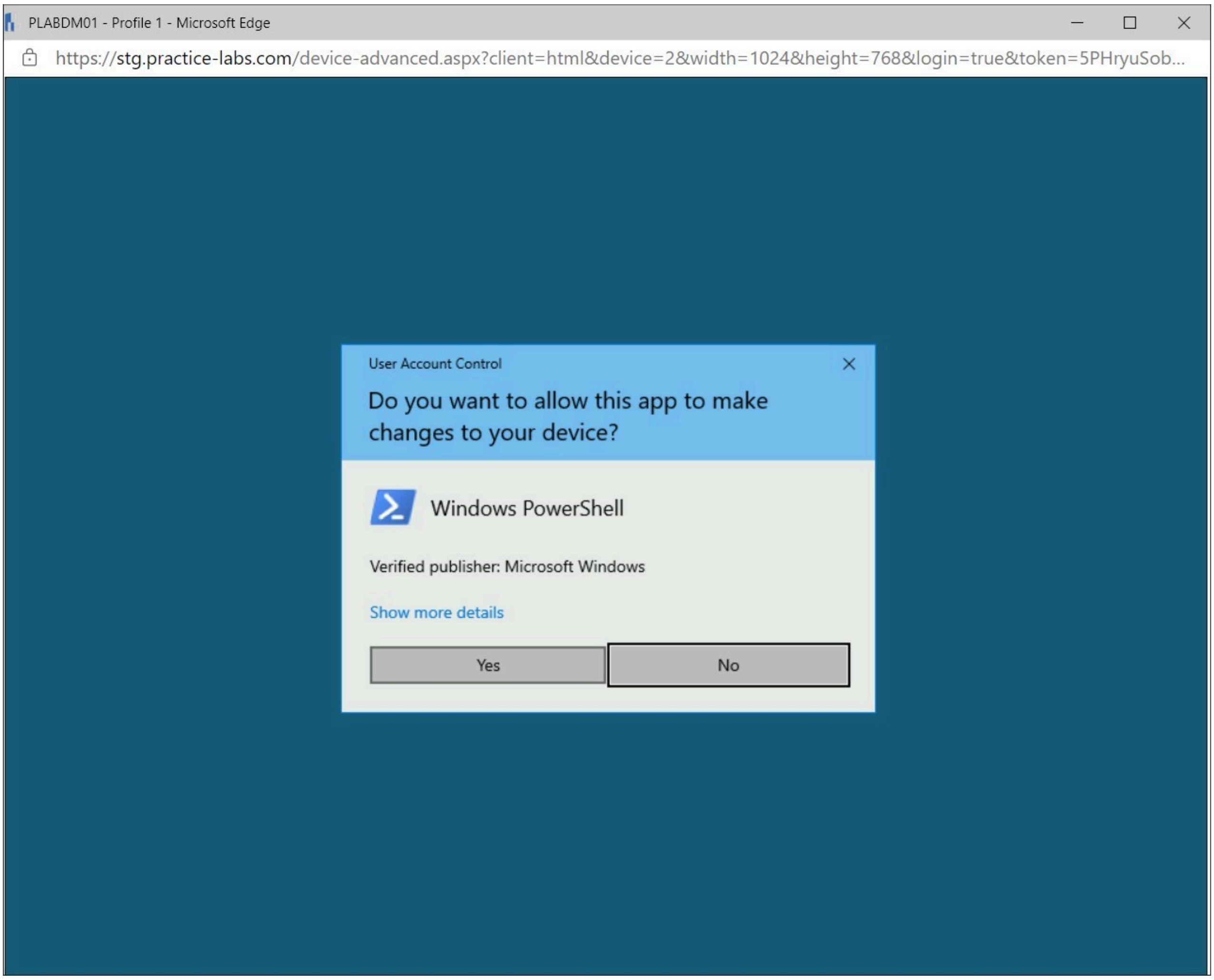

Figure 1.11 Screenshot of PLABDM01: Displaying clicking Yes in the User Account Control pop-up window.

Step 12

Type the following command in the **Administrator: Windows PowerShell** window and press **Enter**.

```
Install-WindowsFeature -Name Web-Server -
IncludeManagementTools
```

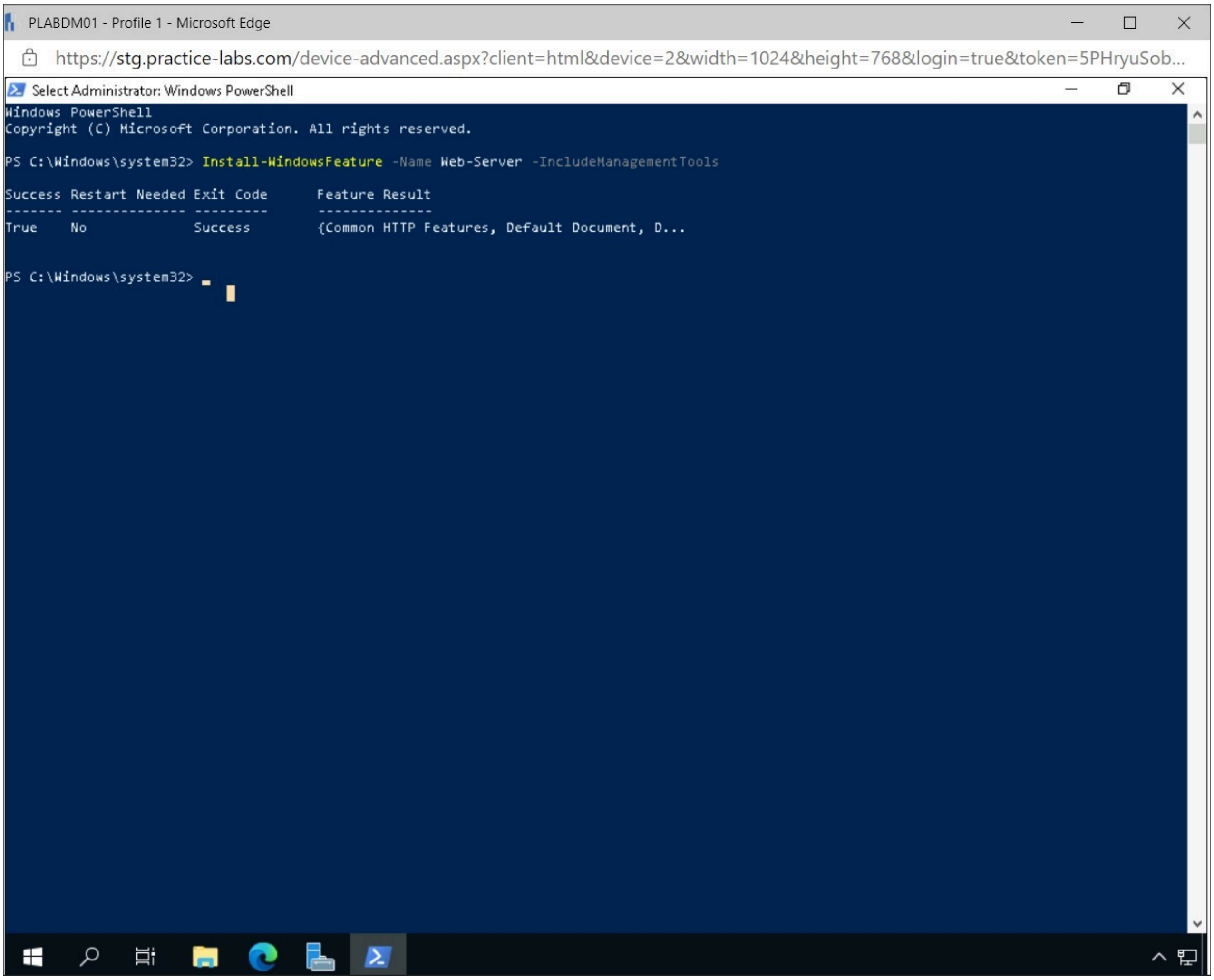

Figure 1.12 Screenshot of PLABDM01: Displaying executing the command in Windows Powershell.

Step 13

Right-click **Start** and select **Settings**.

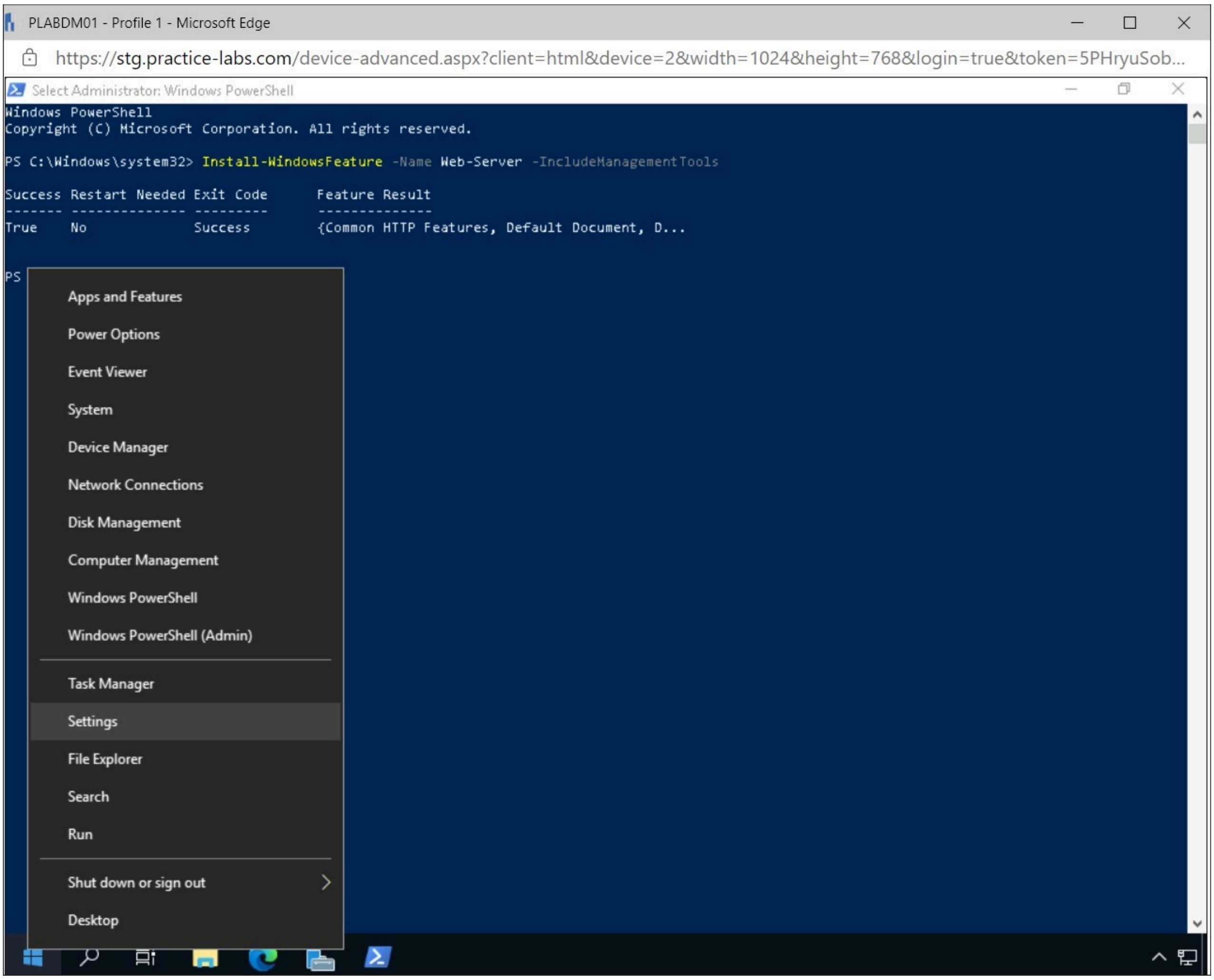

Figure 1.13 Screenshot of PLABDM01: Displaying right-clicking Sart and selecting Settings.

Step 14

In the **Windows Settings** window, select **Network & Internet**.

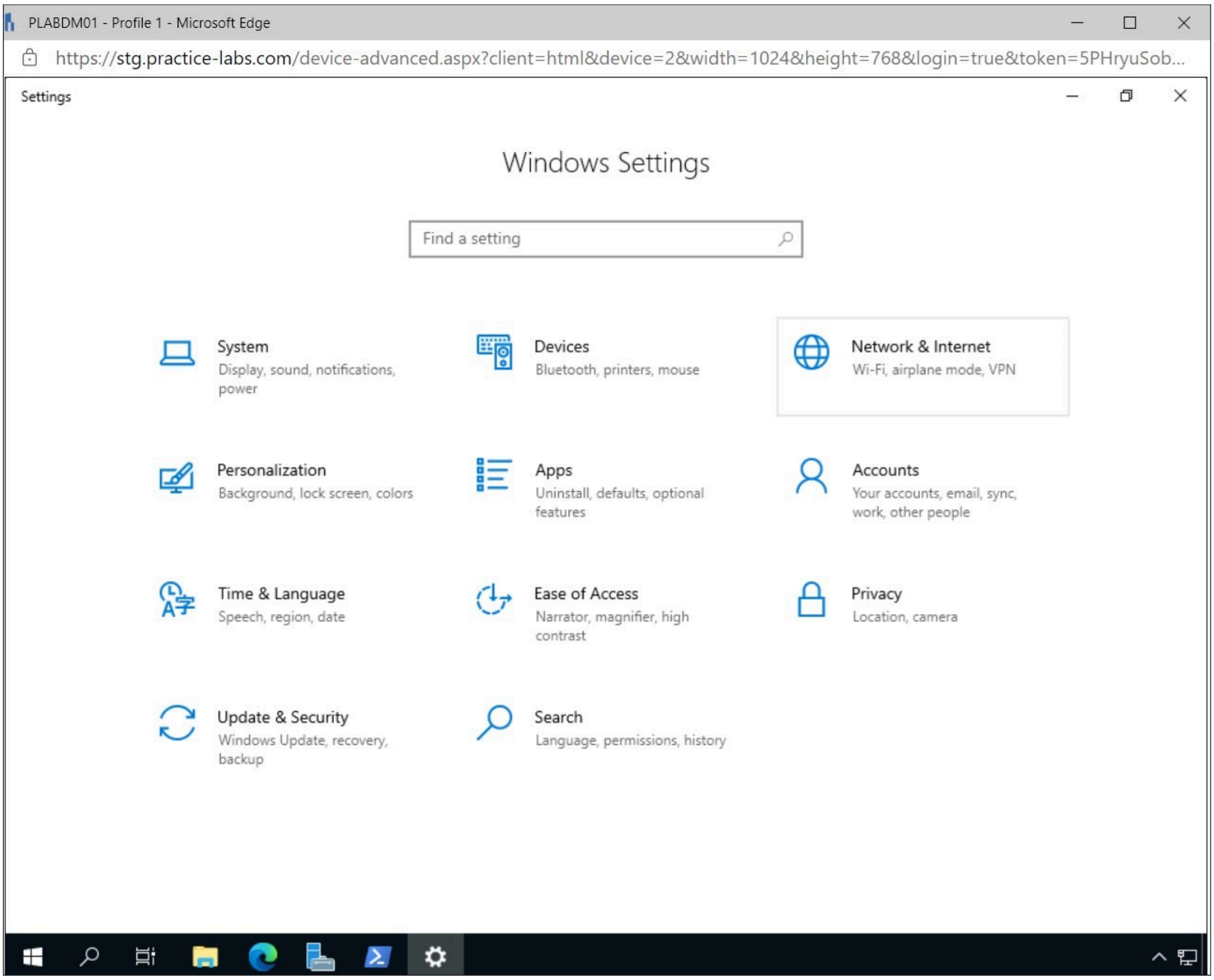

Figure 1.14 Screenshot of PLABDM01: Displaying selecting Network & Internet in the Windows Settings window.

Step 15

Select **Proxy** in the **Settings - Network & Internet** pane.

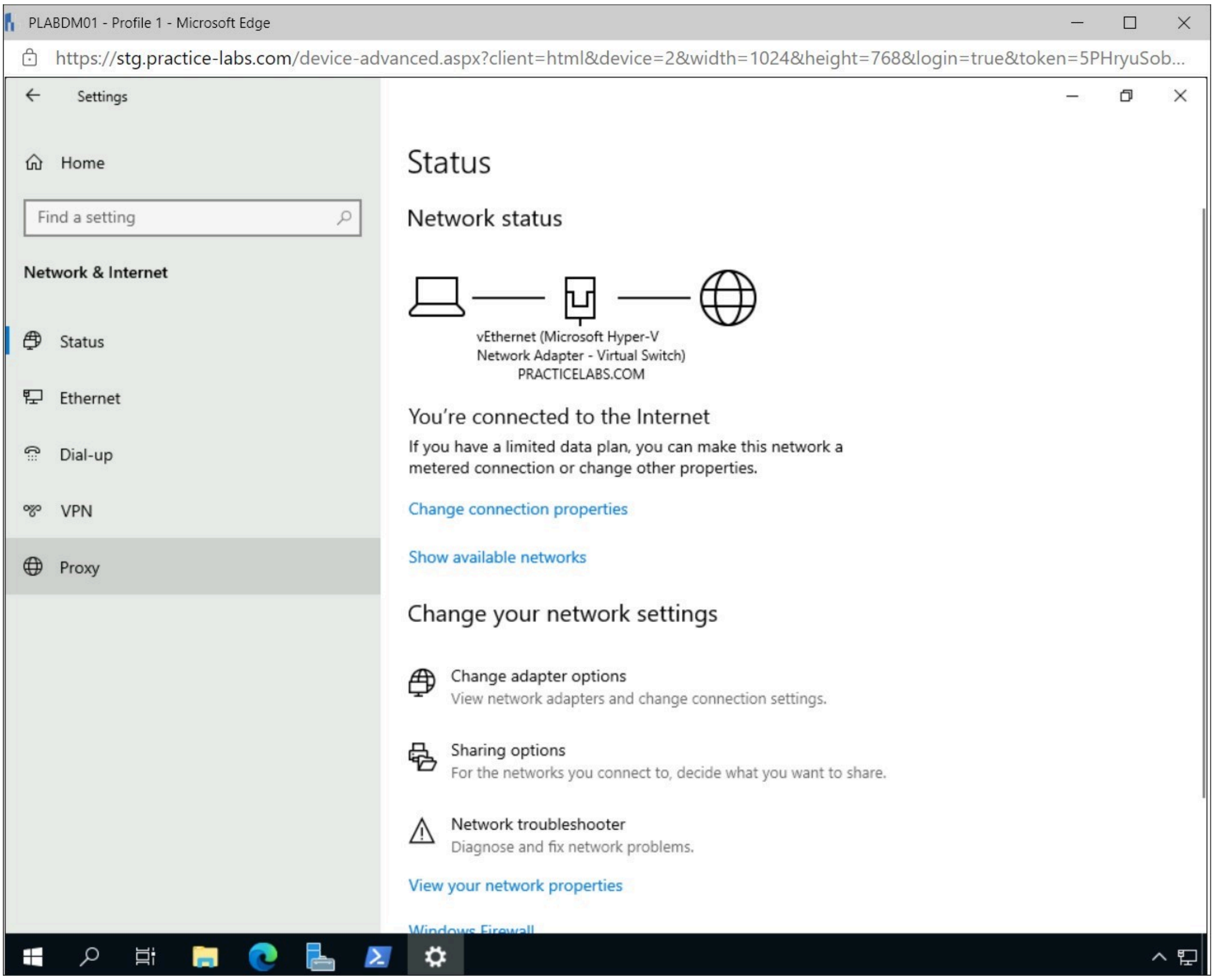

Figure 1.15 Screenshot of PLABDM01: Displaying selecting Proxy in the Settings - Network & Internet window.

Step 16

In the **Manual proxy setup** pane, add the following exception and click **Save**.

```
;plab*
```

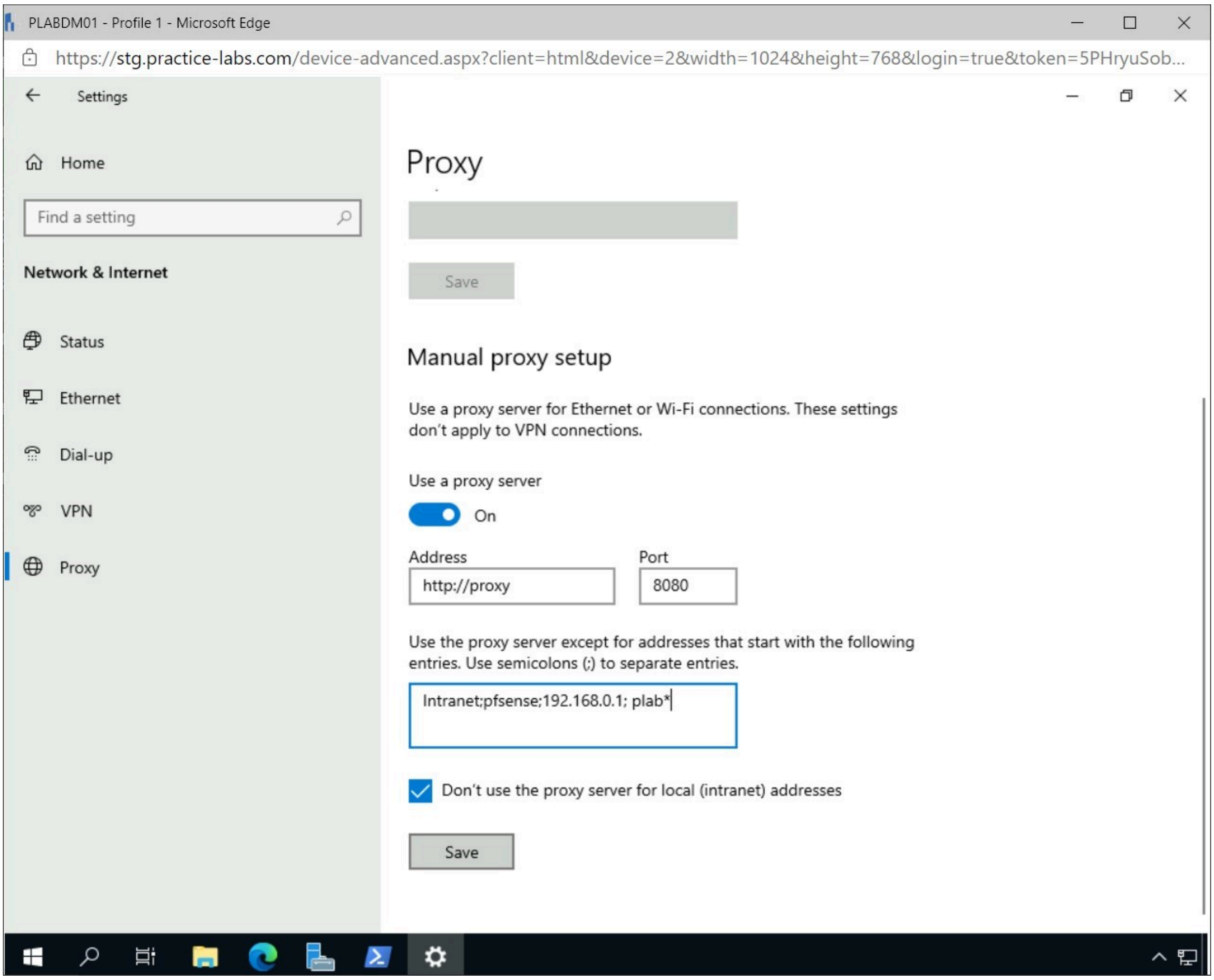

Figure 1.16 Screenshot of PLABDM01: Displaying adding a proxy exception to the Manual proxy setup.

Step 17

Close the **Settings** window.

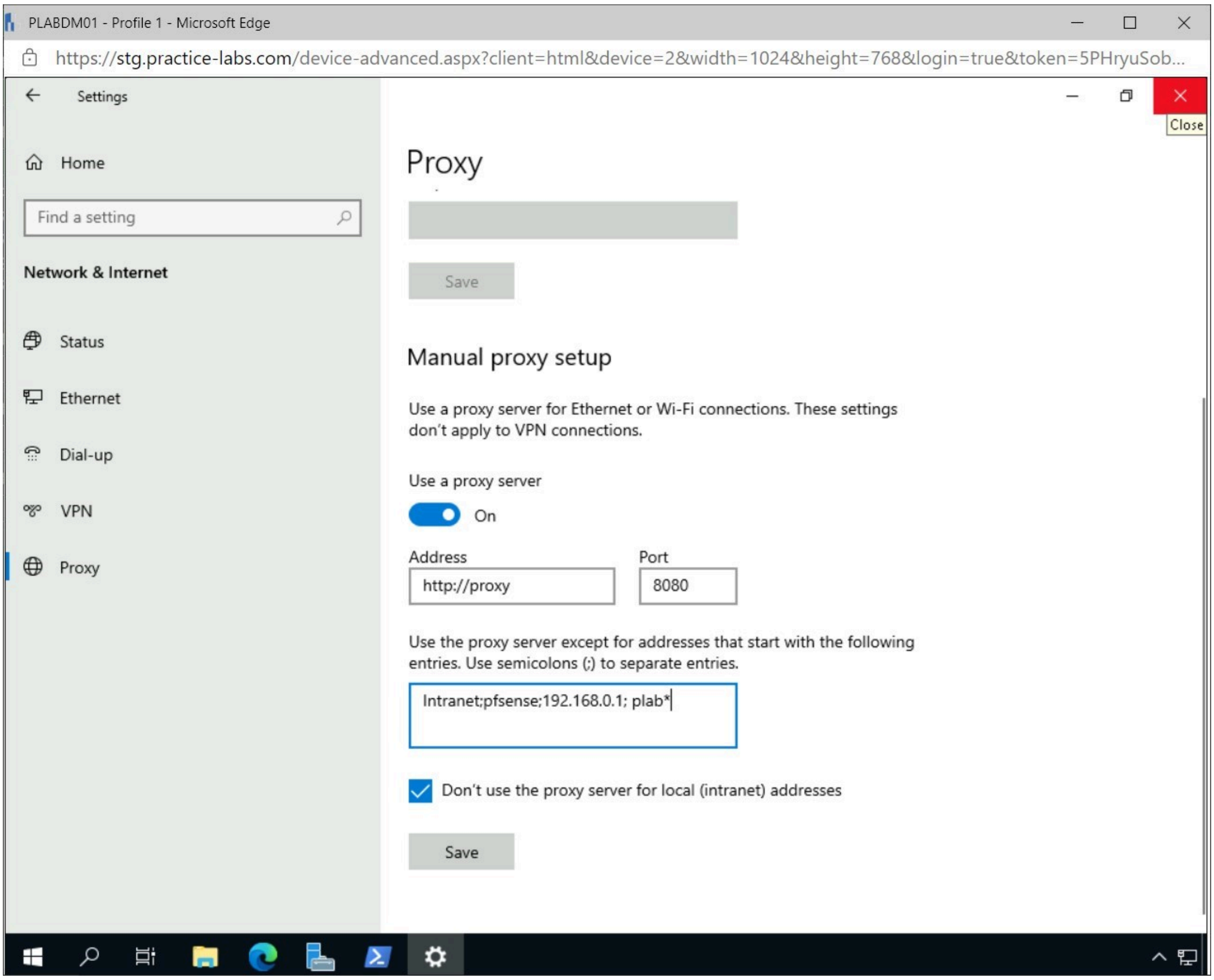

Figure 1.17 Screenshot of PLABDM01: Displaying closing the Settings window.

Step 18

Open **Microsoft Edge** from the **Taskbar** and browse to the following URL:

```
http://plabdm01
```

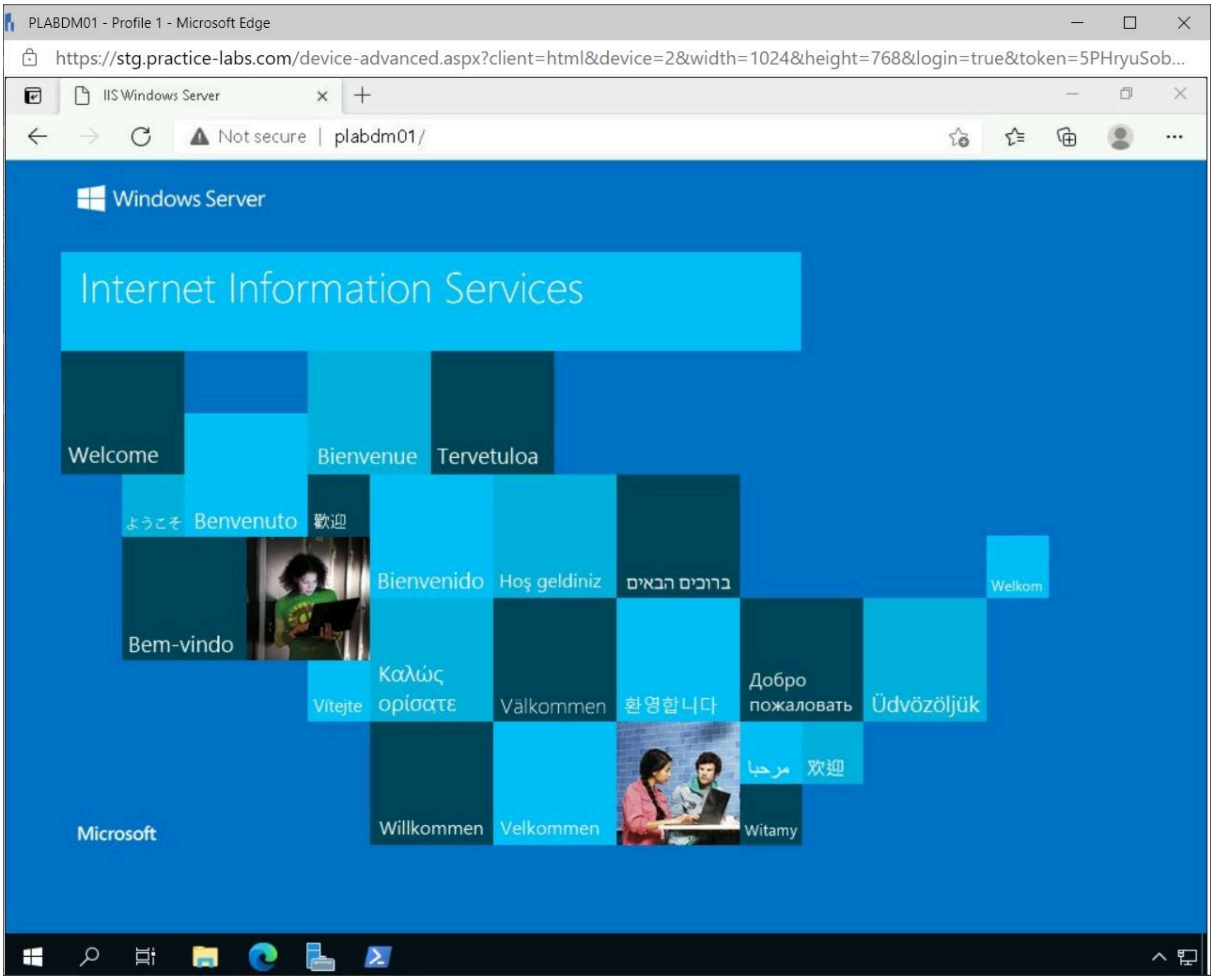

Figure 1.18 Screenshot of PLABDM01: Displaying browsing to the internal website using Microsoft Edge.

Step 19

Close the **Microsoft Edge** browser.

Close the **Windows PowerShell** window.

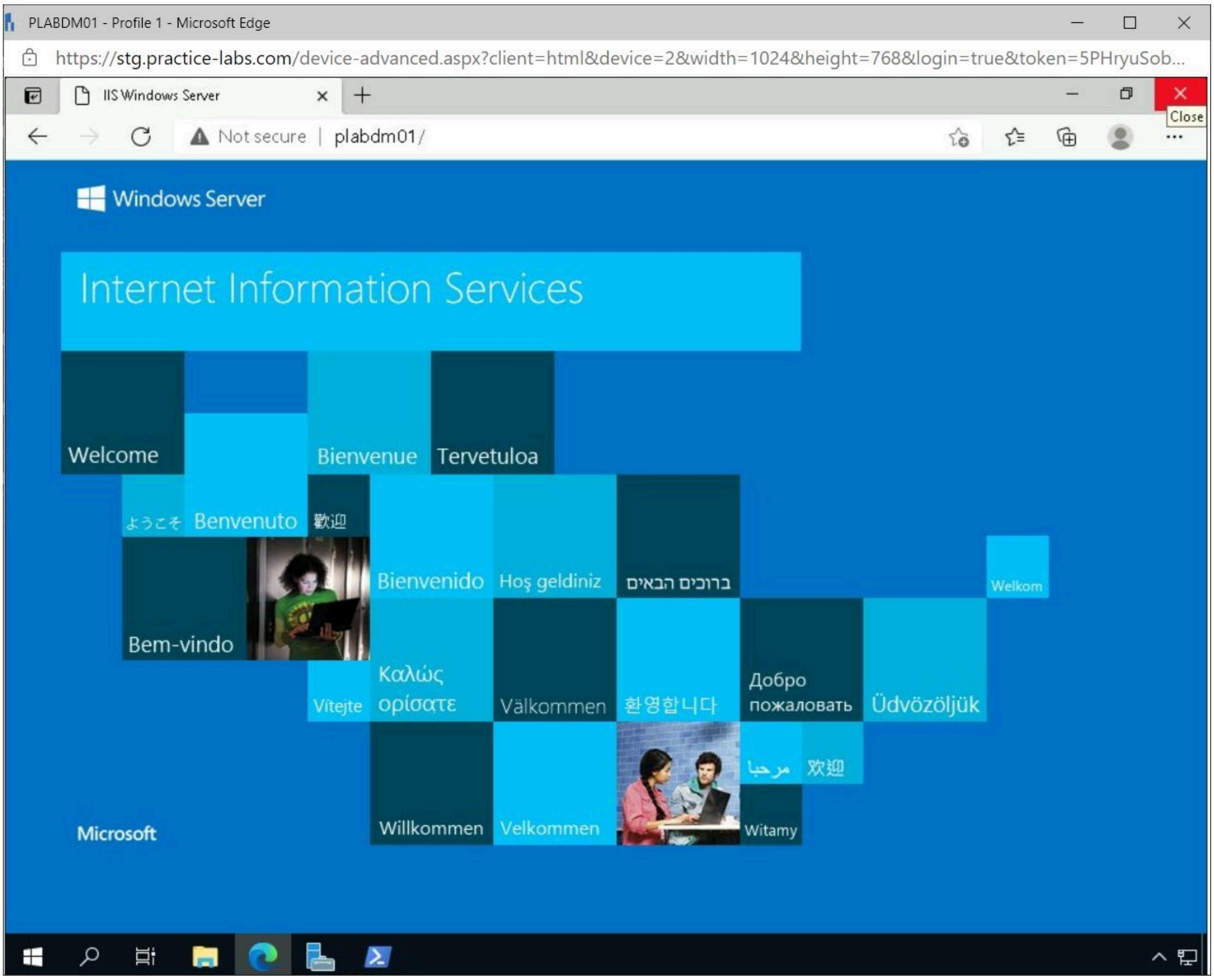

Figure 1.19 Screenshot of PLABDM01: Displaying closing the Microsoft Edge browser.

Task 2 - Install the Network Load Balancing Feature

Before a Network Load Balancing cluster can be configured, the feature must be installed on the devices, which will then be added to the cluster.

In this task, the network load balancing feature will be installed on specific devices.

Step 1

Connect to **PLABDC01**.

Right-click **Start** and select **Windows Powershell (Admin)**.

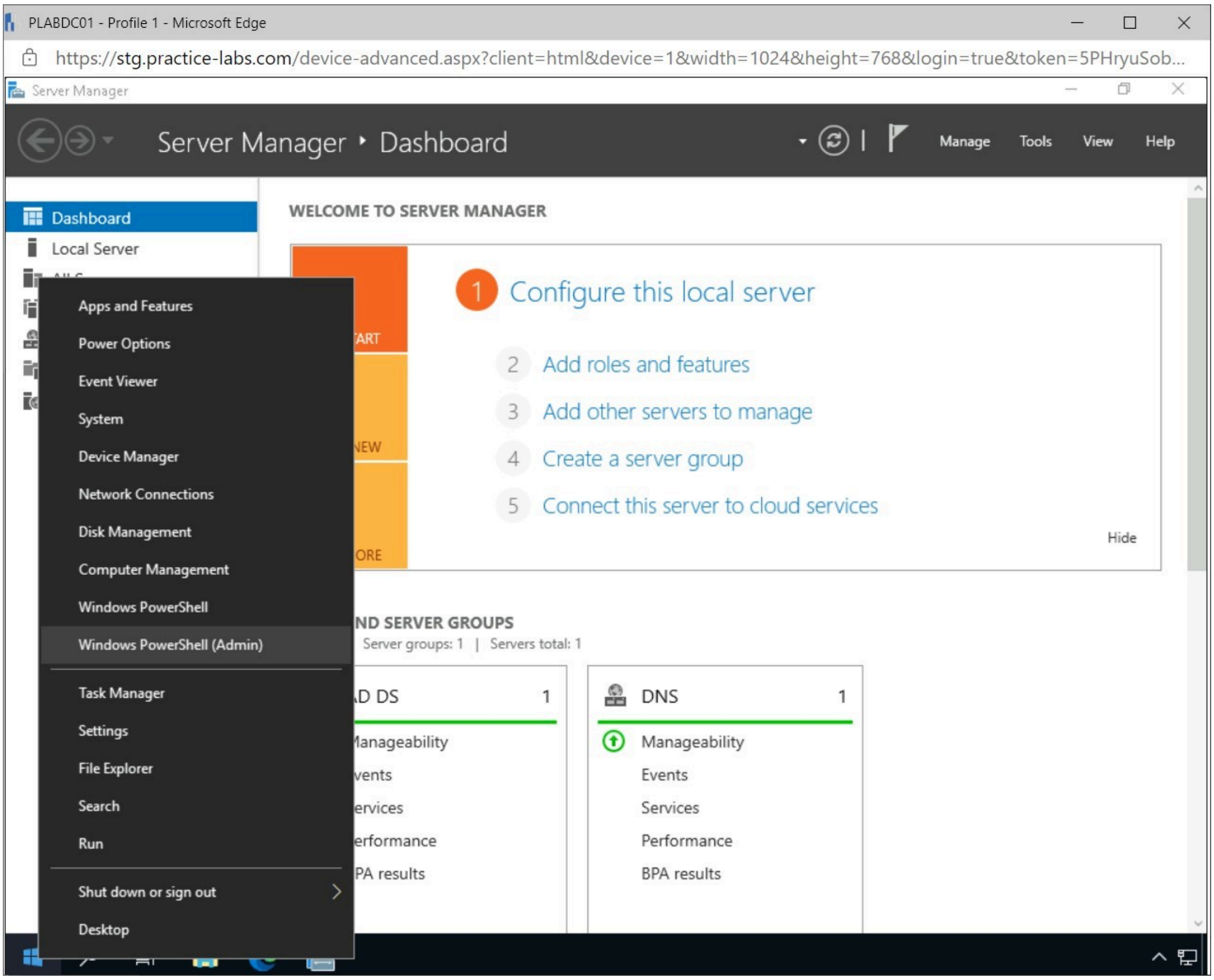

Figure 1.20 Screenshot of PLABDC01: Displaying right-clicking Start and selecting Windows PowerShell (Admin).

Step 2

In the **User Account** pop-up window, click **Yes**.

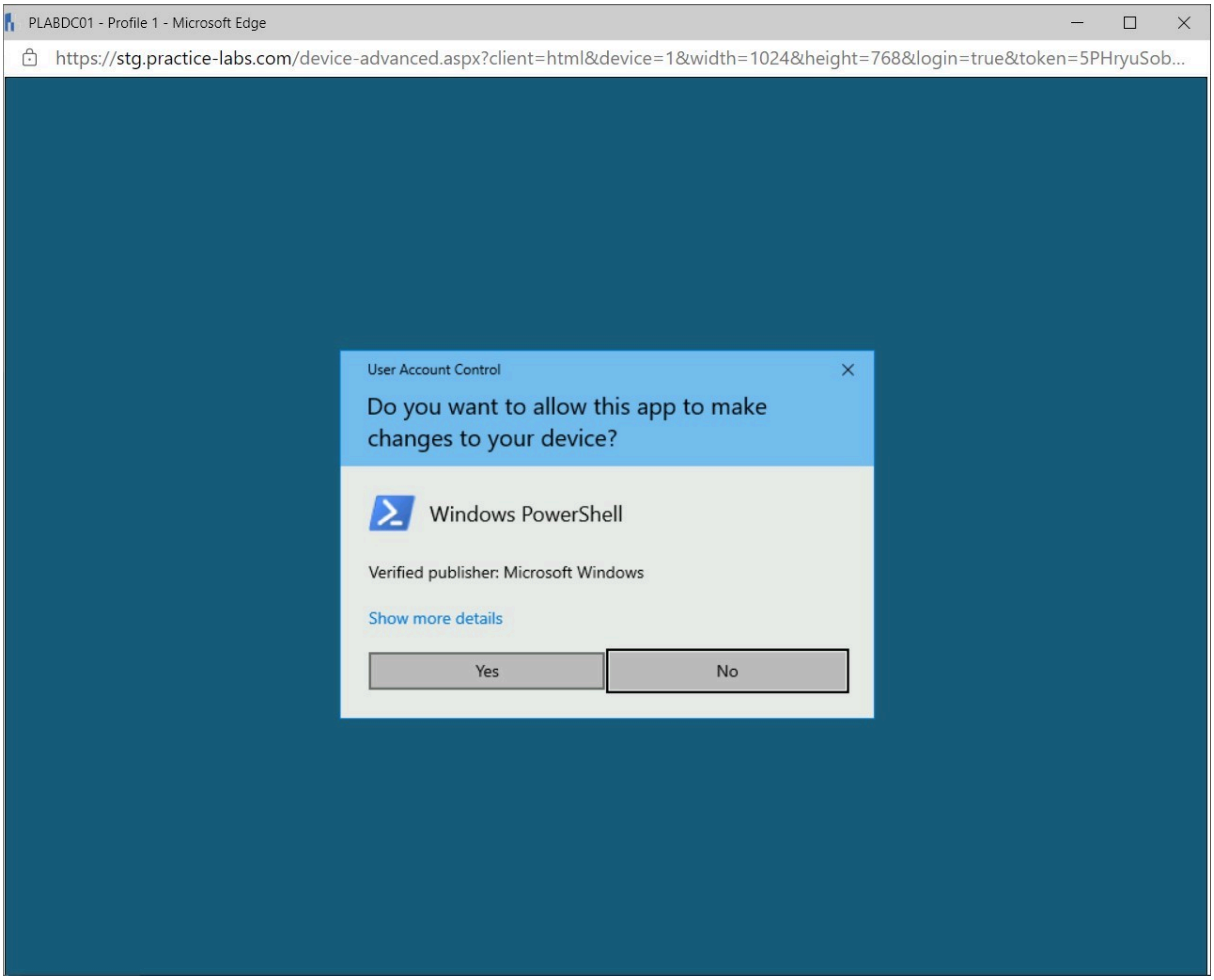

Figure 1.21 Screenshot of PLABDC01: Displaying clicking Yes in the User Account Control pop-up window.

Step 3

In the **Administrator: Windows PowerShell** window, type the following and press **Enter**:

```
Install-WindowsFeature -Name nlb -
IncludeAllSubFeature -IncludeManagementTools
```

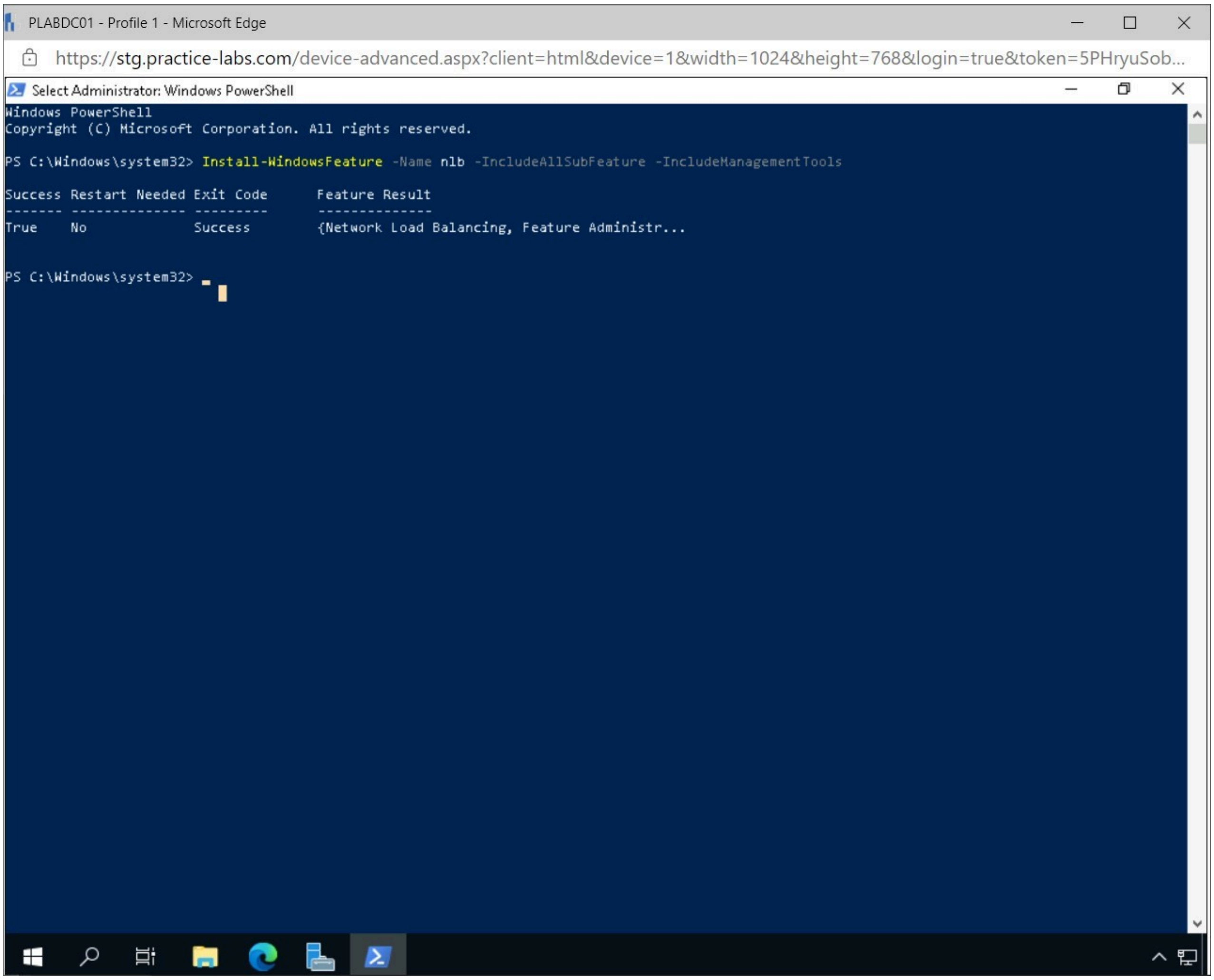

Figure 1.22 Screenshot of PLABDC01: Displaying executing the specified command in Windows PowerShell.

> ***Note:*** *By executing the specified command, the Network Load Balancing feature will be installed on the device. Similarly, Server Manager can be used to install the feature using the graphical user interface.*

Close **Windows PowerShell** window.

Step 4

Connect to **PLABDM01.**

Right-click **Start** and select **Windows PowerShell (Admin**).

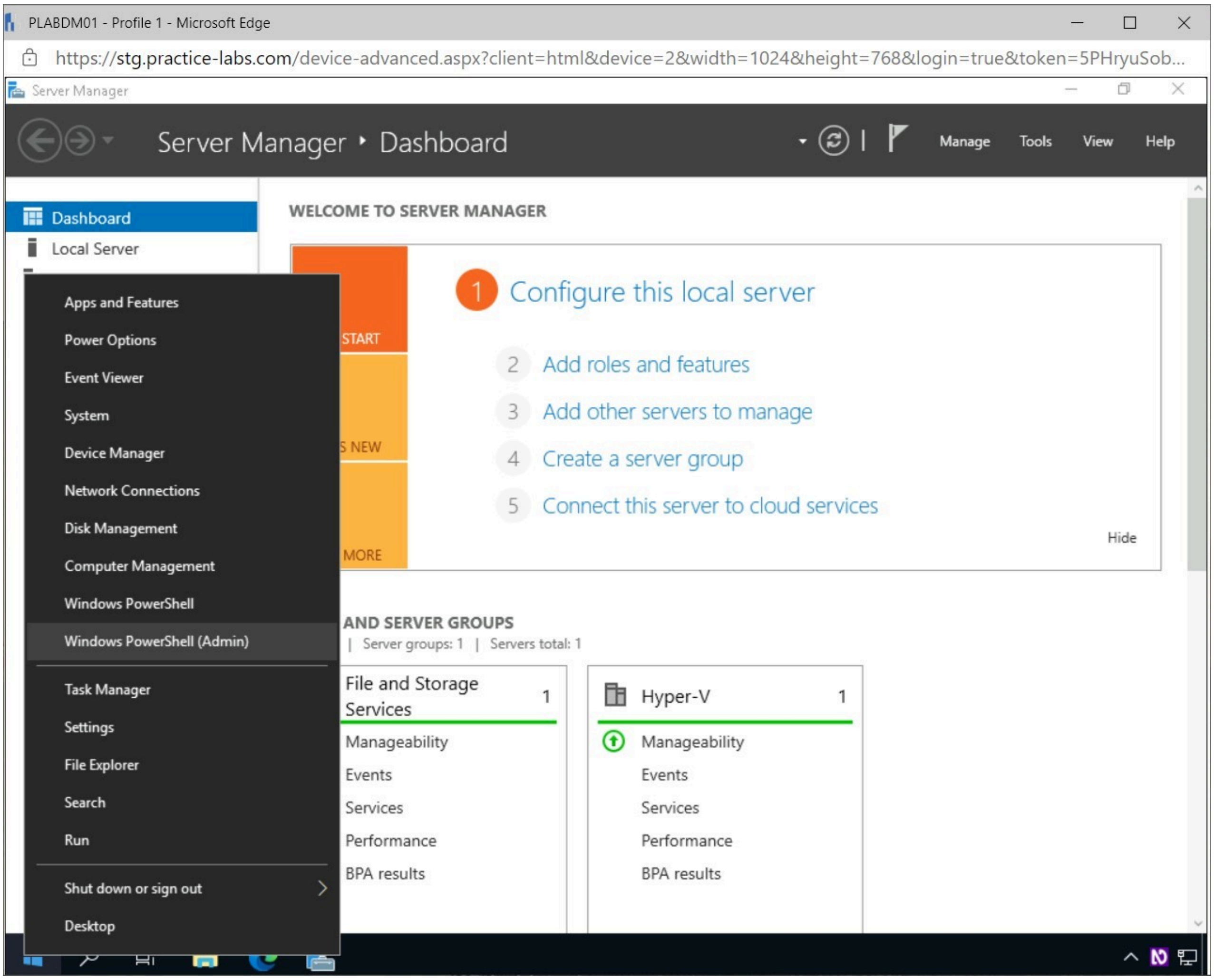

Figure 1.23 Screenshot of PLABDM01: Displaying right-clicking Start and selecting Windows Powershell (Admin).

Step 5

Click **Yes** in the **User Account Control** pop-up window.

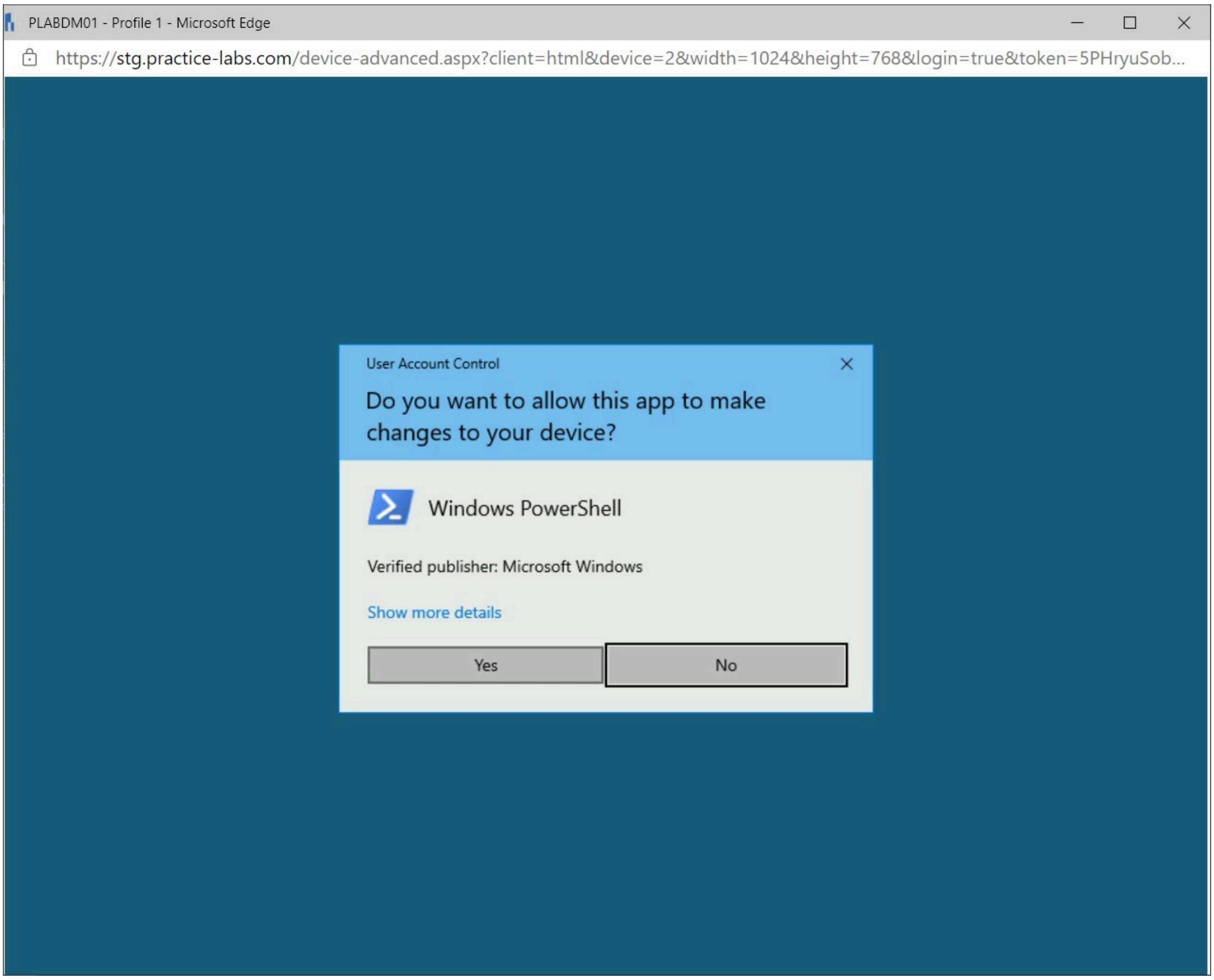

Figure 1.24 Screenshot of PLABDM01: Displaying clicking Yes in the User Account Control pop-up window.

Step 6

In the **Administrator: Windows PowerShell** window, type the following command and press **Enter**:

```
Install-WindowsFeature -Name nlb -
IncludeAllSubFeature -IncludeManagementTools
```

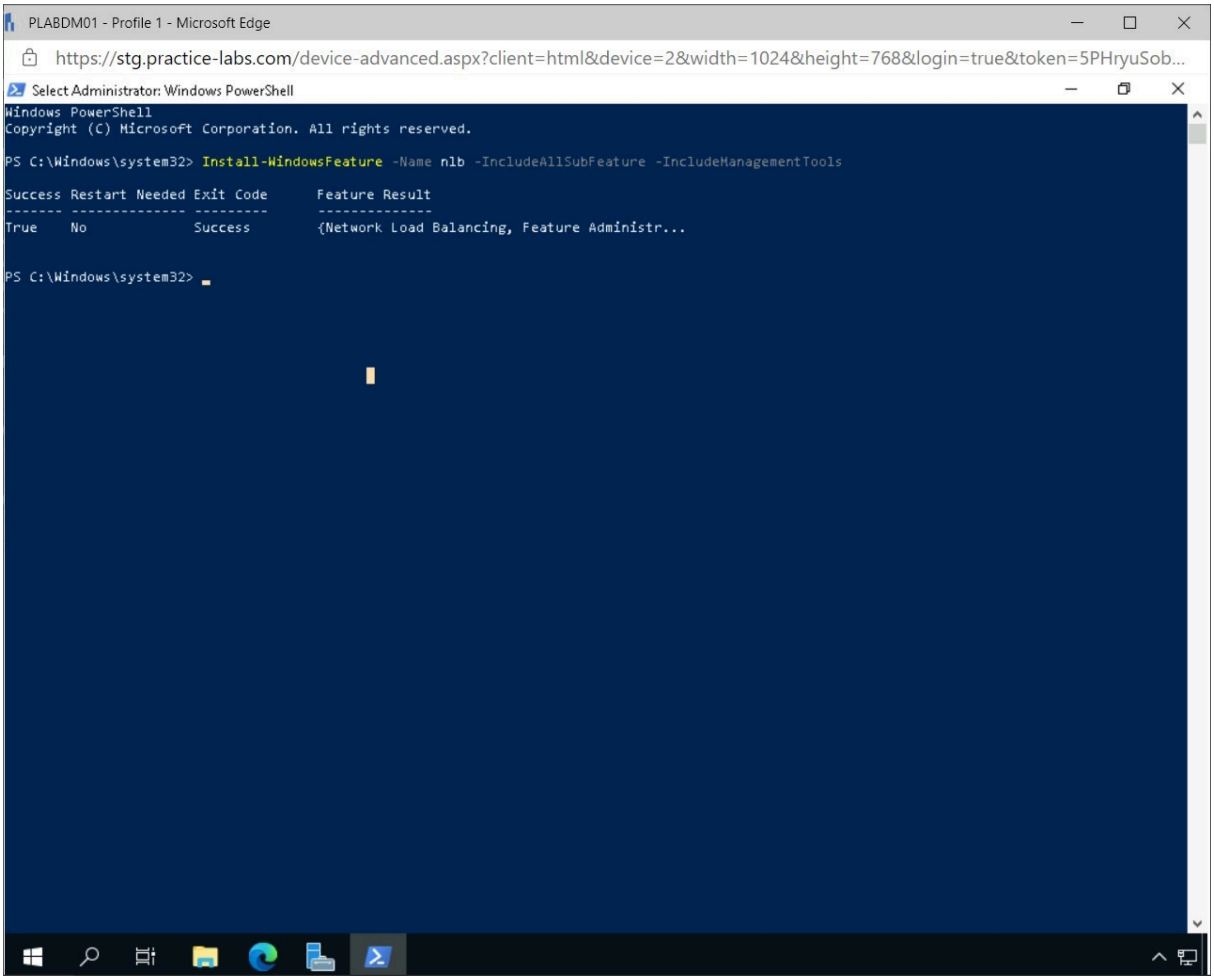

Figure 1.25 Screenshot of PLABDM01: Displaying executing the specified command in Windows PowerShell.

Close **Windows PowerShell** window.

Task 3 - Configure a Network Load Balancing Cluster

After the network load balancing feature has been installed on the servers, a cluster needs to be configured, and the specified devices must be added.

In this task, a network load balancing cluster will be configured.

Step 1

Connect to **PLABDC01.**

In **Server Manager,** click **Tools** and select **Network Load Balancing Manager**.

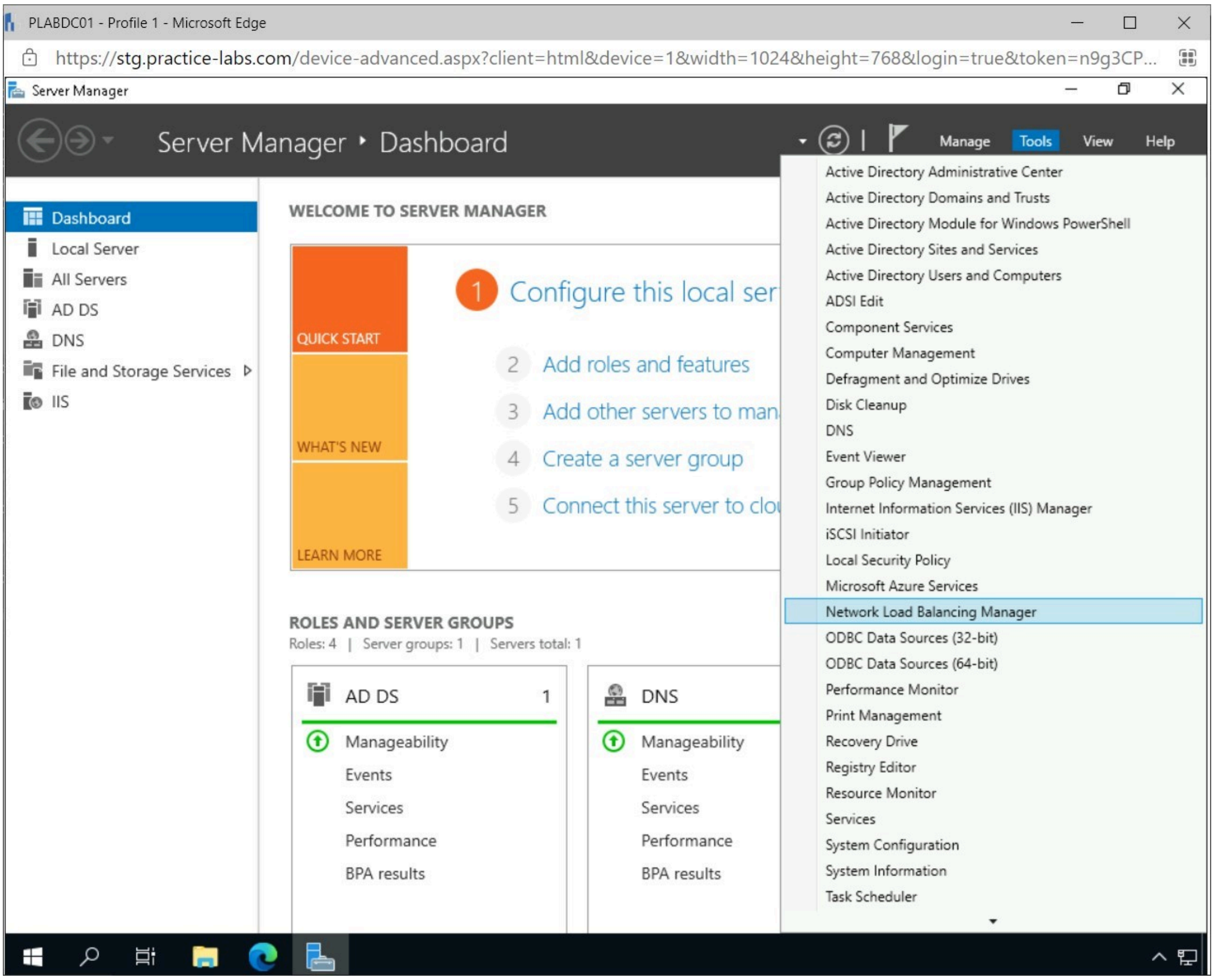

Figure 1.26 Screenshot of PLABDC01: Displaying opening the Network Load Balancing Manager from Server Manager.

Step 2

In the **Network Load Balancing Manager** window, select the **Cluster** menu and click **New**.

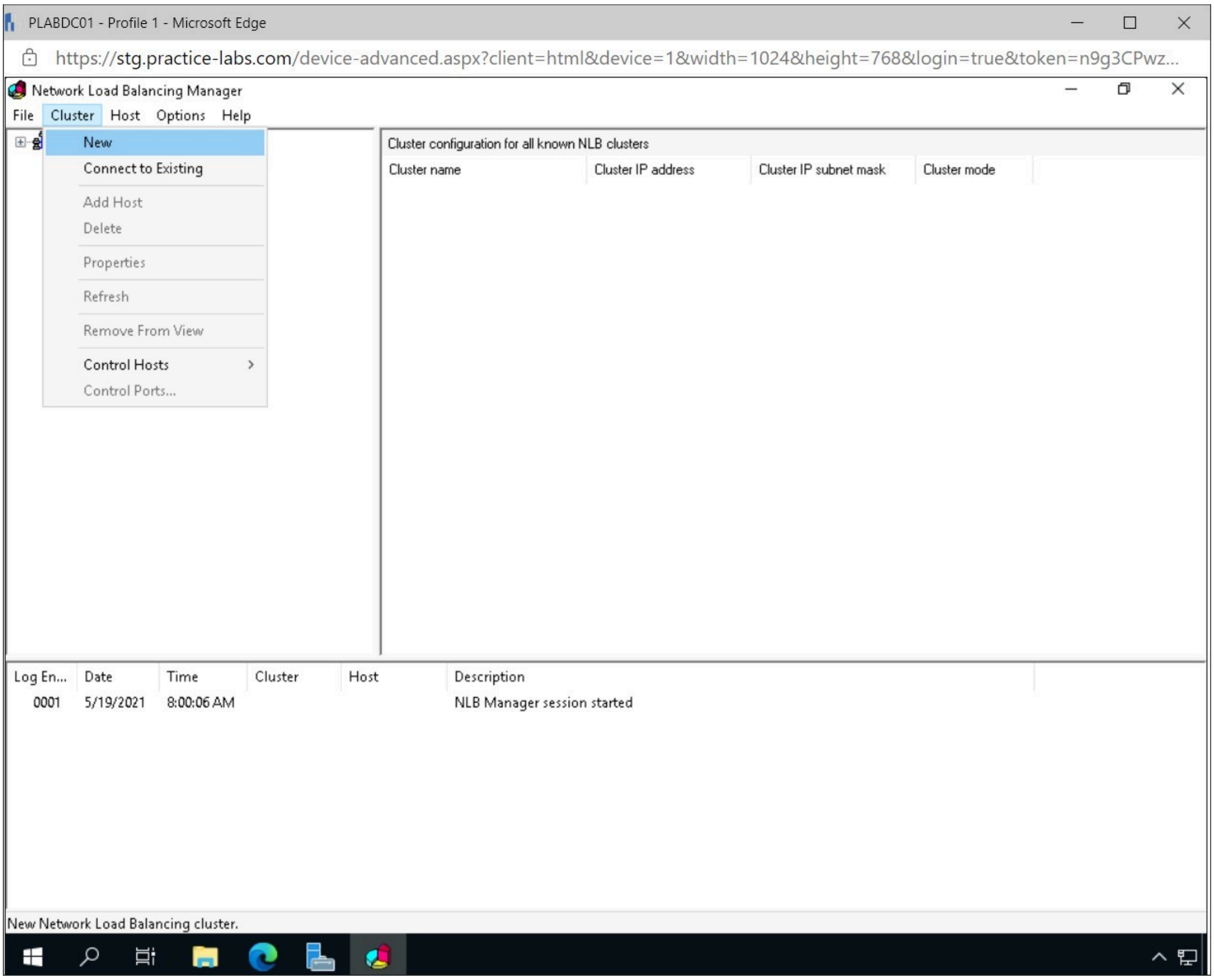

Figure 1.27 Screenshot of PLABDC01: Displaying the Network Load Balancing Manager and selecting New from the Cluster menu.

Step 3

In the **New Cluster: Connect** window enter the following in the **Host** field and click **Connect**.

```
plabdc01
```

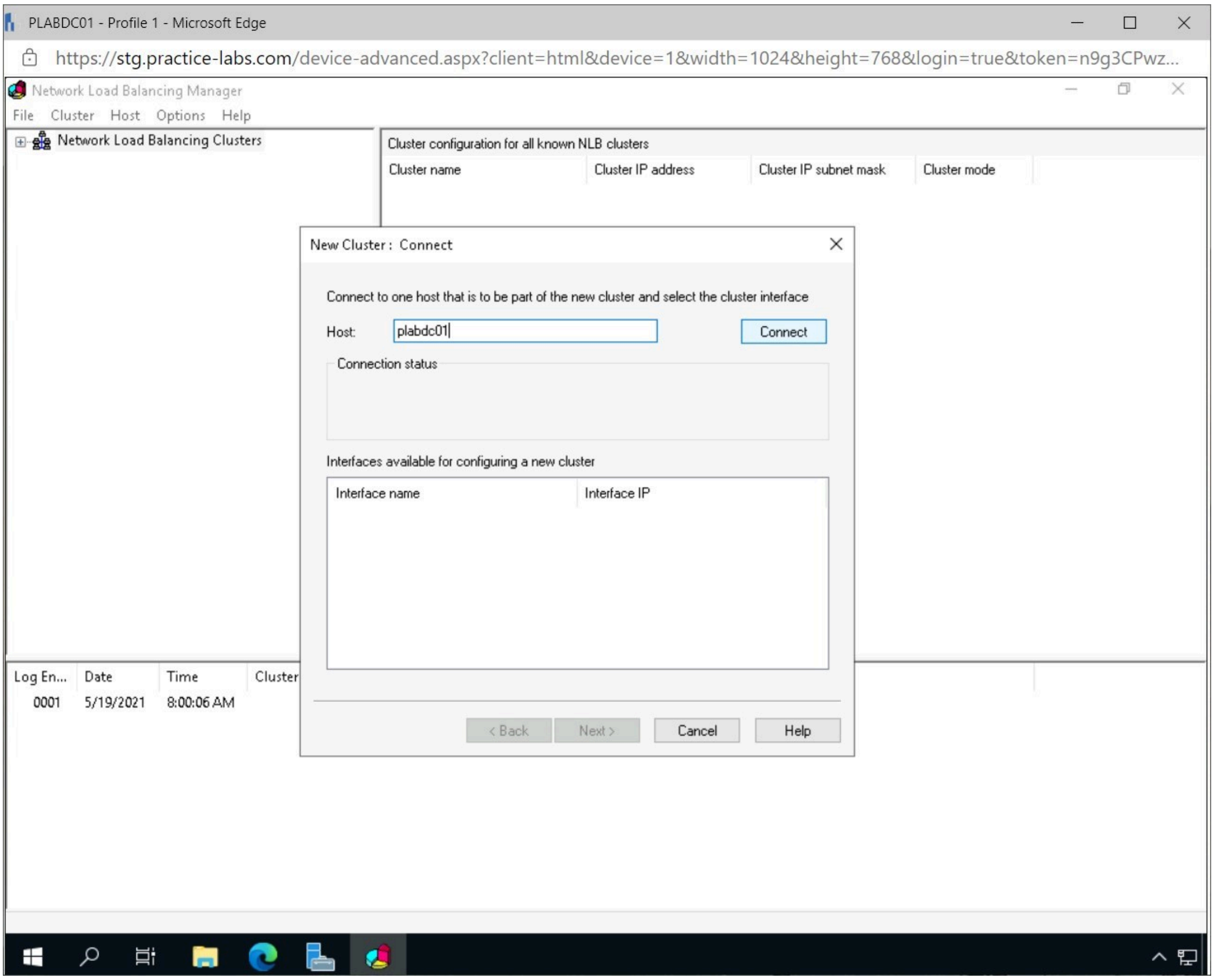

Figure 1.28 Screenshot of PLABDC01: Displaying adding a new host to the cluster.

Step 4

In the **New Cluster: Connect** window, select **Ethernet** and click **Next**.

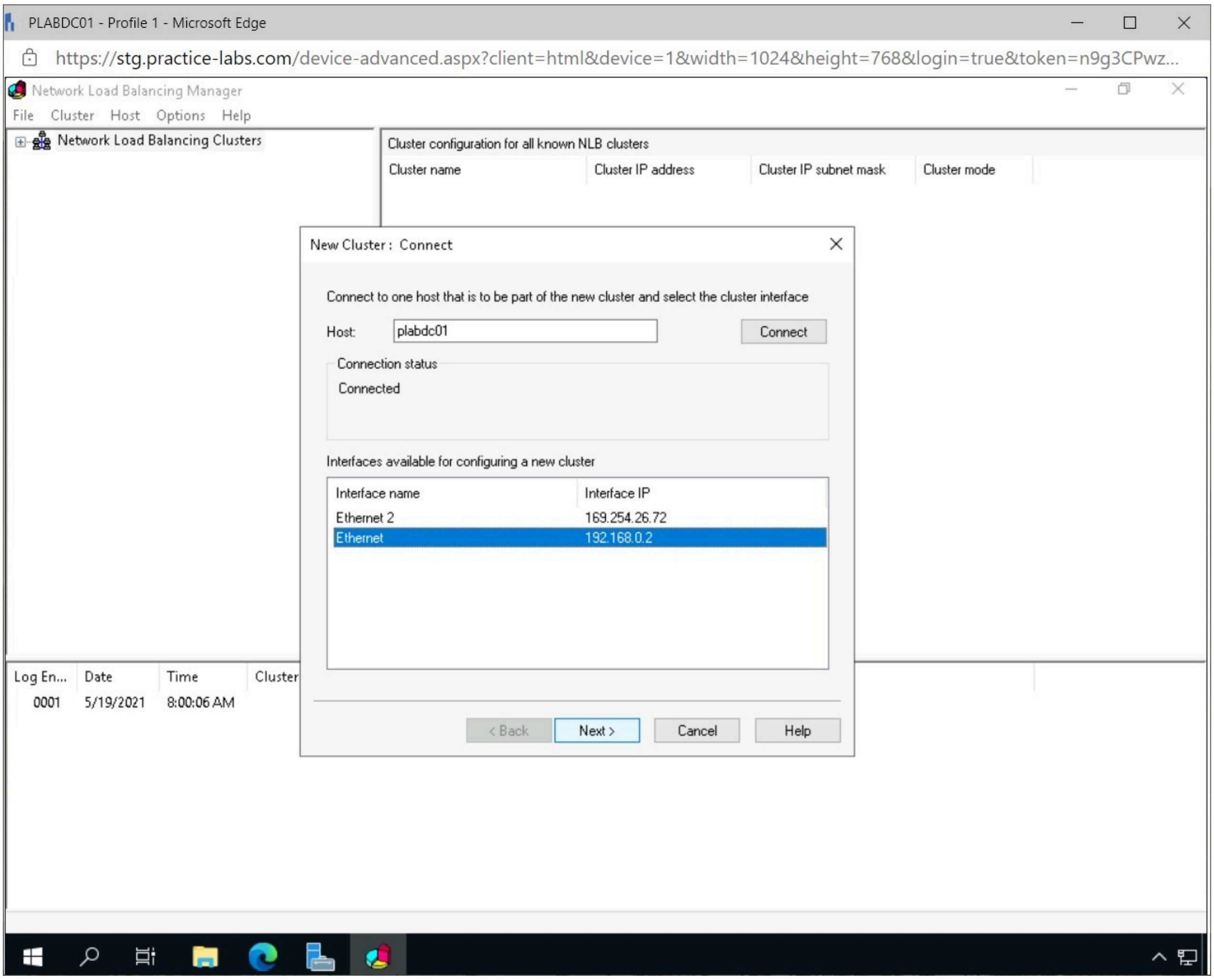

Figure 1.29 Screenshot of PLABDC01: Displaying creating a new cluster and selecting Ethernet as the interface.

Step 5

Leave the default settings in the **New Cluster: Host Parameters** window and click **Next**.

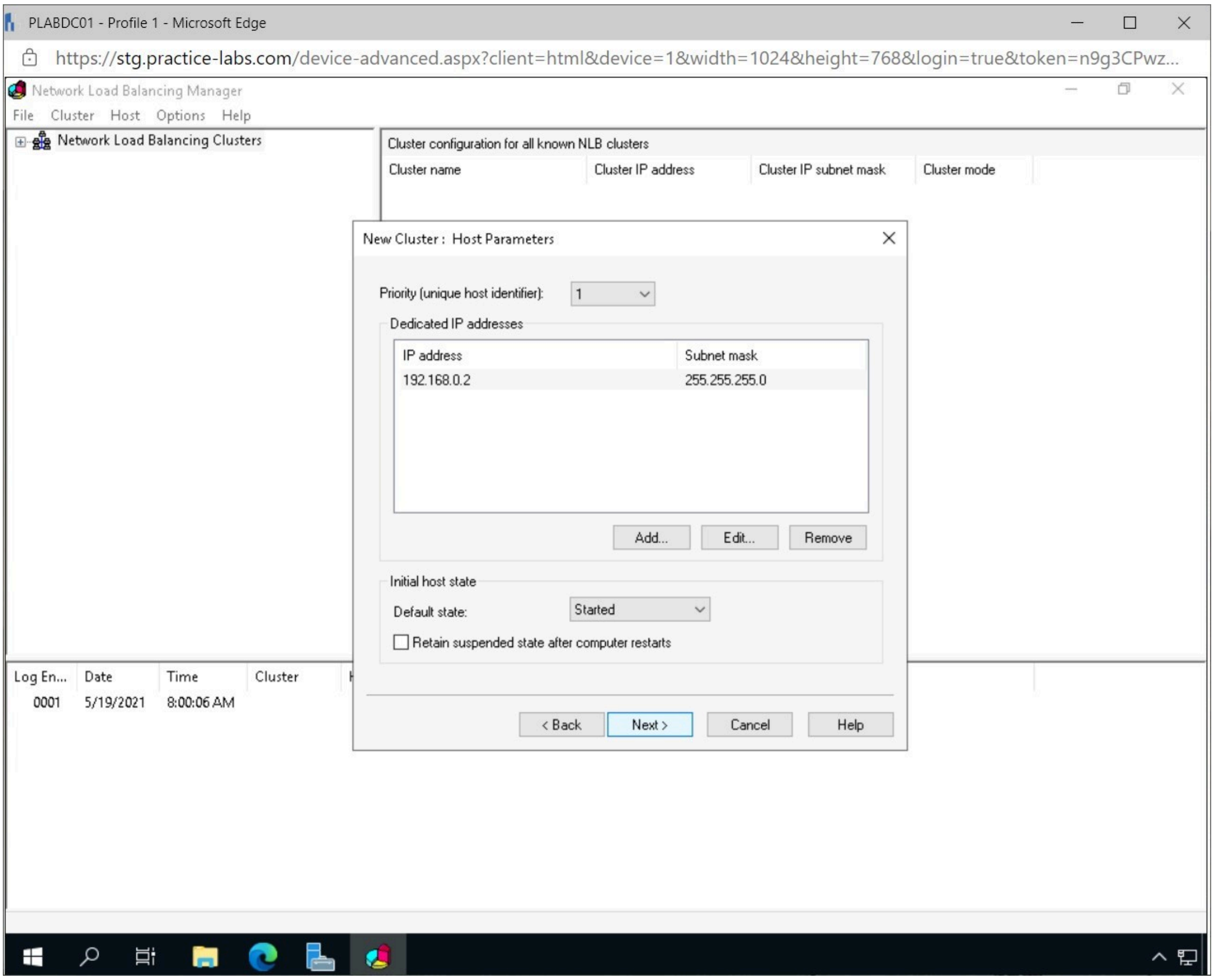

Figure 1.30 Screenshot of PLABDC01: Displaying clicking Next on the New Cluster: Host Parameters window.

Step 6

Click **Add** in the **New Cluster: Cluster IP Addresses** window.

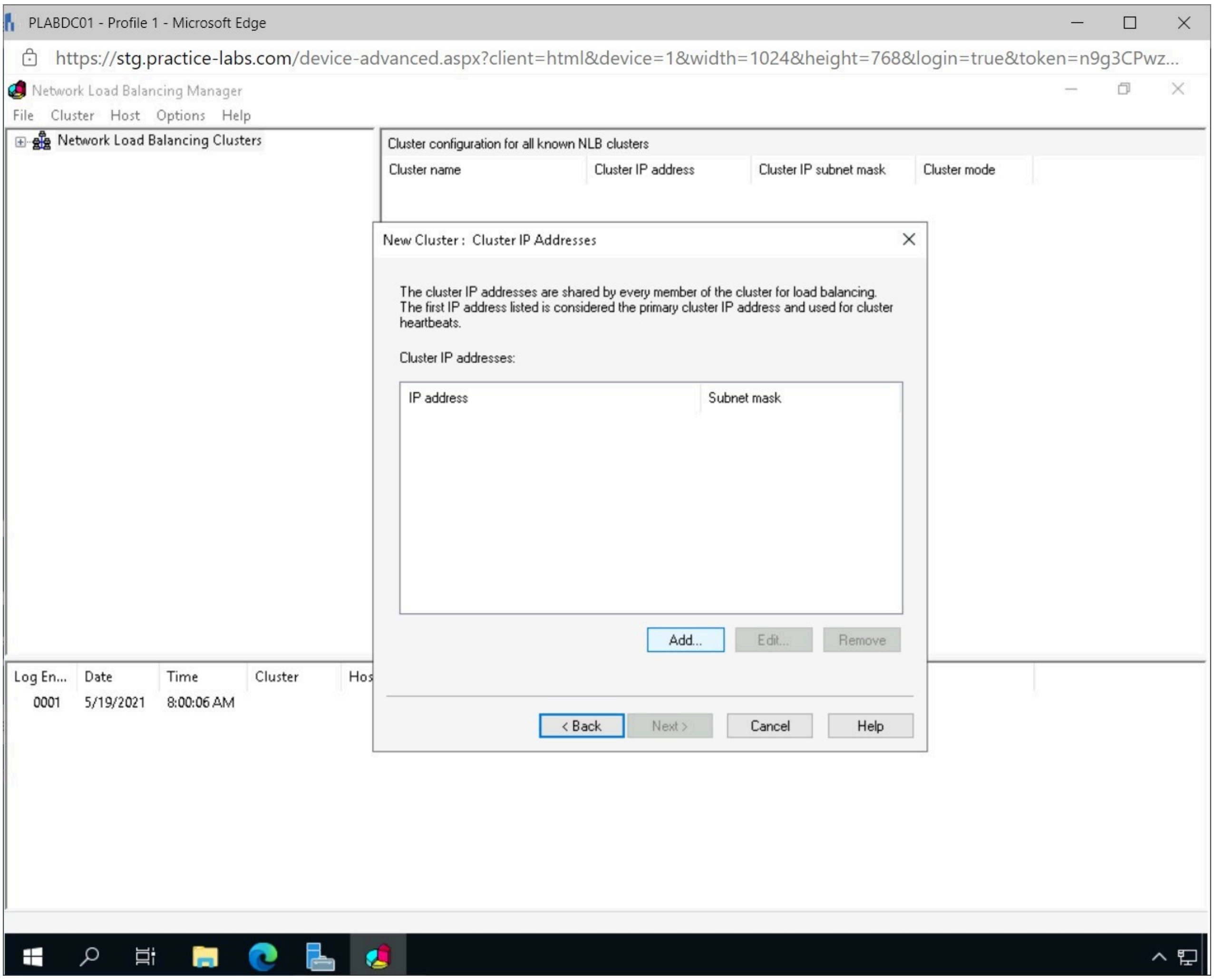

Figure 1.31 Screenshot of PLABDC01: Displaying adding an IP address for the new cluster.

Step 7

In the **Add IP Address** window, enter the following and click **OK**.

```
IPv4 address: 192.168.0.20
Subnet mask: 255.255.255.0
```

Note: *The Subnet mask field gets filled in automatically.*

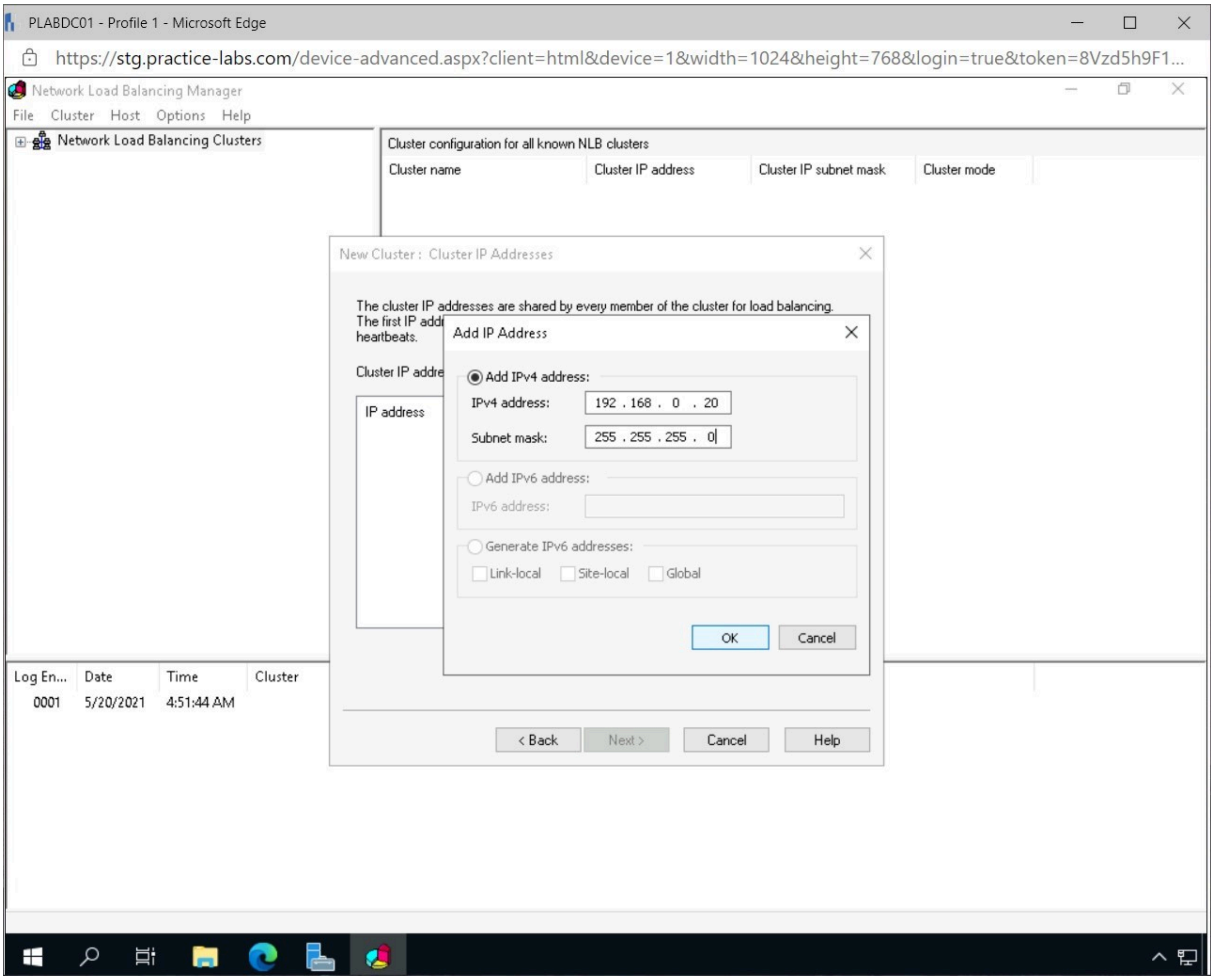

Figure 1.32 Screenshot of PLABDC01: Displaying entering the IP address for the new cluster.

Step 8

Click **Next** in the **New Cluster: Cluster IP Address** window.

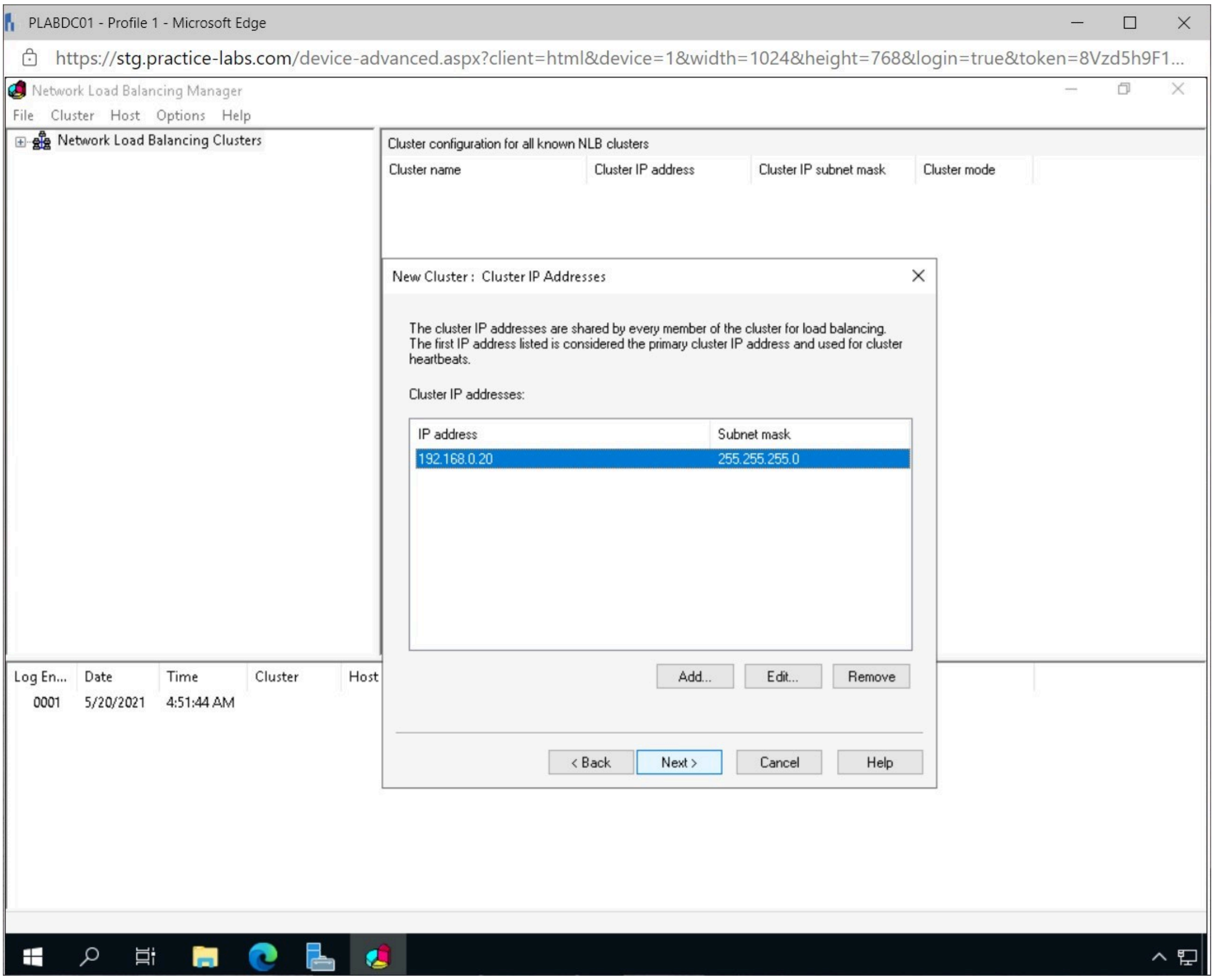

Figure 1.33 Screenshot of PLABDC01: Displaying clicking Next on the New Cluster: Cluster IP Address window.

Step 9

In the **New Cluster: Cluster Parameters** window, select **Multicast,** enter the following in the **Full Internet name** and click **Next:**

```
PLABCluster
```

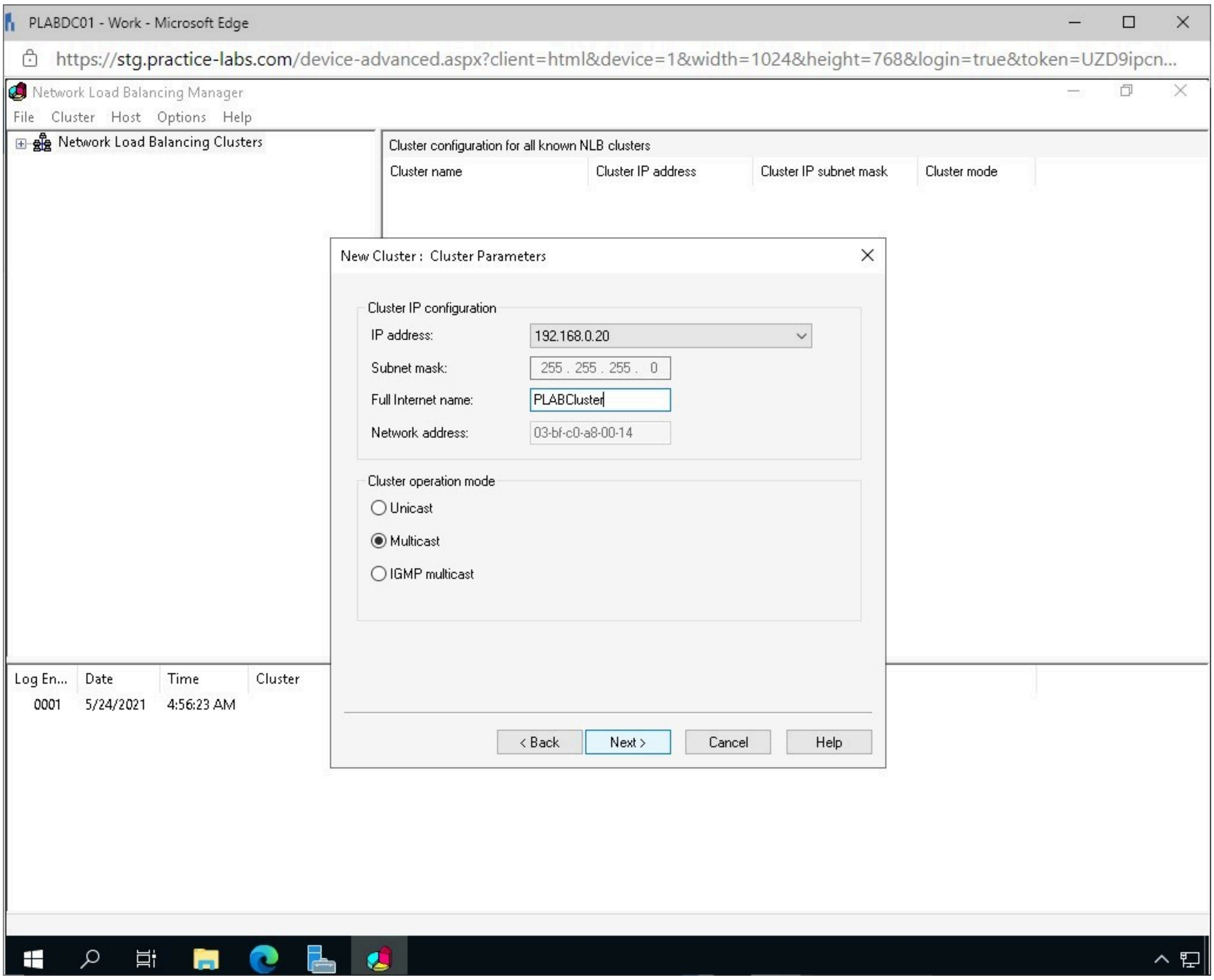

Figure 1.34 Screenshot of PLABDC01: Displaying enabling Multicast and entering the name for the new cluster.

Step 10

Click **Finish** in the **New Cluster: Port Rules** window.

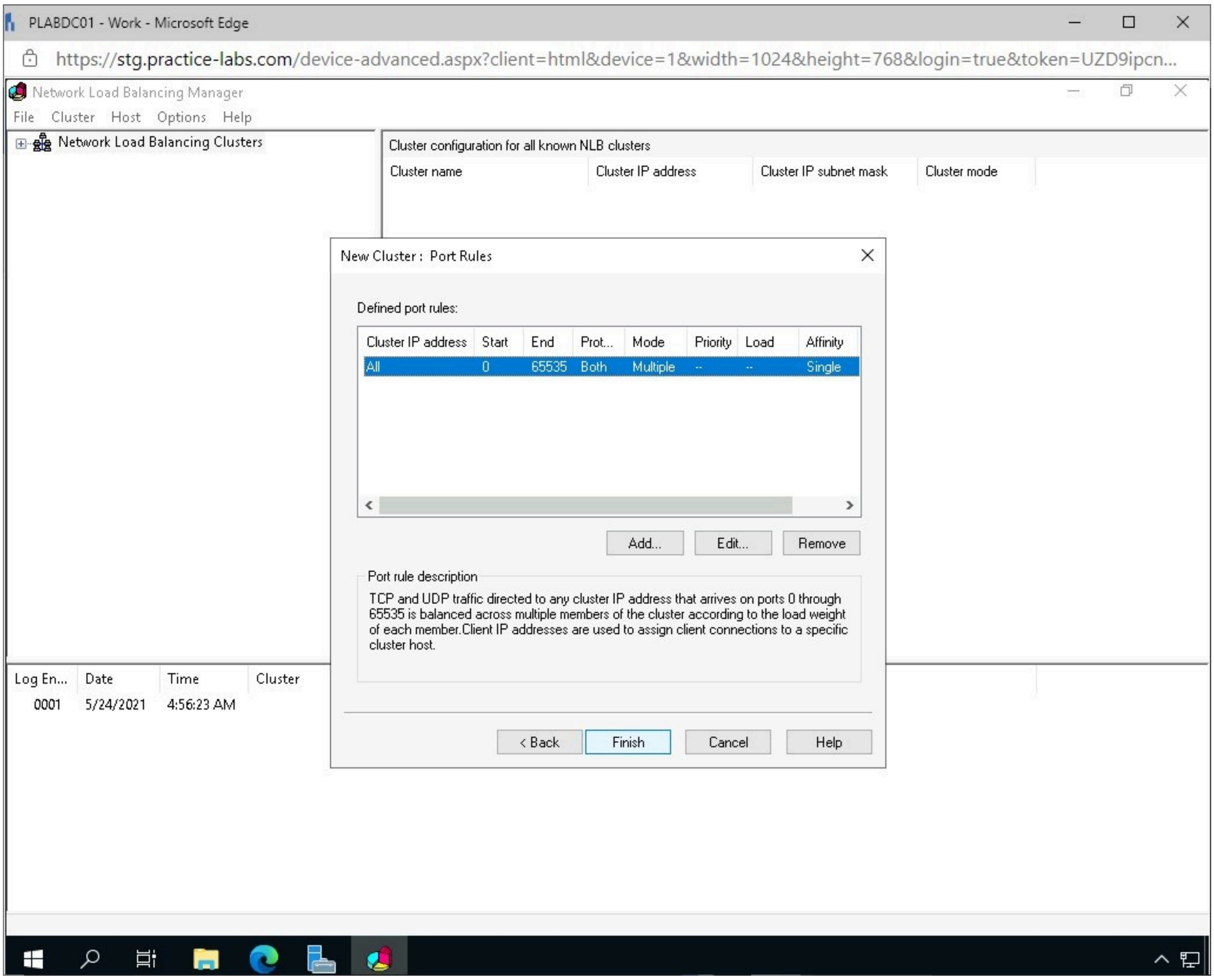

Figure 1.35 Screenshot of PLABDC01: Displaying clicking Finish on the New Cluster: Port Rules window.

Step 11

On the **Network Load Balancing Manager** window, select and right-click **PLABCluster** and select **Add Host To Cluster**.

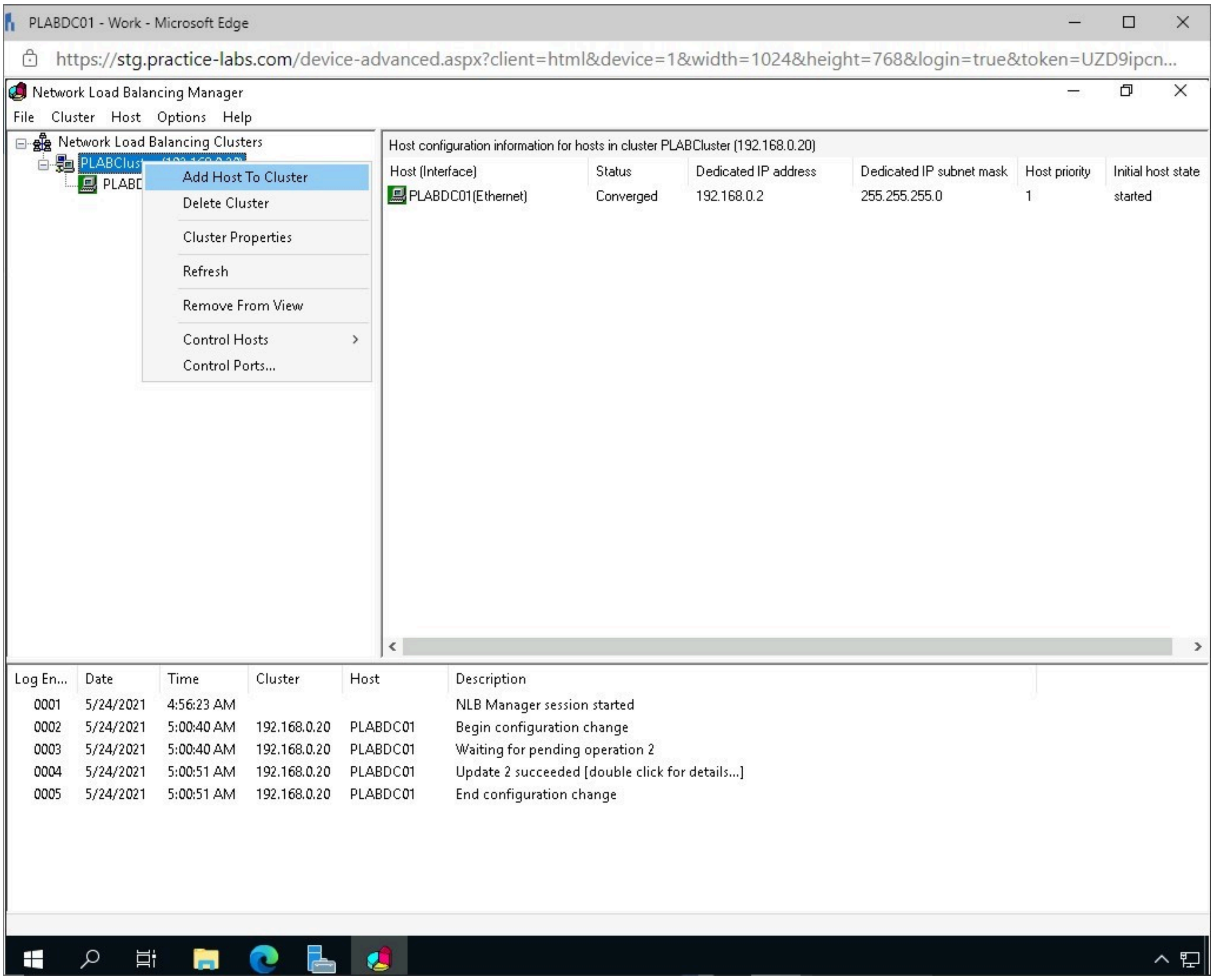

Figure 1.36 Screenshot of PLABDC01: Displaying adding a host to the PLABCluster in Network Load Balancing Manager.

Step 12

In the **Add Host to Cluster : Connect** window, type the following and click **Connect**.

```
plabdm01
```

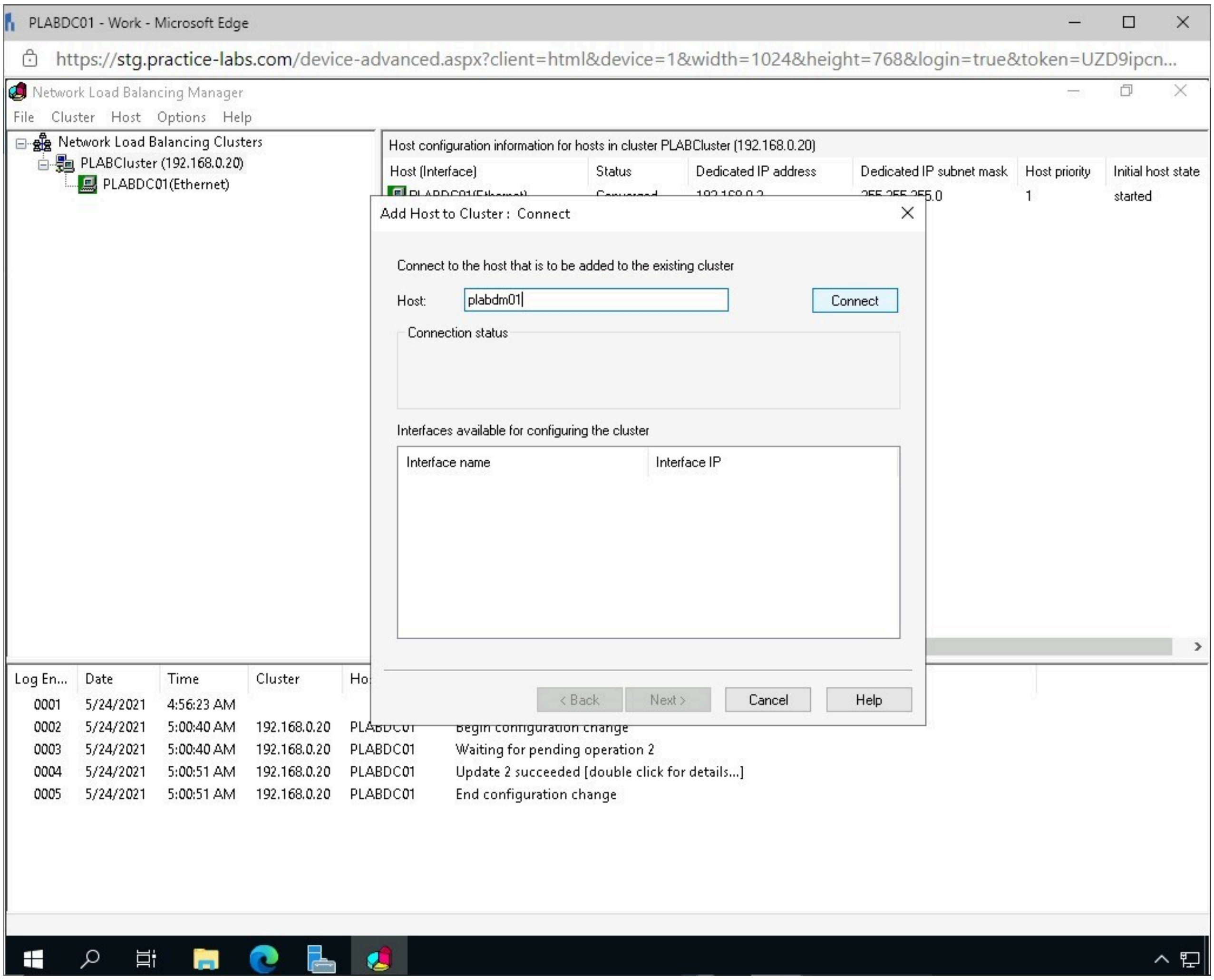

Figure 1.37 Screenshot of PLABDC01: Displaying adding a new host to the PLABCluster.

Step 13

In the **Add Host to Cluster: Connect** window, ensure **Ethernet** is selected and click **Next**.

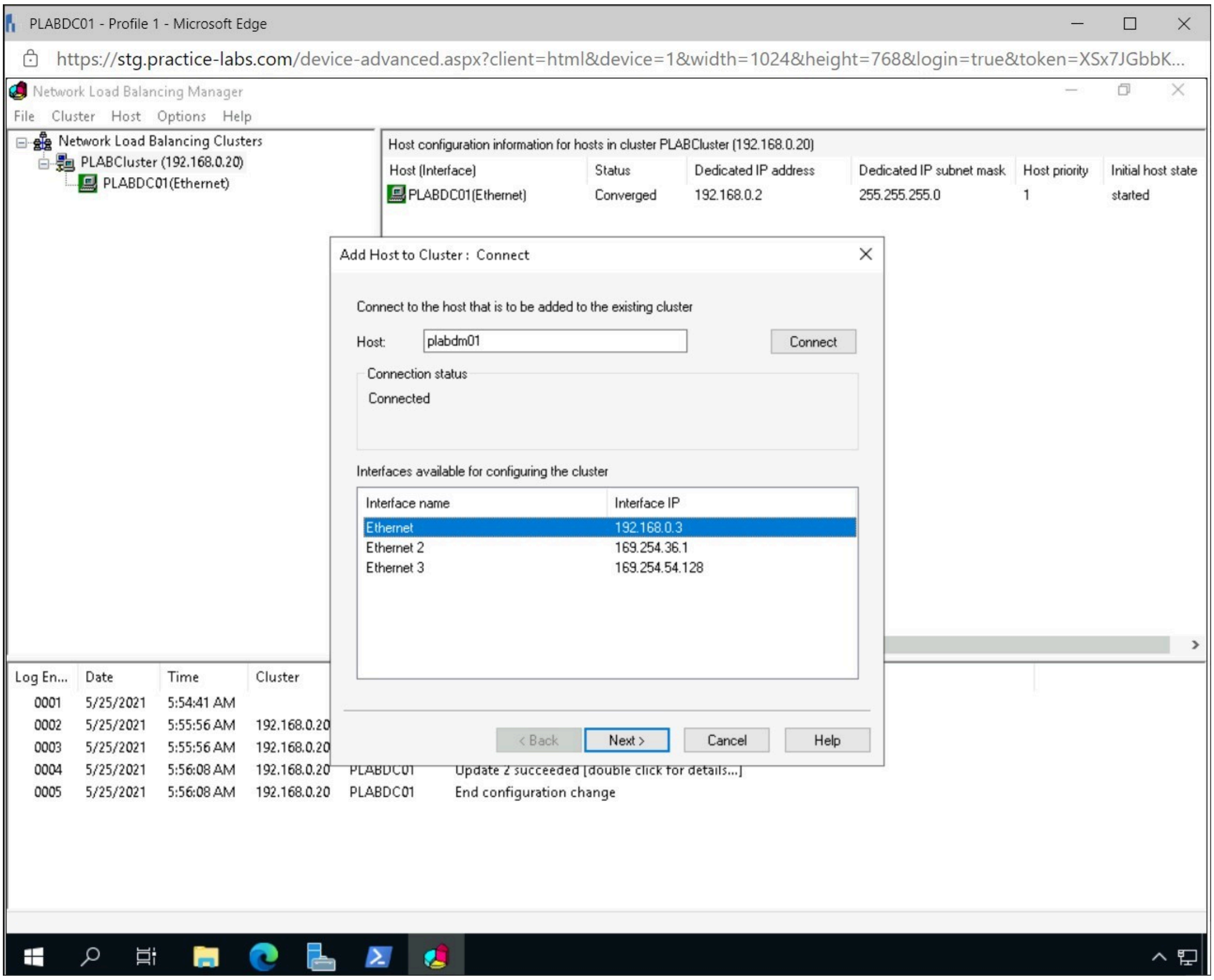

Figure 1.38 Screenshot of PLABDC01: Displaying selecting Ethernet and clicking Next on the Add Host to Cluster: Connect window.

Step 14

In the **Add Host to Cluster: Host Parameters** window, click **Next**.

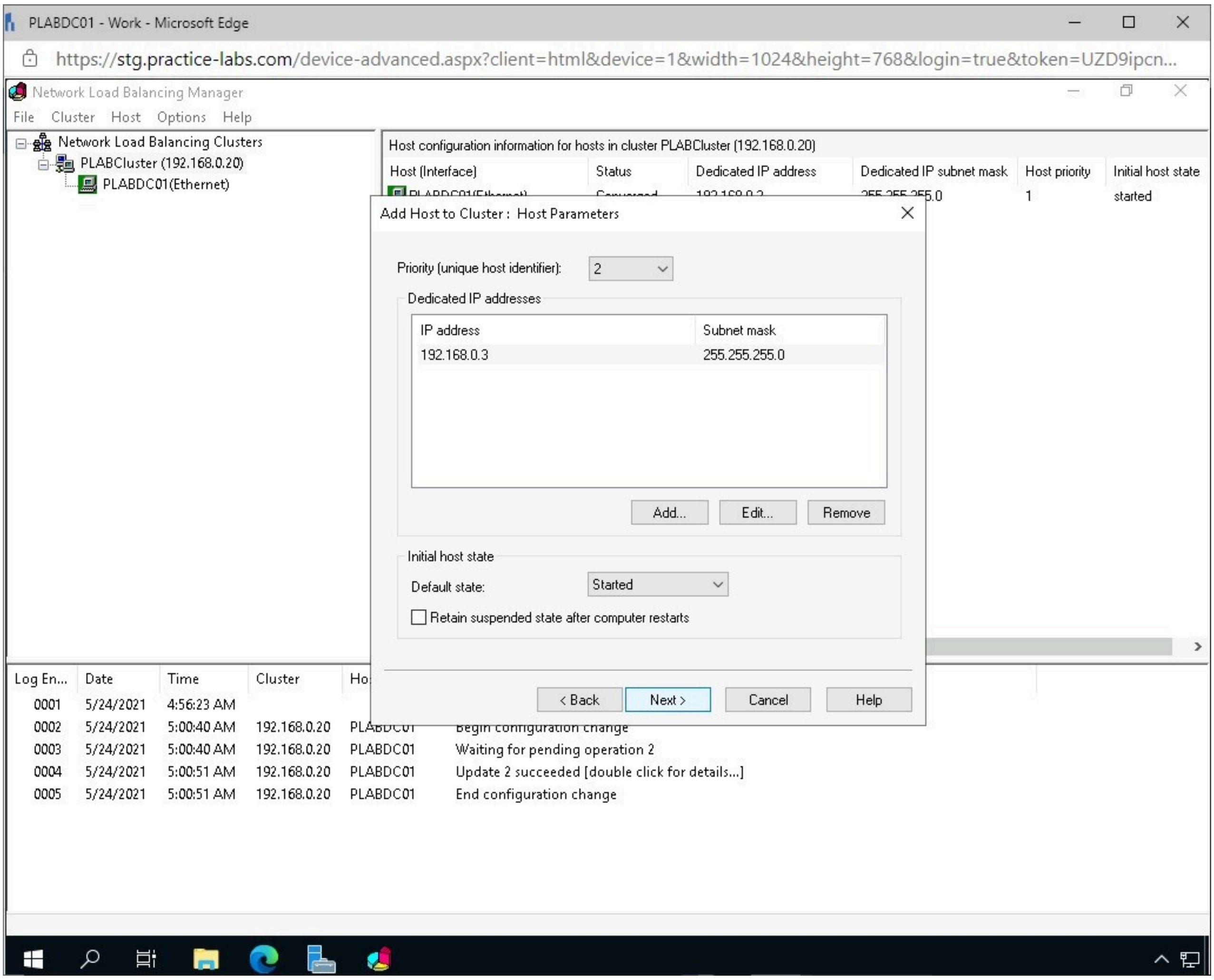

Figure 1.39 Screenshot of PLABDC01: Displaying clicking Next on the Add Host to Cluster: Host Parameters window.

Step 15

In the **Add Host to Cluster: Port Rules** window, click **Finish**.

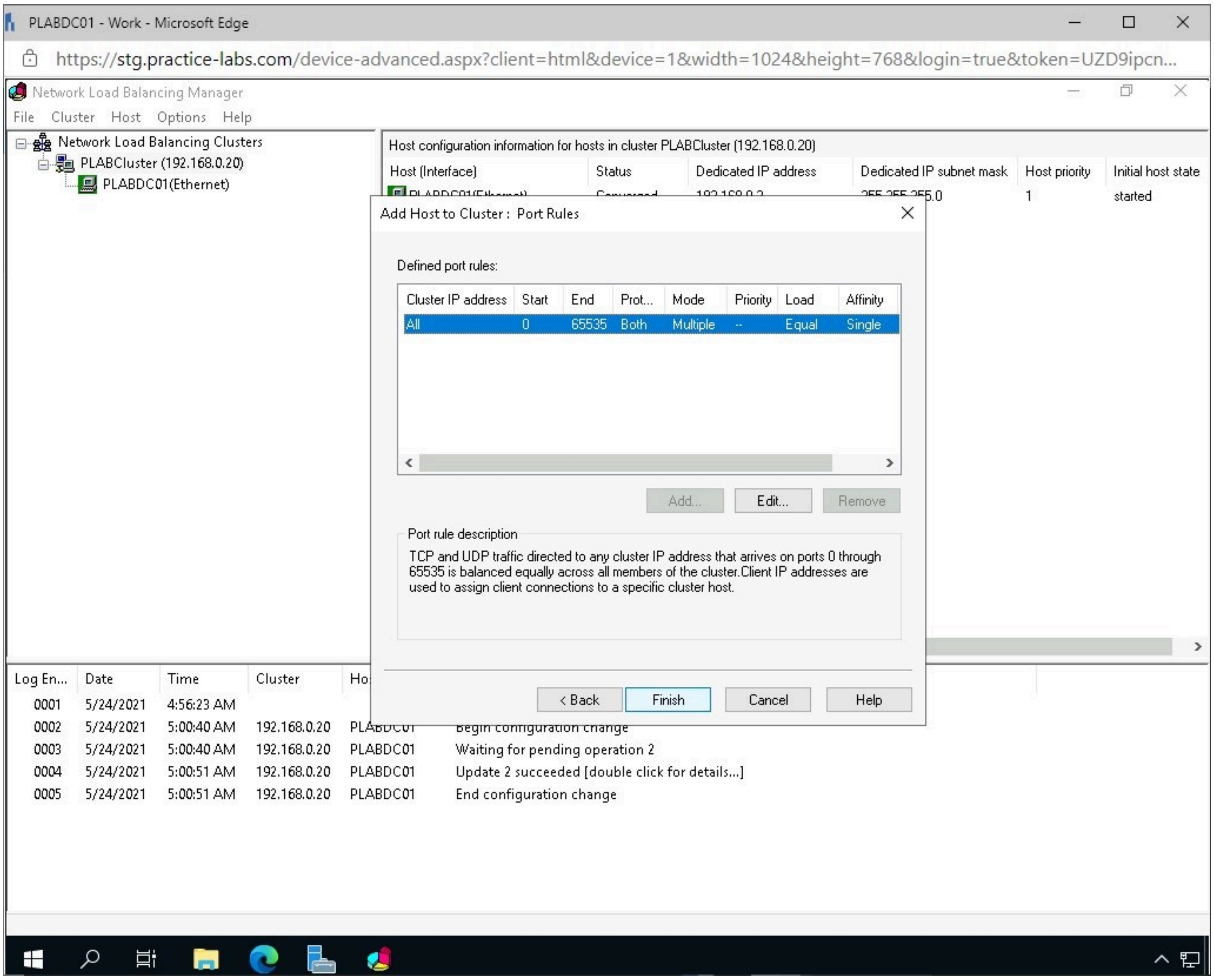

Figure 1.40 Screenshot of PLABDC01: Displaying clicking Finish on the Add Host to Cluster: Port Rules window.

Task 4 - Configure a Forward Lookup Zone for Network Load Balancing Cluster

To ensure that the newly created Network Load Balancing Cluster will be contactable by devices on the network, a new forward lookup zone needs to be created.

In this task, a forward lookup zone will be created in the DNS of the network.

Step 1

Connect to **PLABDC01**.

Open **Server Manager** from the **Taskbar**.

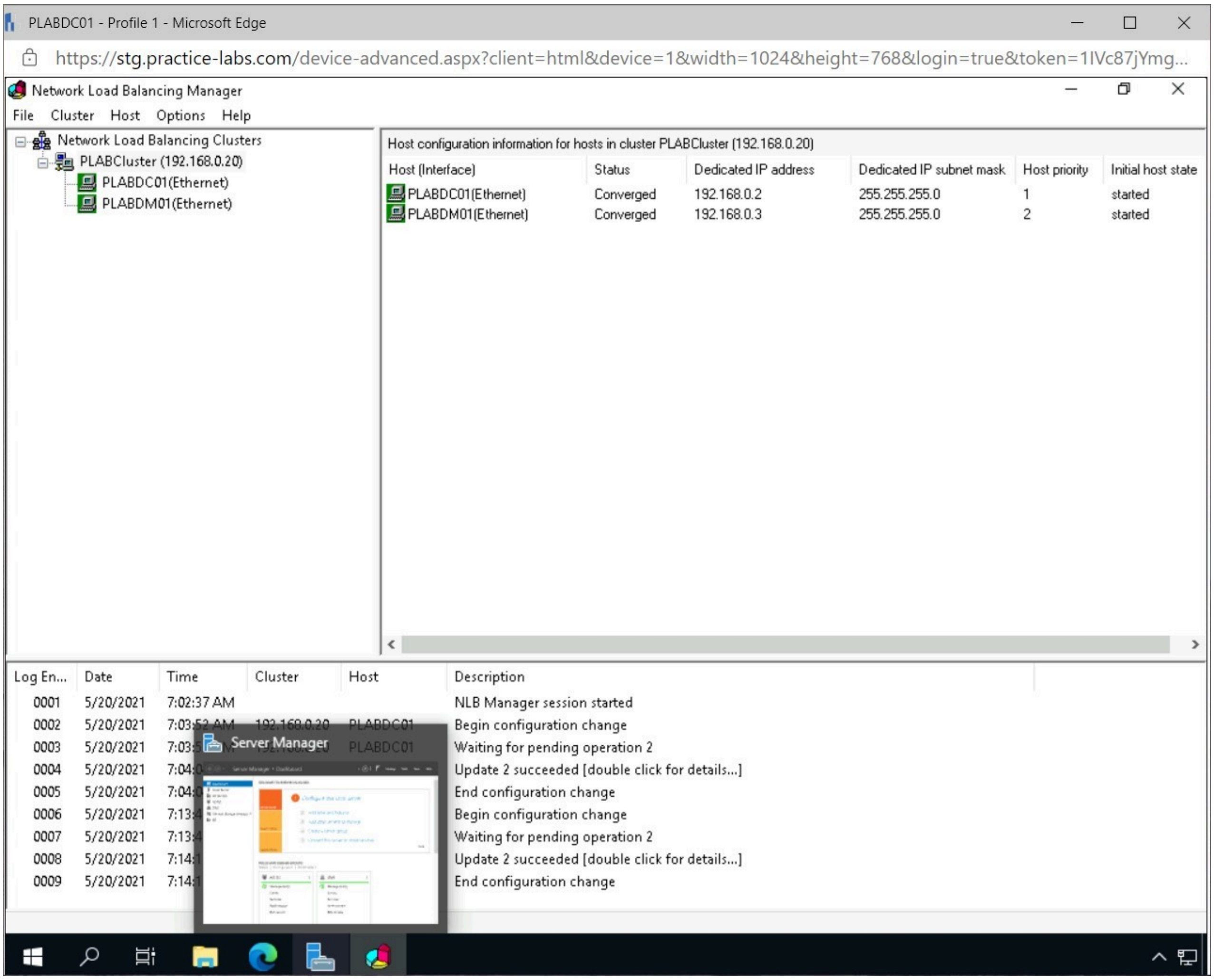

Figure 1.41 Screenshot of PLABDC01: Displaying opening Server Manager from the Taskbar.

Step 2

On the **Server Manager** window, click **Tools** and select **DNS**.

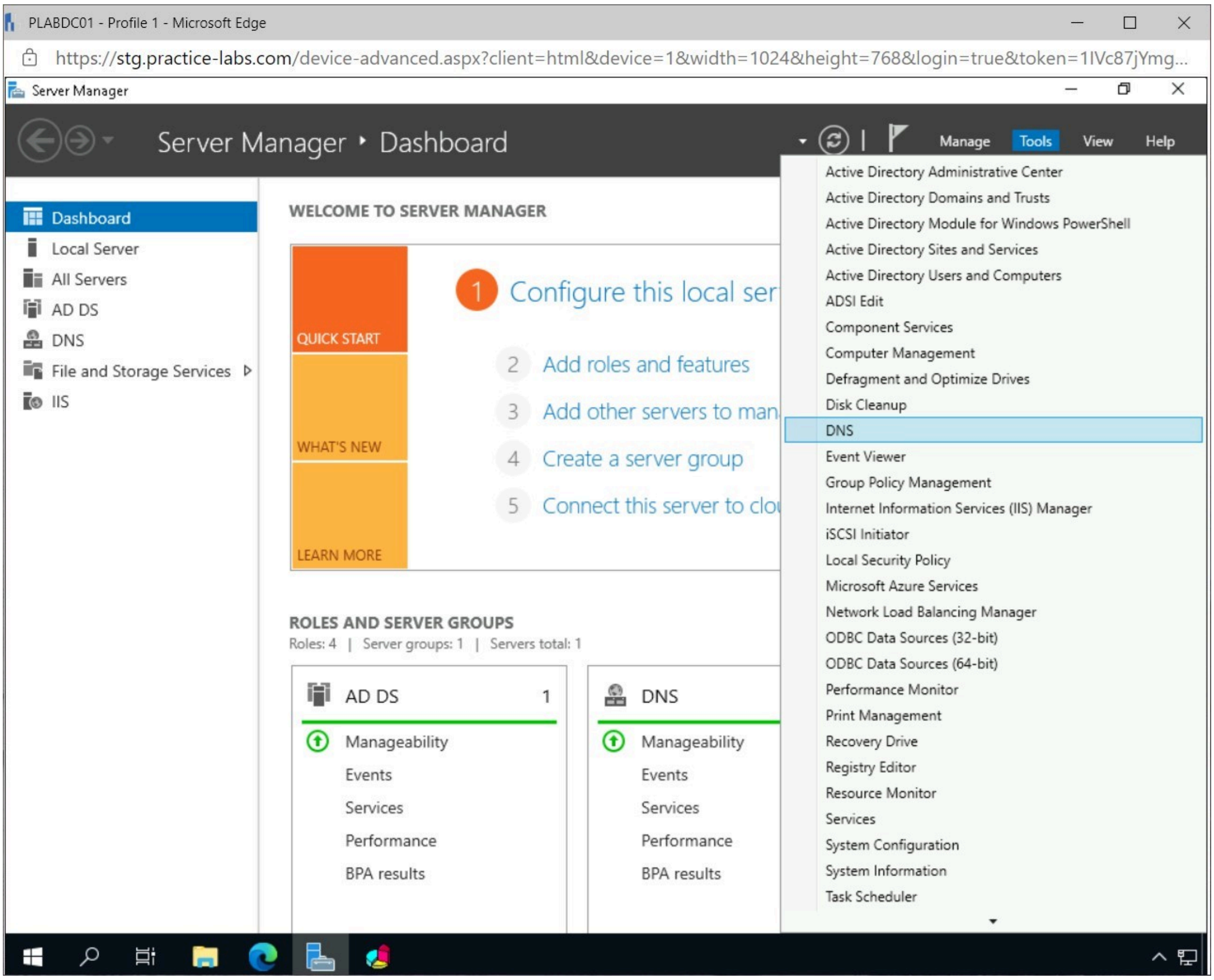

Figure 1.42 Screenshot of PLABDC01: Displaying opening DNS from the Tools menu in Server Manager.

Step 3

From the **DNS Manager** window, expand the nodes, **PLABDC01** >> **Forward Lookup Zones** and select **PRACTICELABS.COM**.

Right-click **PRACTICELABS.COM** and select **New Host (A or AAAA**).

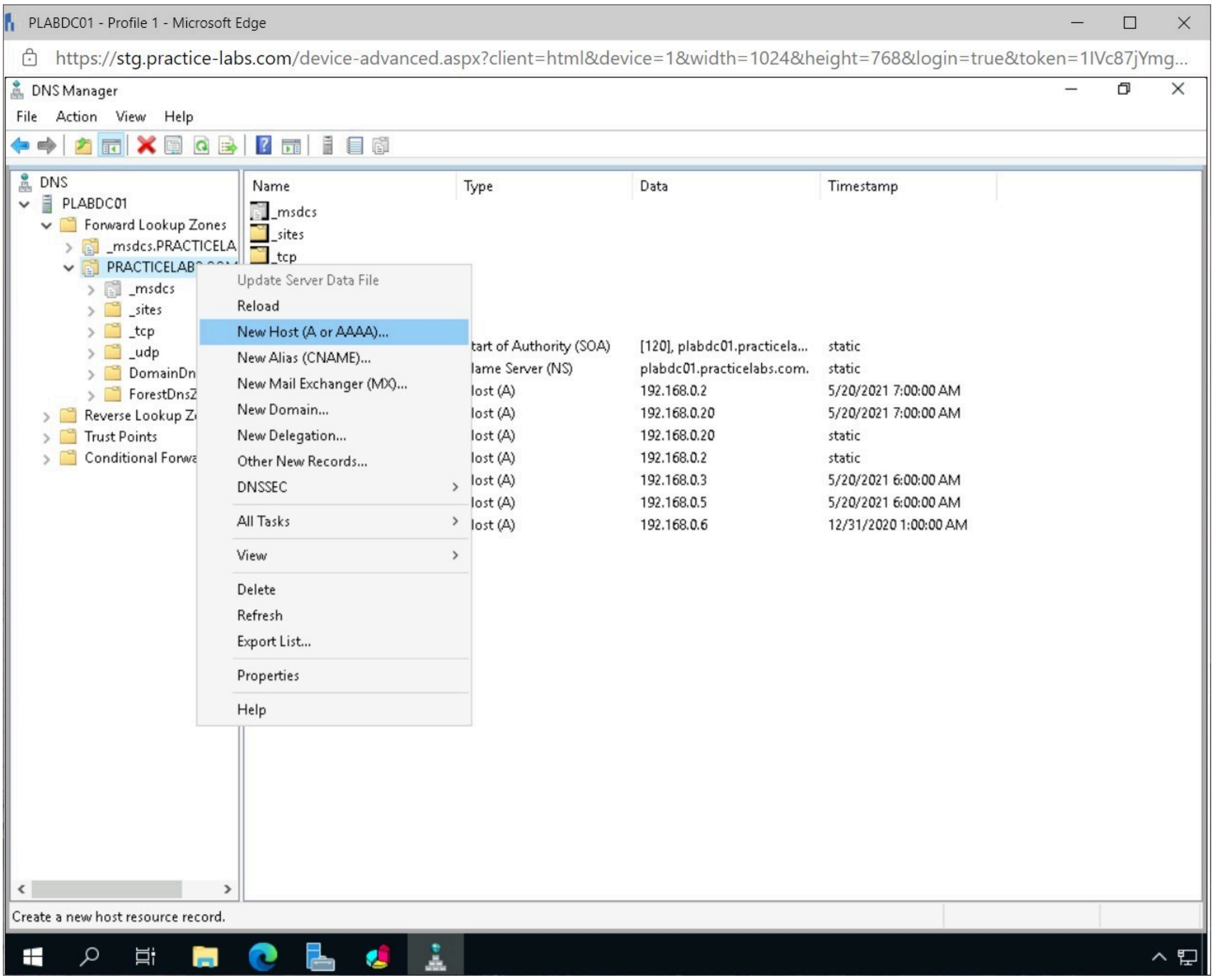

Figure 1.43 Screenshot of PLABDC01: Displaying right-clicking PRACTICELABS.COM and selecting New Host (A or AAAA) on the DNS Manager window.

> ***Note:*** *A forward lookup zone in DNS is created to ensure resources on the network can be accessed using a network name instead of an IP address.*

Step 4

On the **New Host** dialog box, complete the following fields and click **Add Host**.

```
Name(uses parent domain name if blank): PLABcluster
IP Address: 192.168.0.20
```

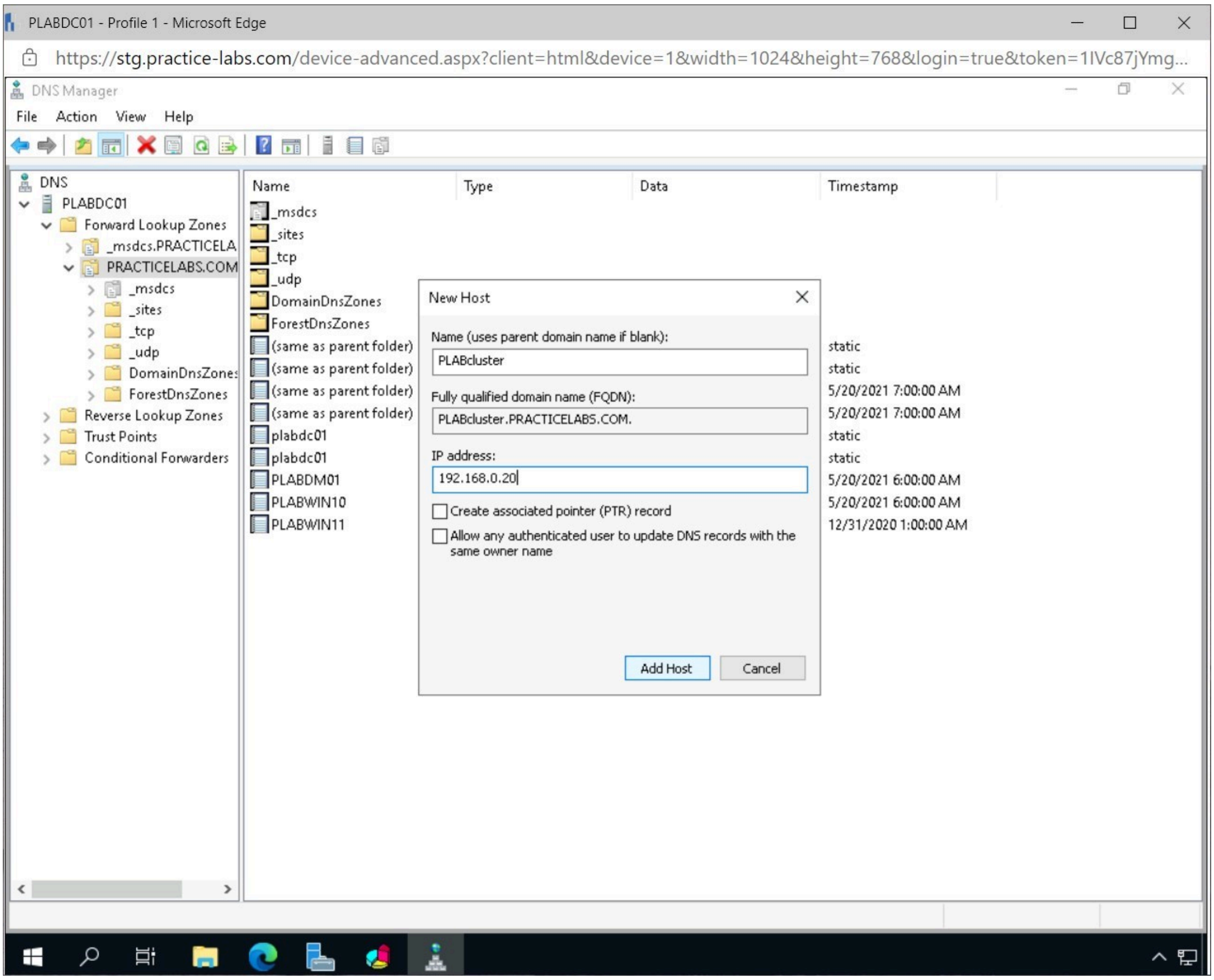

Figure 1.44 Screenshot of PLABDC01: Displaying creating a forward lookup zone for the PLABcluster in the New Host dialog box.

> ***Note:*** *The IP address that was used when creating the new cluster is used to create the forward lookup zone to point to the correct resource.*

Step 5

In the **DNS** pop-up window, click **OK**.

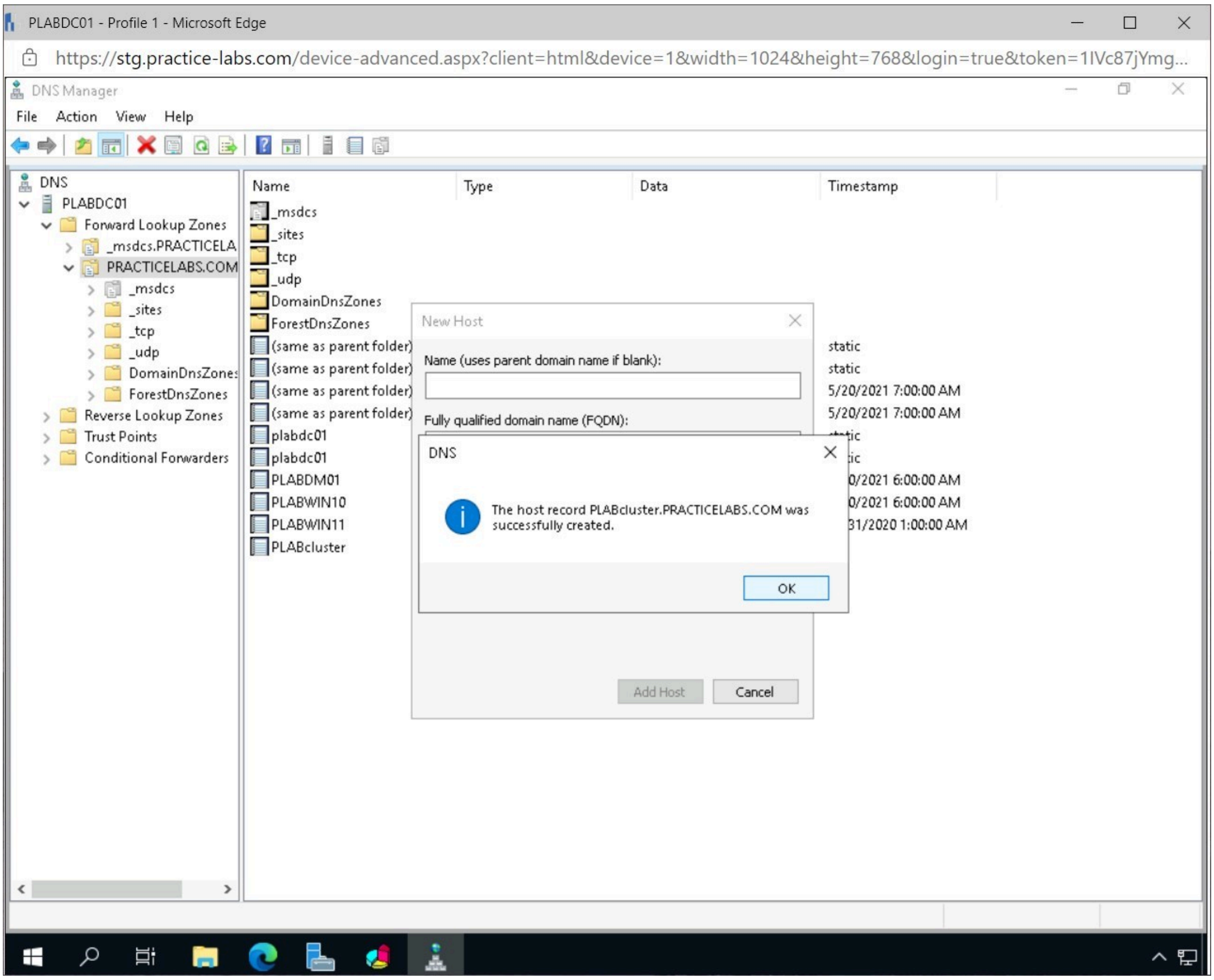

Figure 1.45 Screenshot of PLABDC01: Displaying clicking OK on the DNS pop-up window.

Step 6

Click **Done** in the **New Host** window.

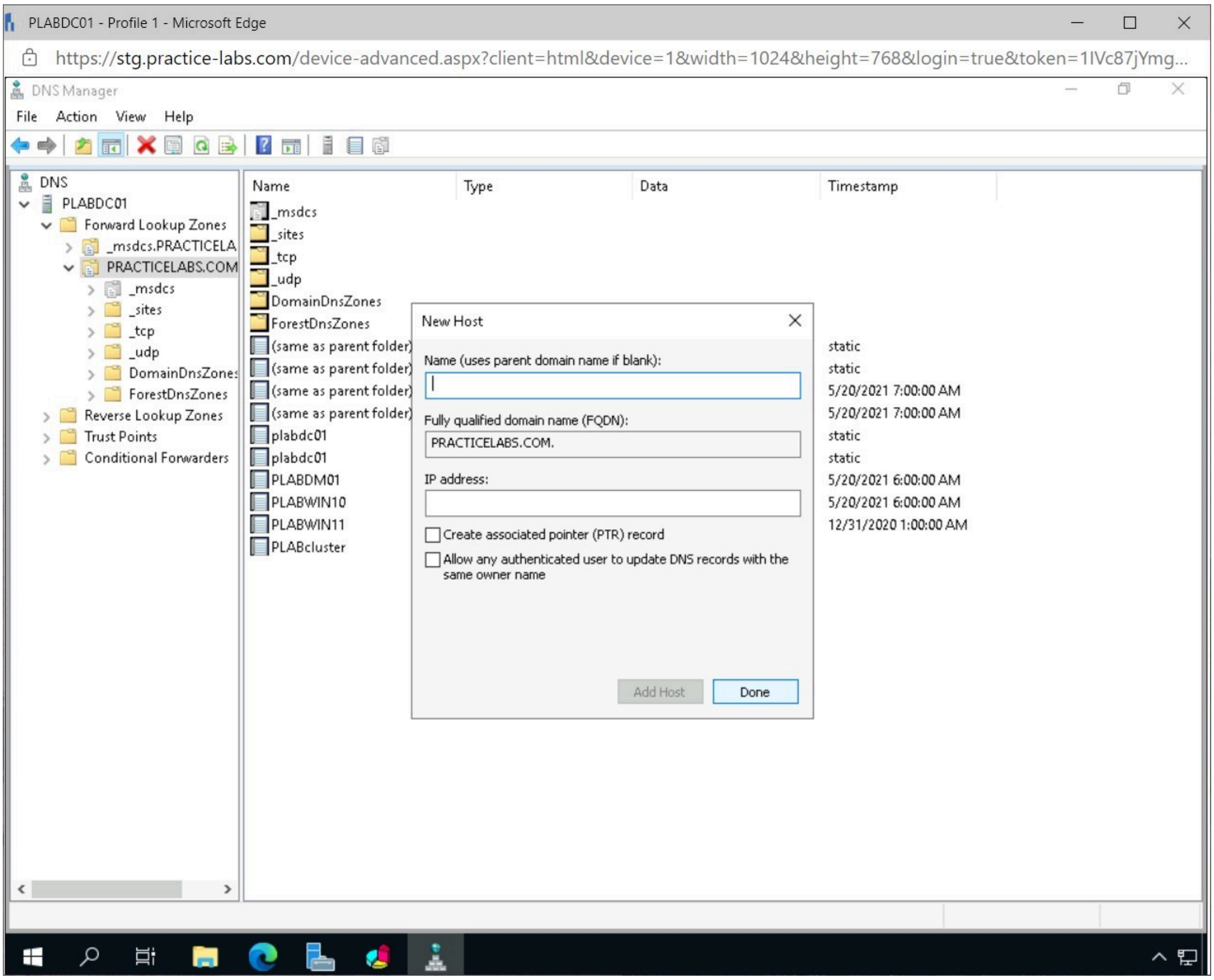

Figure 1.46 Screenshot of PLABDC01: Displaying clicking Done on the New Host window.

Step 7

Close the **DNS Manager** window.

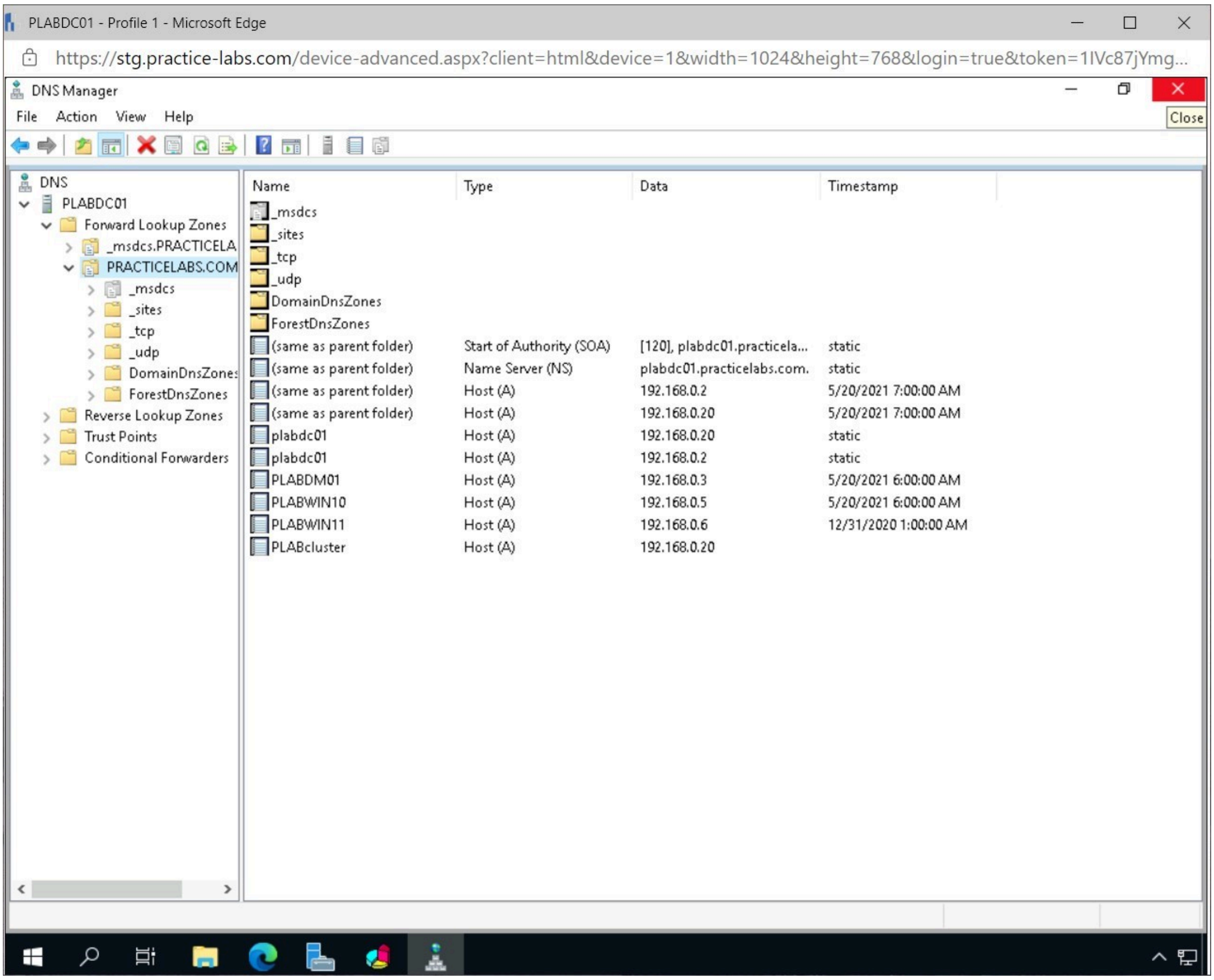

Figure 1.47 Screenshot of PLABDC01: Displaying closing the DNS Manager window.

Task 5 - Verify the Created Network Load Balancing Cluster

After creating a network load balancing cluster, access to the resource, for example, an internal website, will be balanced between the different nodes in the cluster.

In this task, the created network load balancing cluster will be verified.

Step 1

Connect to **PLABWIN10.**

From the **Taskbar,** click **Microsoft Edge** and browse to the following URL:

```
http://plabcluster
```

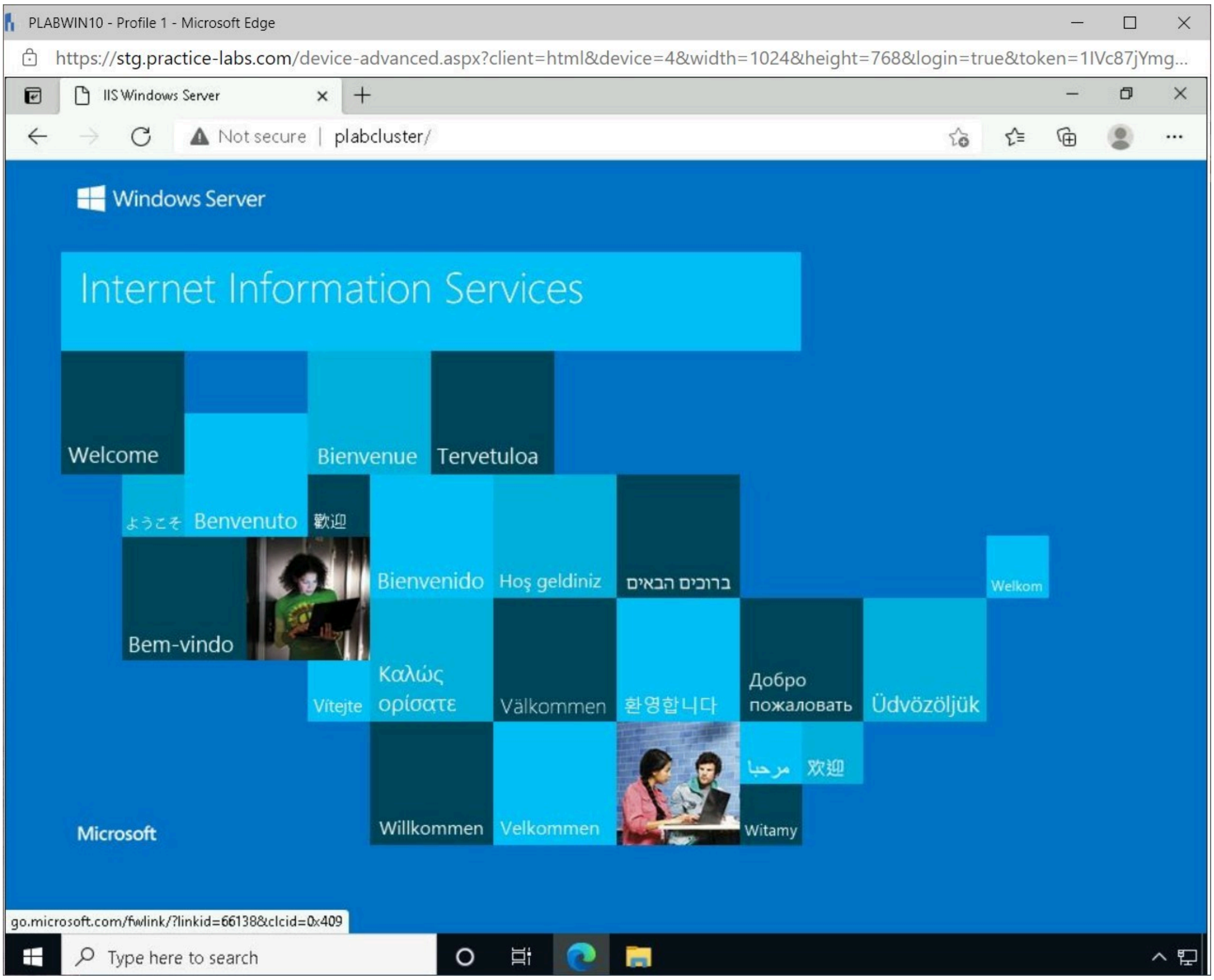

Figure 1.48 Screenshot of PLABWIN10: Displaying Microsoft Edge with the specified URL open.

> ***Note:*** *The website that was created is accessible using the network load balancing cluster that was created.*

Step 2

Connect to **PLABDM01.**

Right-click **Start**, point to **Shut down or sign out** and select **Shut down**.

In the pop-up window, click **Continue**.

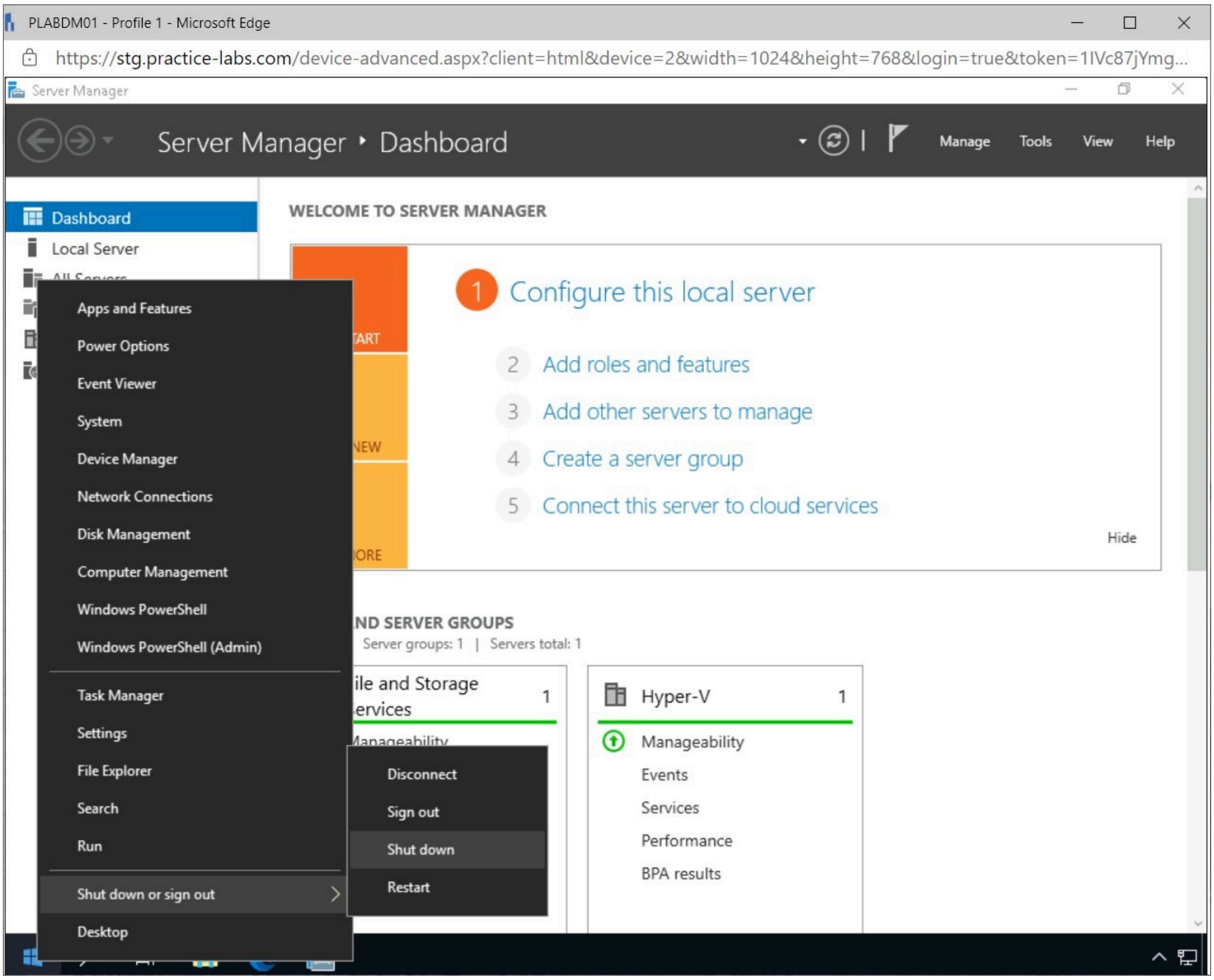

Figure 1.49 Screenshot of PLABDDM01: Displaying shutting down the device.

> ***Note:*** *Wait a minute for the device to shut down before continuing to the next step.*

Step 3

Connect to **PLABWIN10** and refresh the **Microsoft Edge** browser.

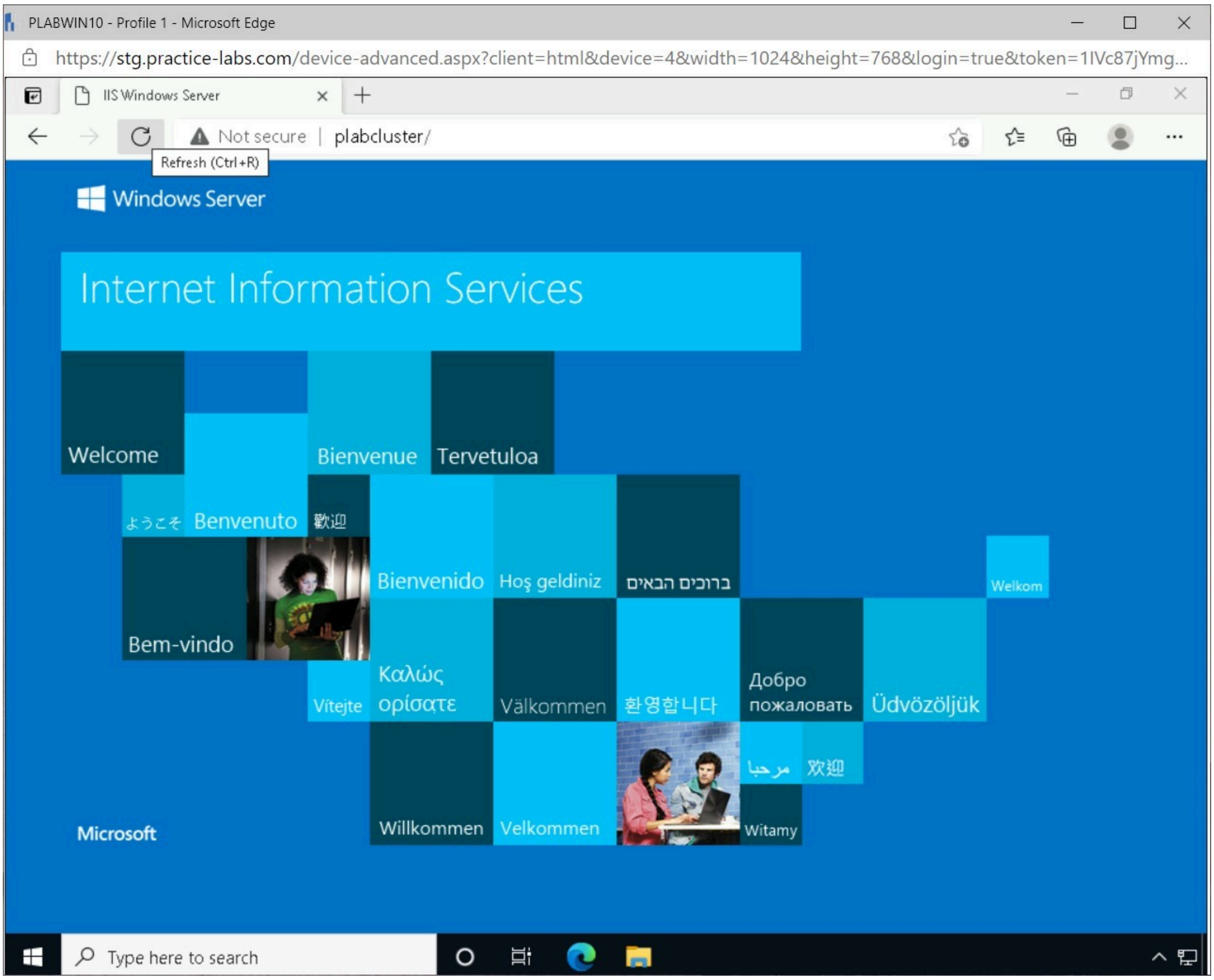

Figure 1.50 Screenshot of PLABWIN10: Displaying refreshing the page in Microsoft Edge.

> ***Note:*** *After shutting down the* ***PLABDM01*** *device, the internal website is still available, indicating that the Network load balancing works and that the resource is still available.*

Step 4

Close the **Microsoft Edge** browser.

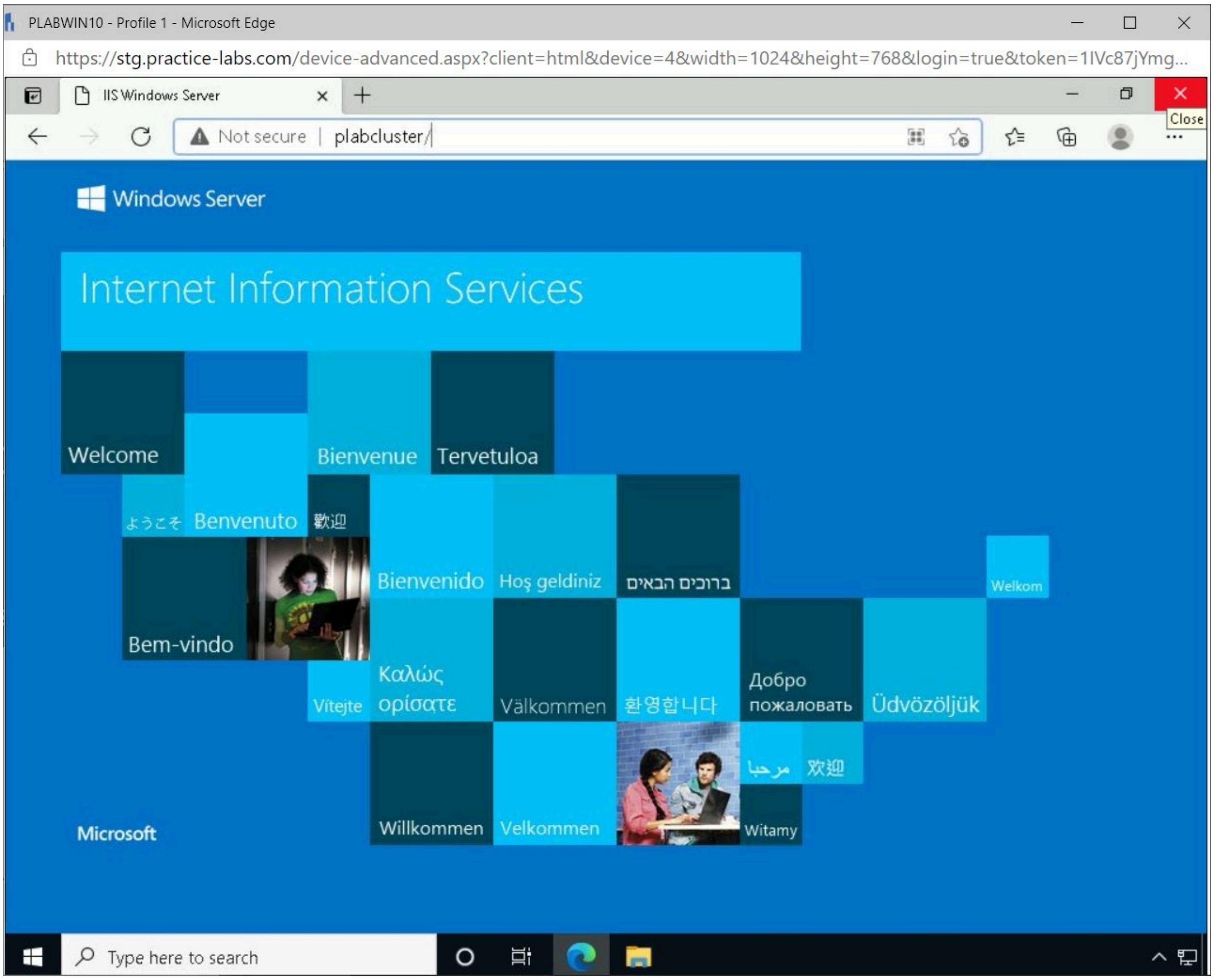

Figure 1.51 Screenshot of PLABWIN10: Displaying closing the Microsoft Edge browser.

Exercise 2 - Configuring NIC Teaming on a Server

Servers on a network need to be highly available for requests from client devices. To ensure these servers can manage the network traffic, additional network interface cards can be added to these servers. These additional network interface cards can then be teamed together to increase the capacity for network traffic.

In this exercise, a newly added Network Interface Card will be configured and teamed with another NIC on the device.

Learning Outcomes

After completing this exercise, you should be able to:

- Configure a Secondary Network Interface Card (NIC)

- Configure NIC Teaming

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01 -** (Windows Server 2019 - Domain Controller)
- **PLABDM01 -** (Windows Server 2019 - Domain Member)

Task 1 - Configure a Secondary Network Interface Card (NIC)

After an additional Network Interface Card has been added, the network card needs to be configured with the correct IP configuration to ensure communication on the network.

In this task, an additional network card will be configured with the correct configuration.

Step 1

Connect to **PLABDM01**.

On the **Taskbar,** right-click the network icon and select **Open Network & Internet settings**.

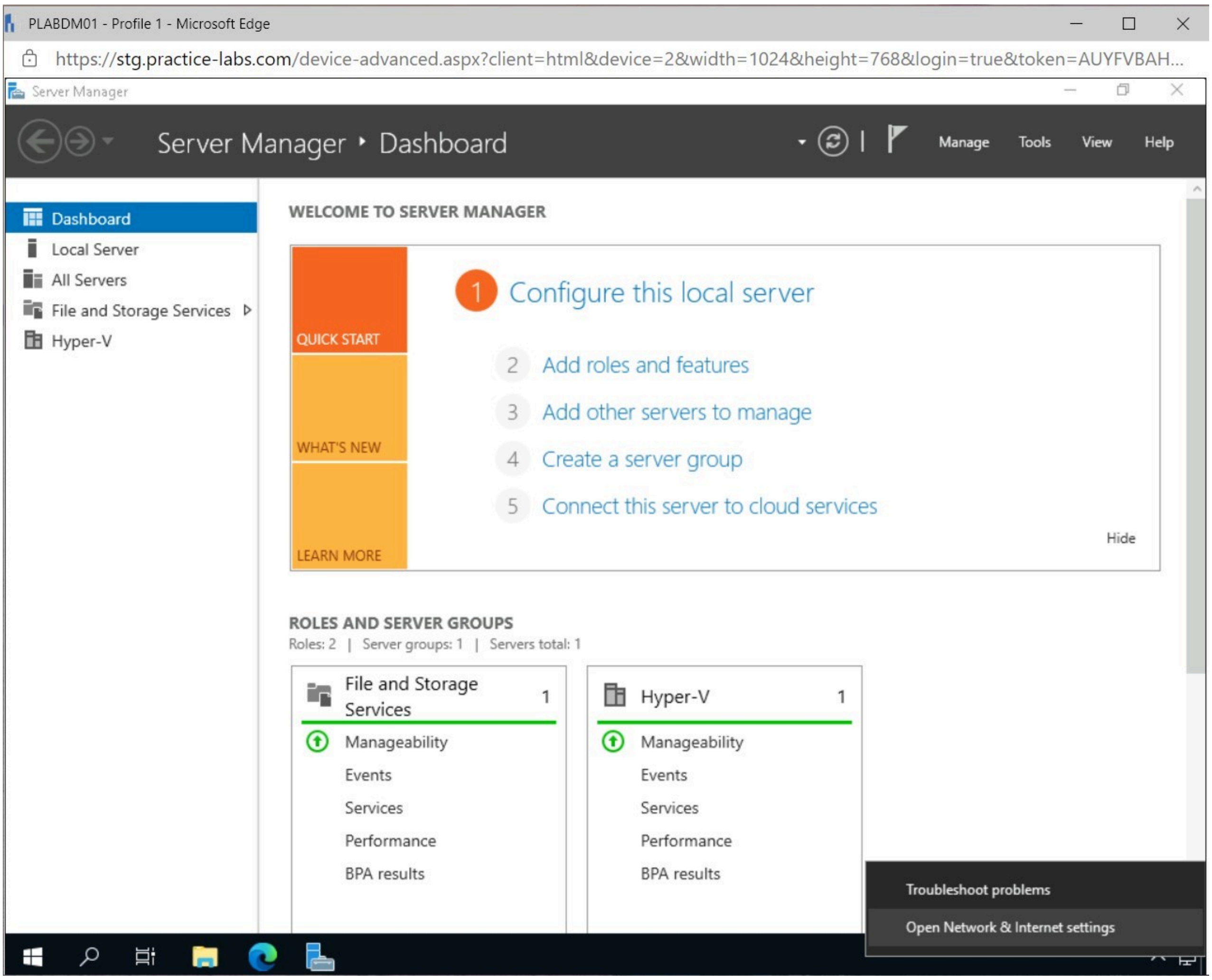

Figure 2.1 Screenshot of PLABDM01: Displaying opening Network & Internet settings from the Taskbar.

Step 2

In the **Settings** window, select **Change adapter options** in the **Network status** pane.

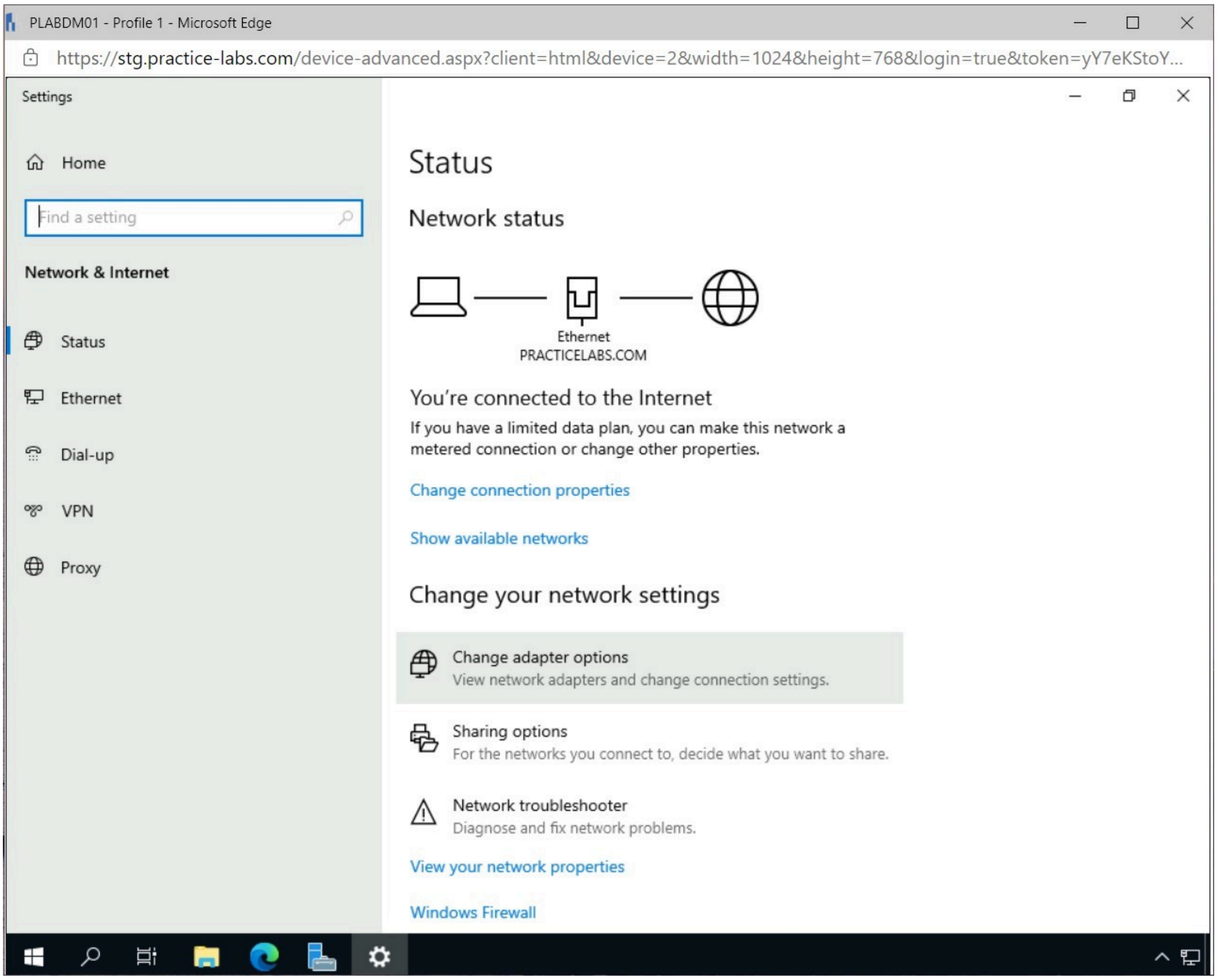

Figure 2.2 Screenshot of PLABDM01: Displaying the Settings window and selecting Changer adapter options.

Step 3

In the **Network Connections** window, right-click **Ethernet 2** and select **Properties**.

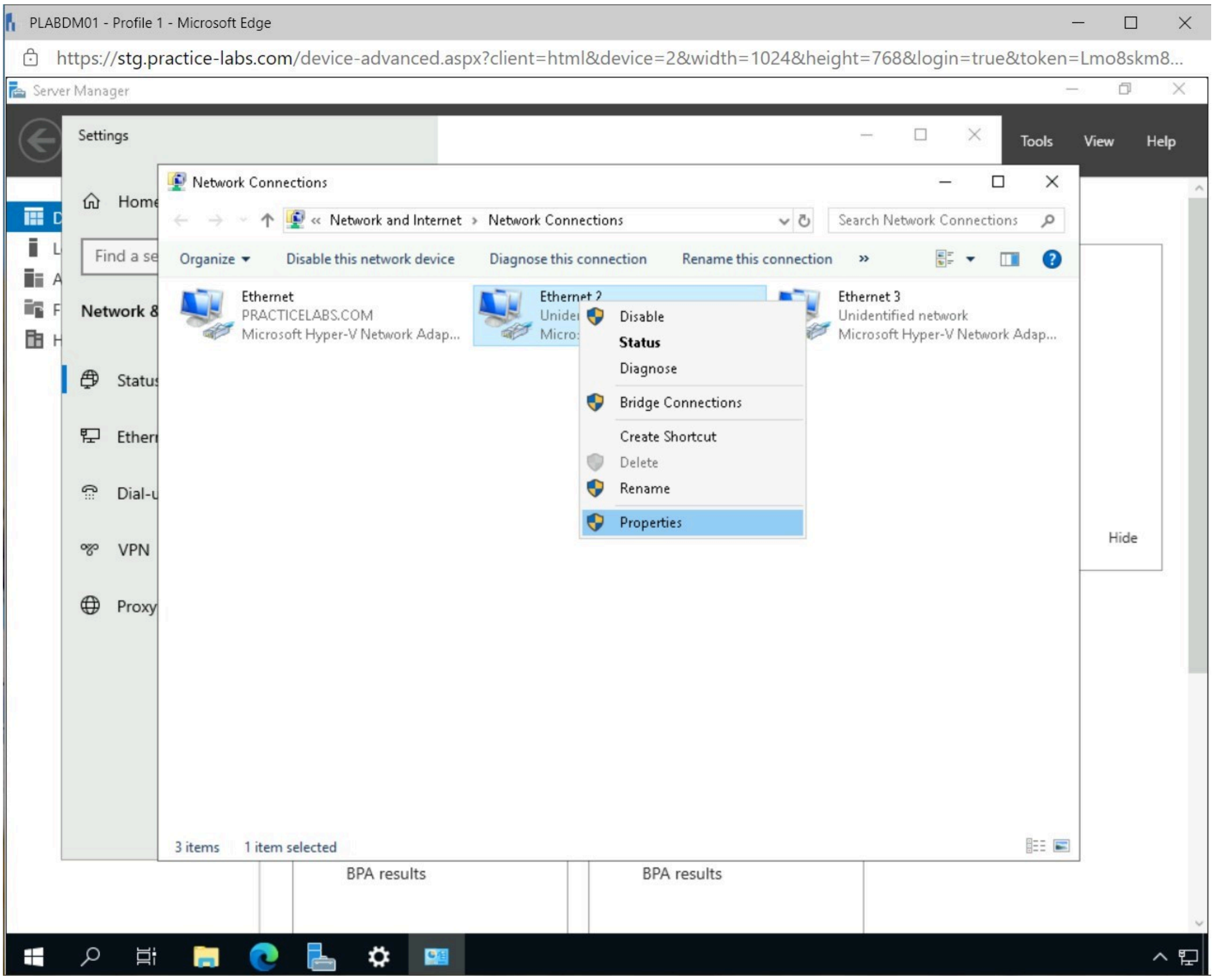

Figure 2.3 Screenshot of PLABDM01: Displaying opening the Properties of the Ethernet 2 network connection.

Step 4

In the **Ethernet 2 Properties** window, select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**.

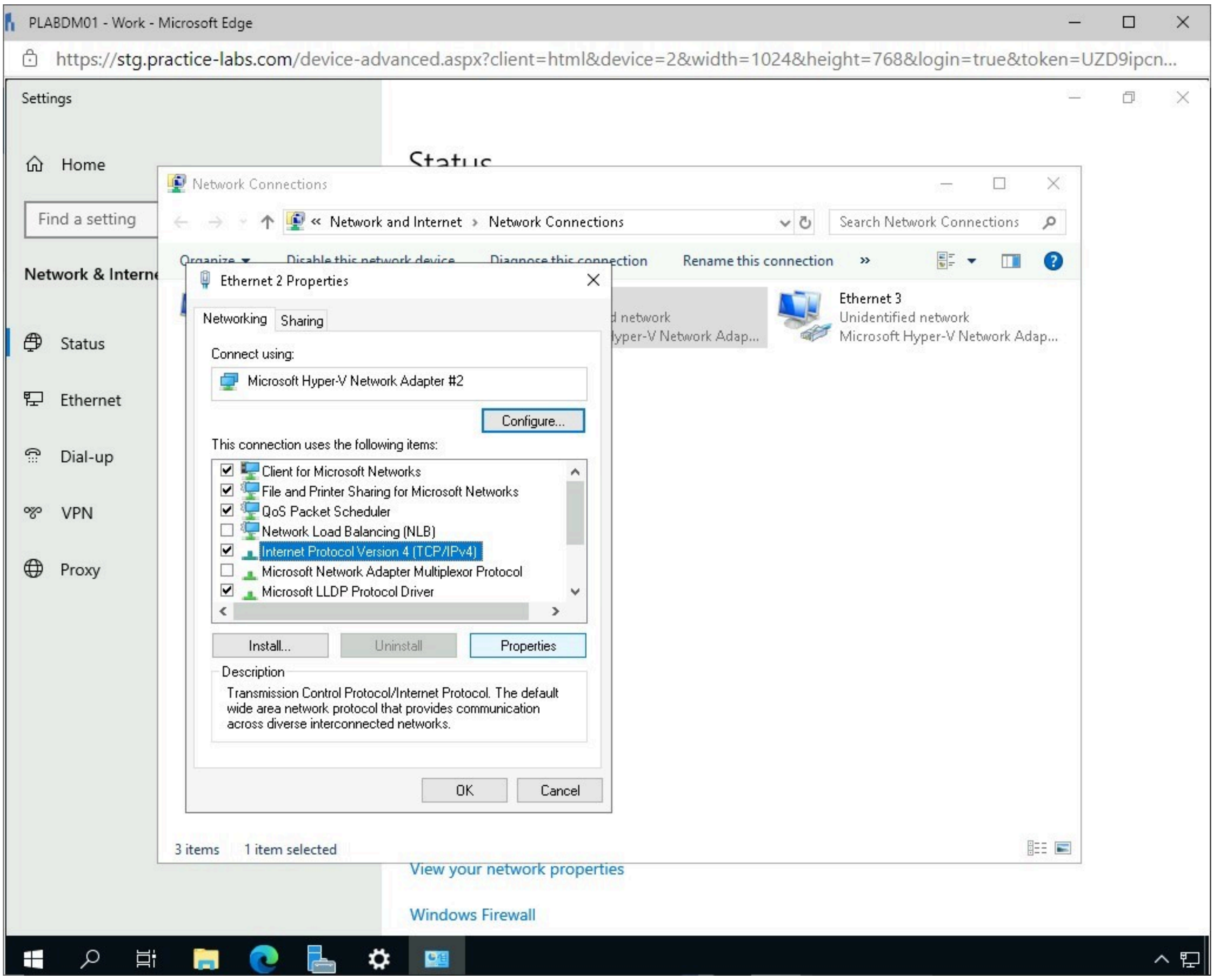

Figure 2.4 Screenshot of PLABDM01: Displaying opening the IPv4 Properties of the Ethernet 2 network connections.

Step 5

Click **Use the following IP address** radio button in the **Internet Protocol Version 4 (TCP/IPv4) Properties** window.

Enter the following and click **OK**.

```
IP address: 192.168.0.7
Subnet mask: 255.255.255.0
Default gateway: 192.168.0.250
Preferred DNS Server: 192.168.0.2
```

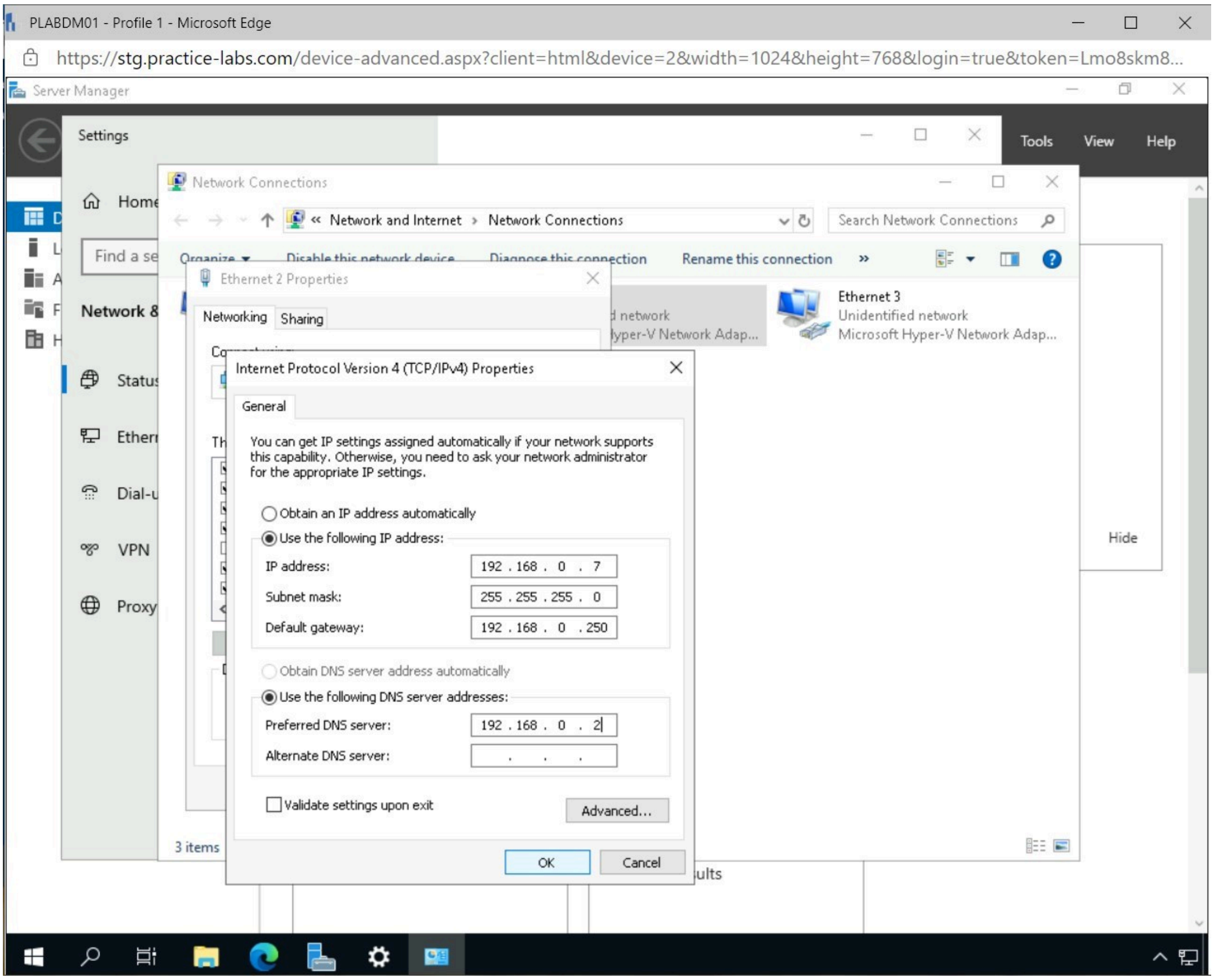

Figure 2.5 Screenshot of PLABDM01: Displaying configuring a static IP address for the Ethernet 2 network connection on the Internet Protocol Version 4 (TCP/IPv4) Properties window.

Step 6

In the **Microsoft TCP/IP** pop-up window, click **Yes**.

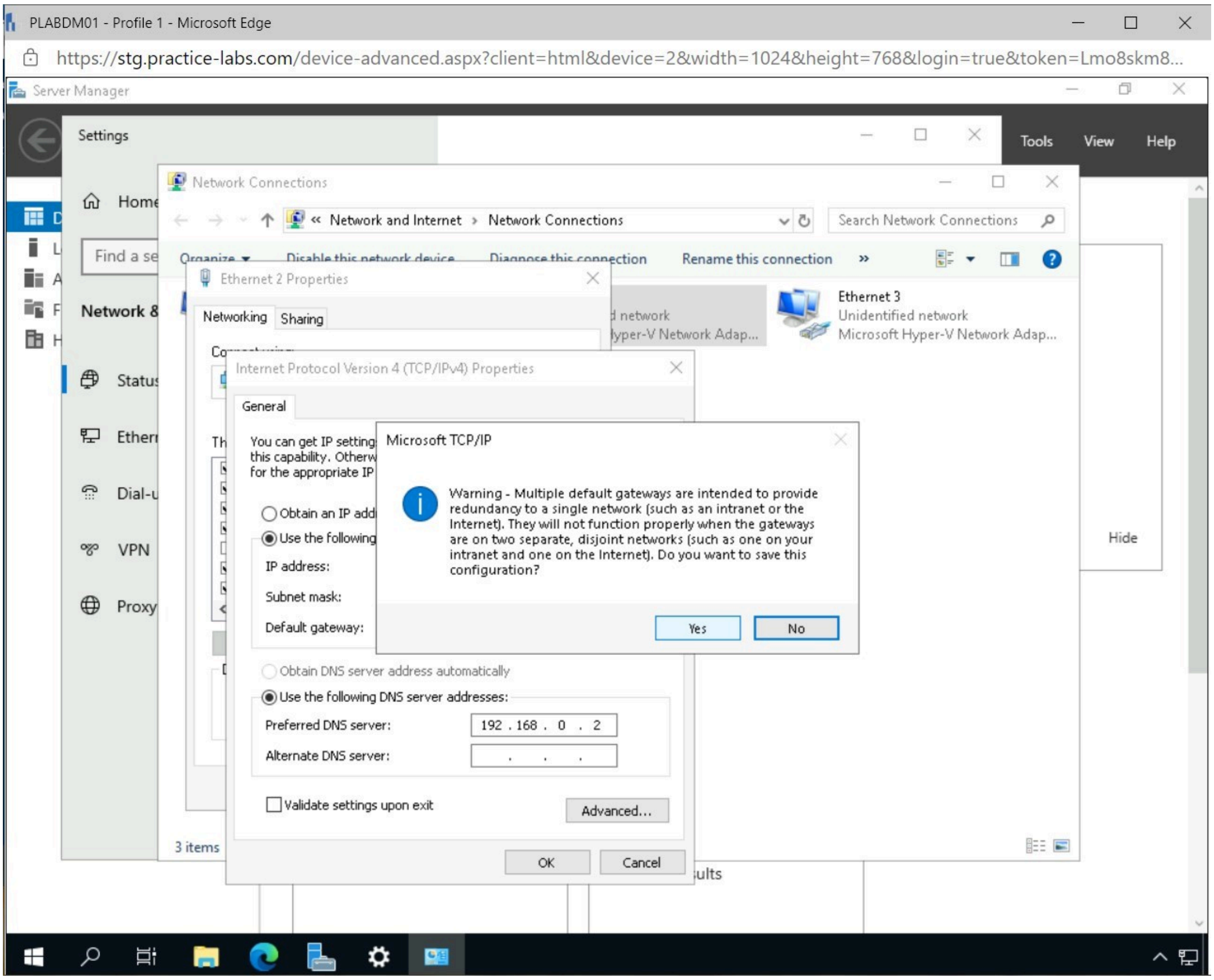

Figure 2.6 Screenshot of PLABDM01: Displaying clicking Yes on the Microsoft TCP/IP pop-up window.

Step 7

Click **Close** in the **Ethernet 2 Properties** window.

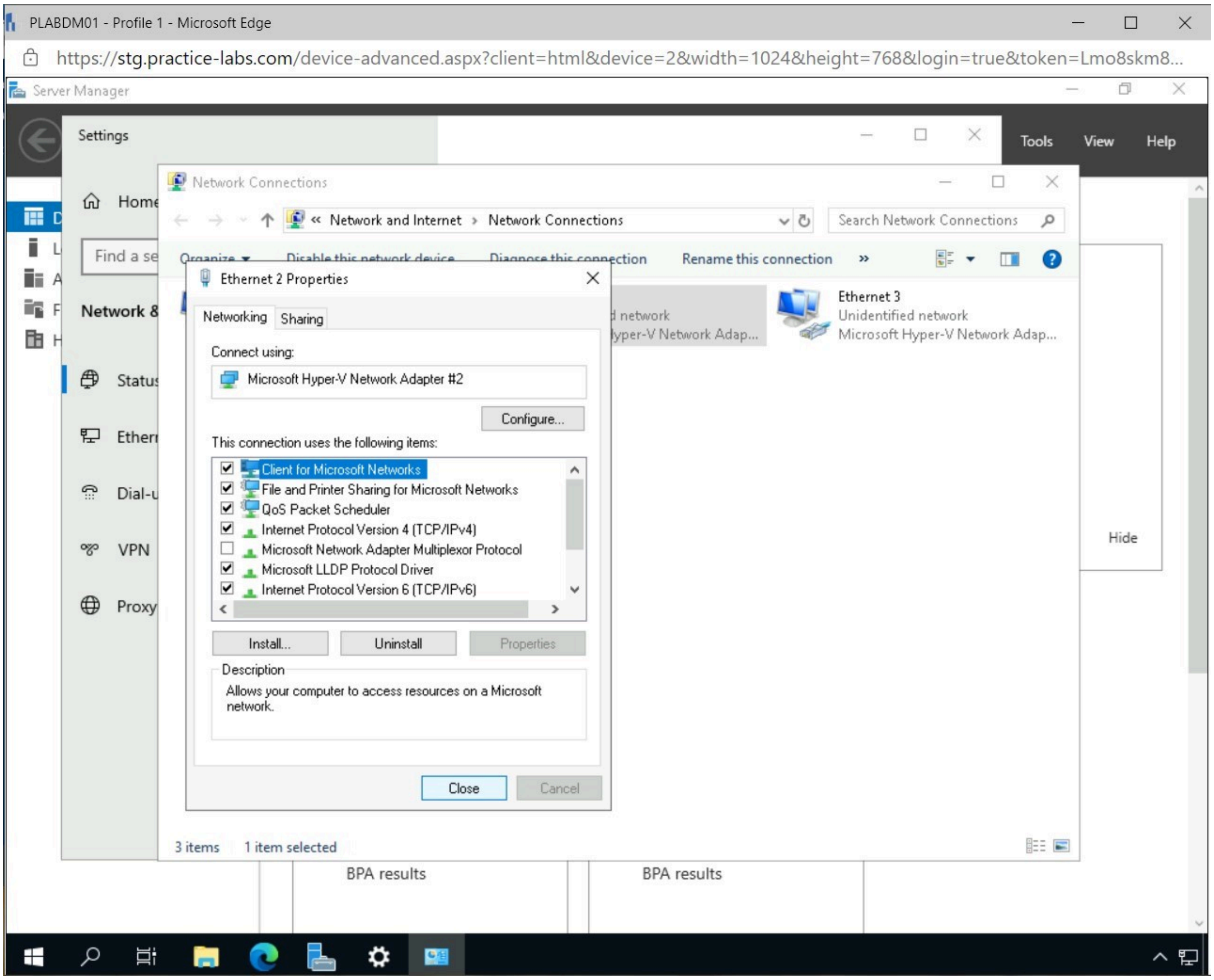

Figure 2.7 Screenshot of PLABDM01: Displaying closing the Ethernet 2 Properties window.

> ***Note:*** *Repeat Steps 3-7 to configure the* ***Ethernet 3*** *network adapter with an IP Address of* ***192.168.0.8****.*

Task 2 - Configuring NIC Teaming

To enable a server to share the network communication across multiple network interfaces, these devices need to be teamed together.

In this task, two preconfigured network interface cards will be teamed together to share the network traffic.

Step 1

Connect to **PLABDM01** and open **Server Manager** from the **Taskbar**.

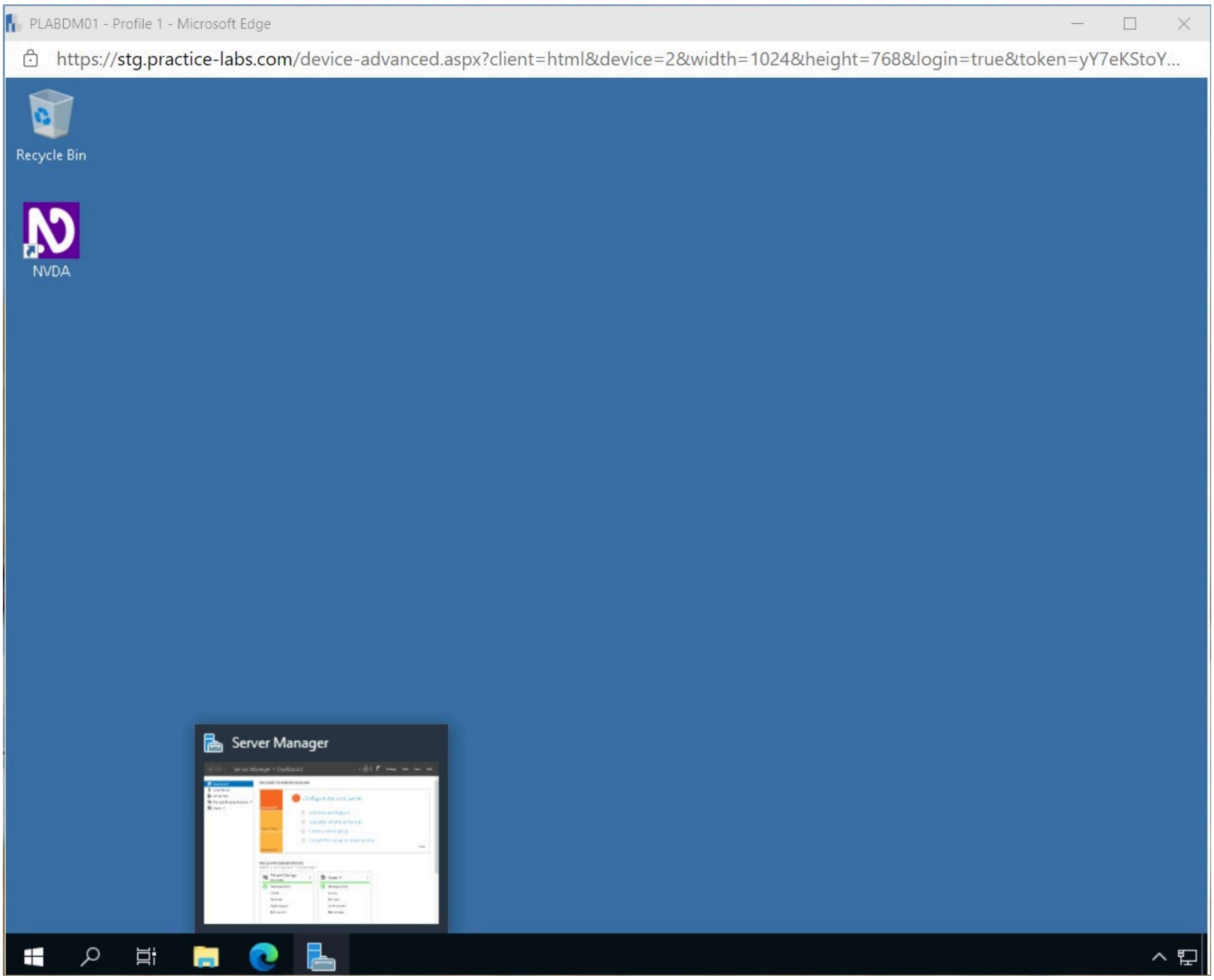

Figure 2.8 Screenshot of PLABDM01: Displaying opening Server Manager from the Taskbar.

Step 2

In **Server Manager,** click **All Servers** on the left pane.

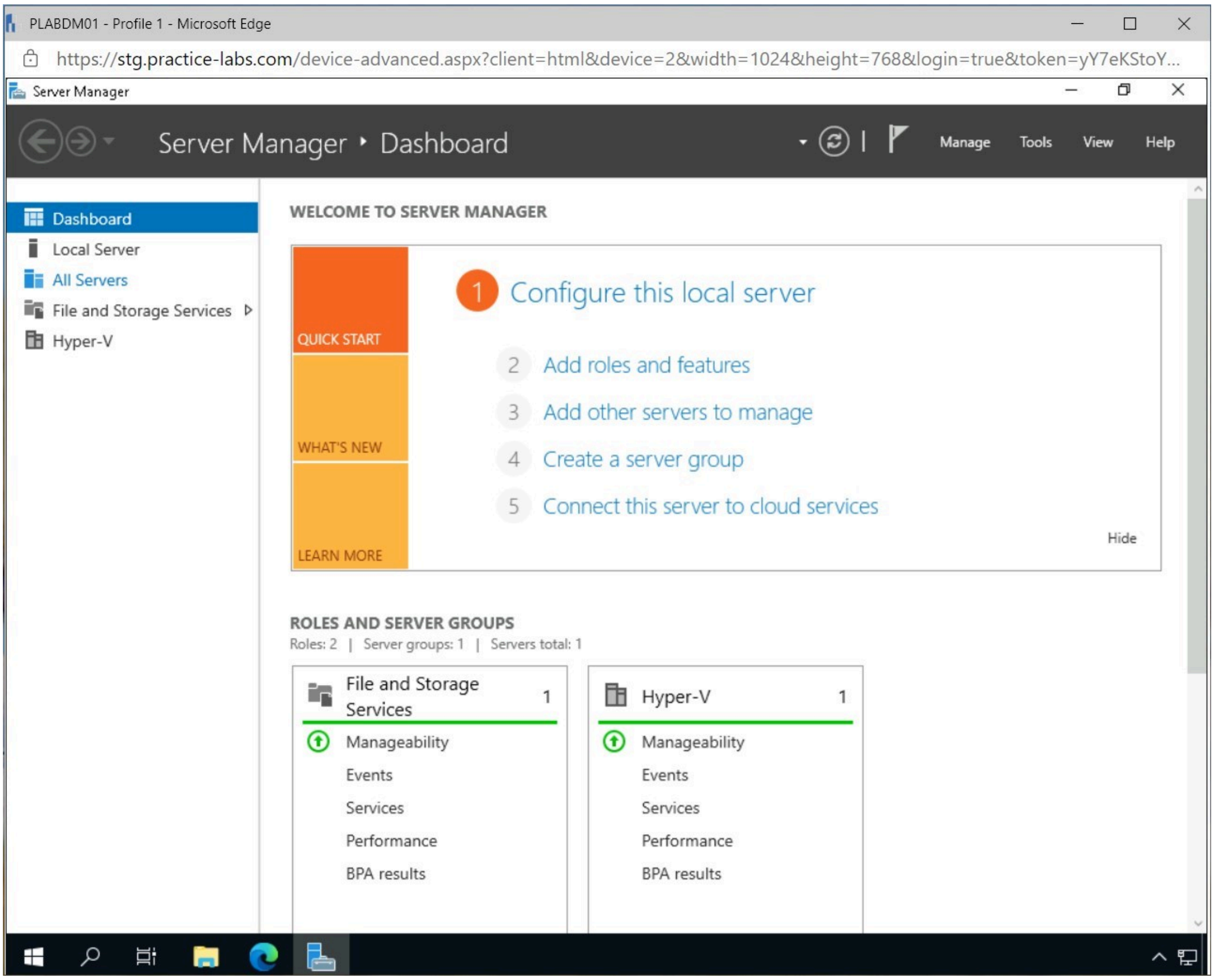

Figure 2.9 Screenshot of PLABDM01: Displaying selecting All Servers in Server Manager.

Step 3

In the **All Servers** window, right-click **PLABDM01** and select **Configure NIC Teaming**.

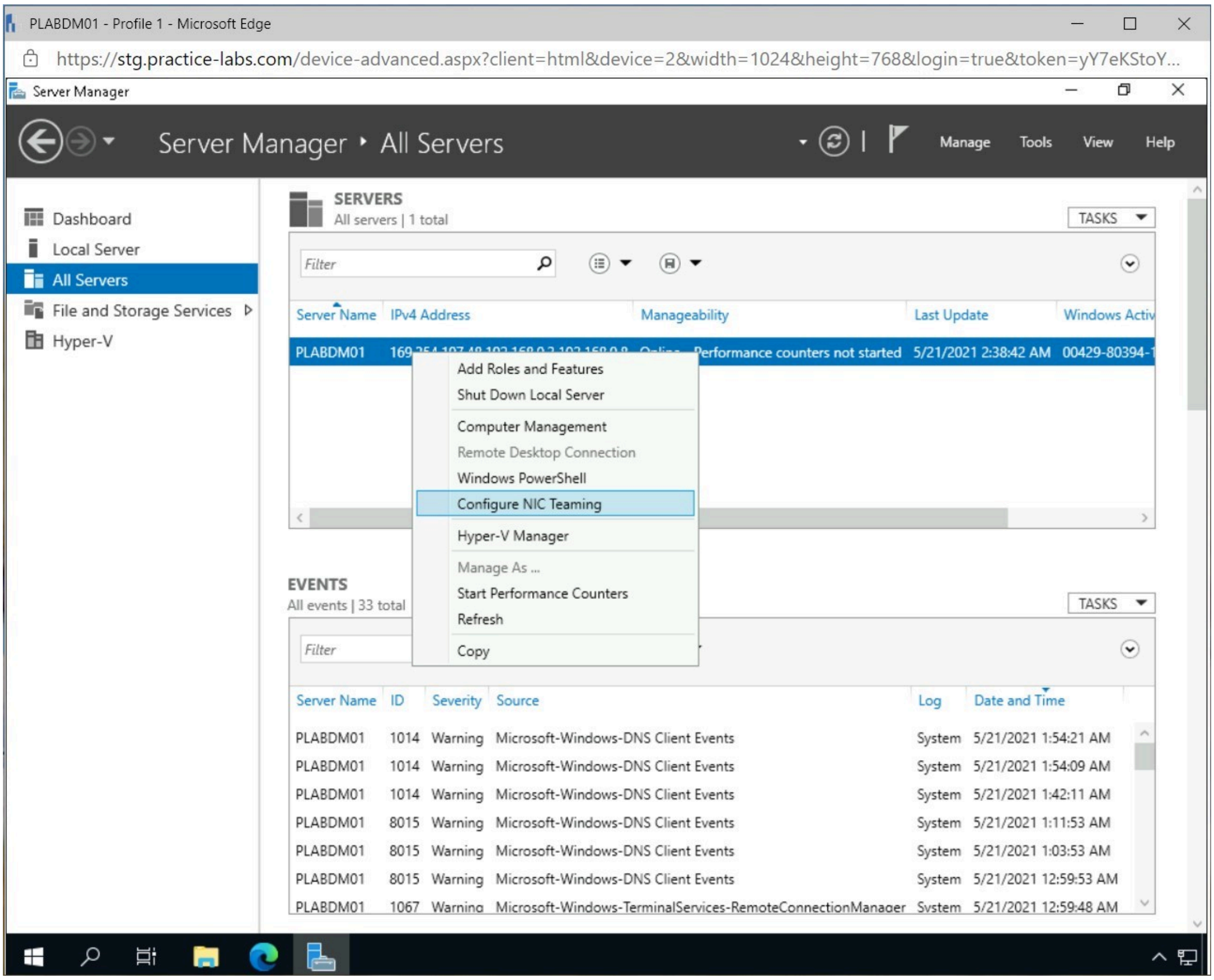

Figure 2.10 Screenshot of PLABDM01: Displaying right-clicking PLABDM01 and selecting Configure NIC Teaming on the All Servers window.

Step 4

In the **NIC Teaming** window, click the **TASKS** drop-down in the **TEAMS** pane and select **New Team**.

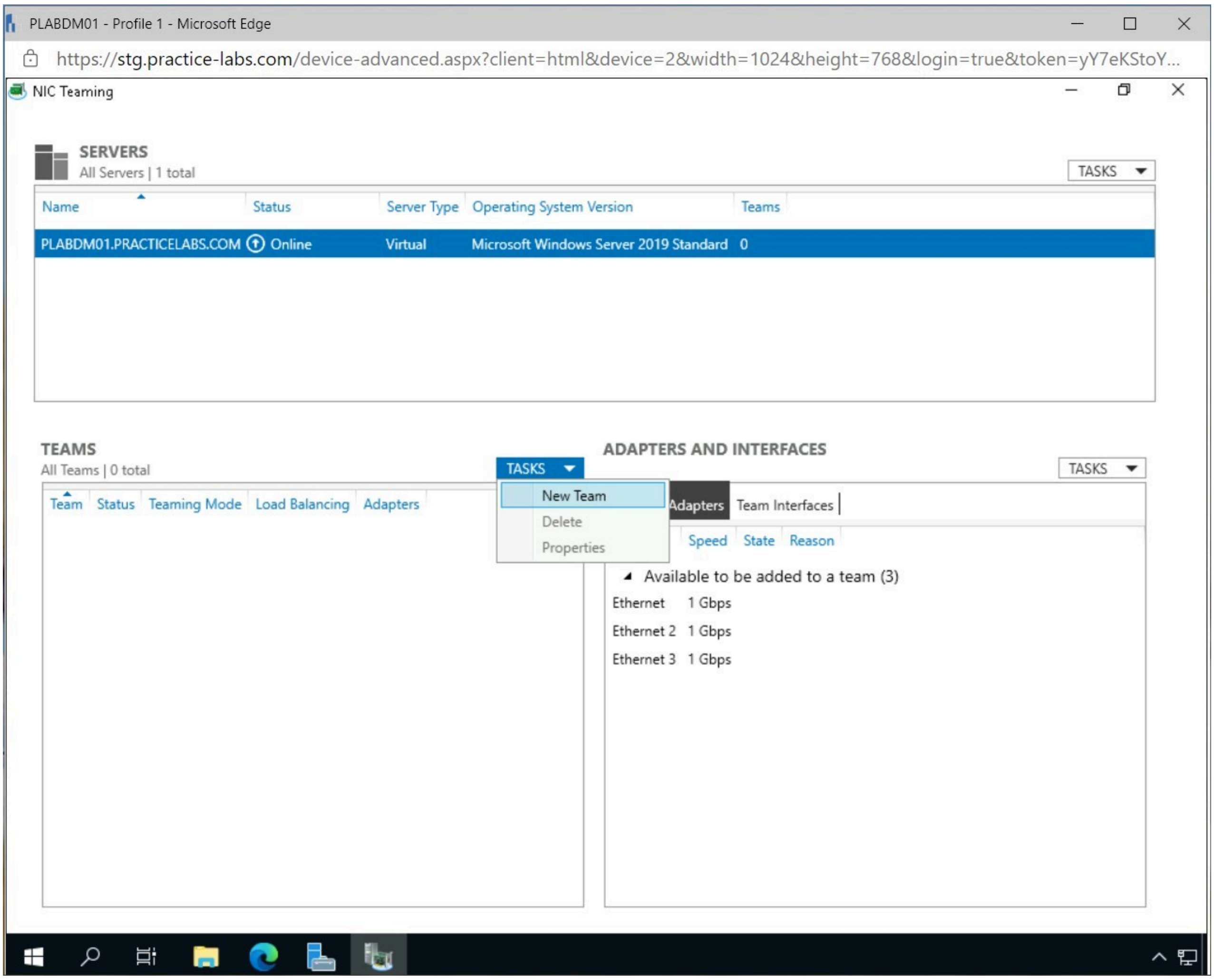

Figure 2.11 Screenshot of PLABDM01: Displaying the NIC Teaming window and selecting New Team from the TASKS drop-down menu.

Step 5

On the **NIC Teaming - New team** window, select **Ethernet 2** and **Ethernet 3,** type the following in the **Team name** field and click **OK.**

```
PLAB NIC Team
```

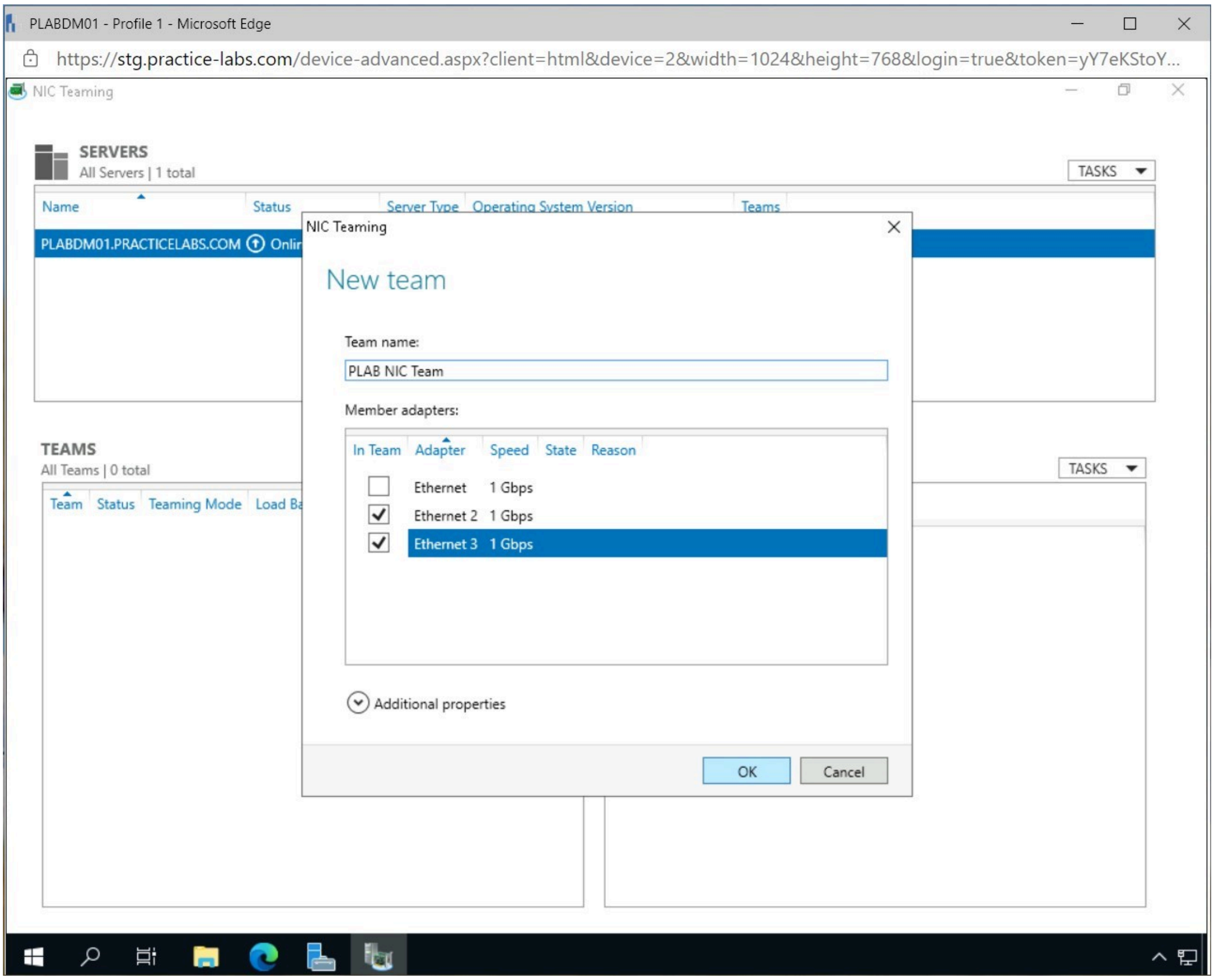

Figure 2.12 Screenshot of PLABDM01: Displaying the NIC Team window and selecting the Ethernet adapters, and entering the name for the NIC Team.

Step 6

Close the **NIC Teaming** window.

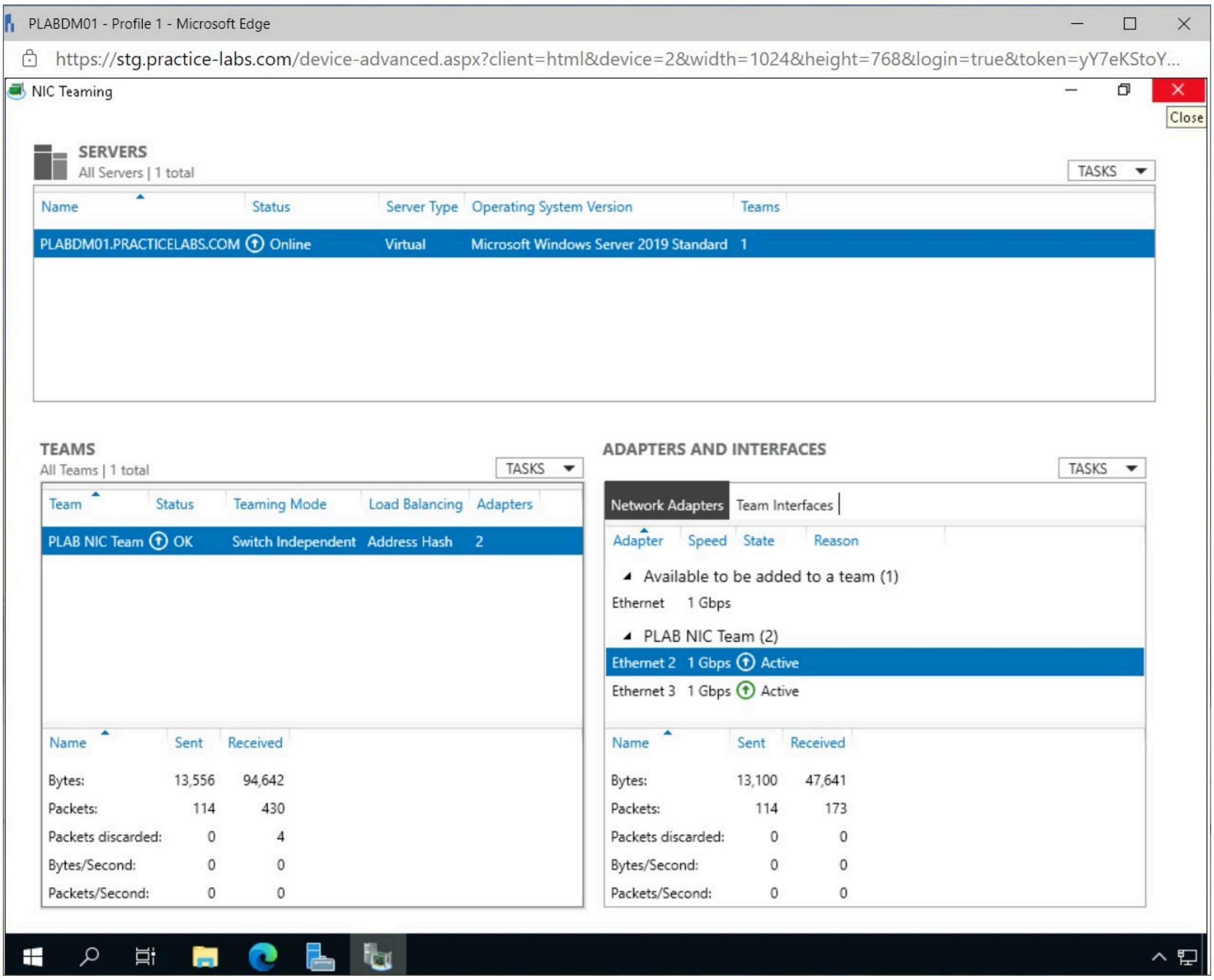

Figure 2.13 Screenshot of PLABDM01: Displaying closing the NIC Teaming window.

Alert: The Ethernet 2 will display an error, please wait for the error to resolve before moving on.

Note: *The Network Interface cards were successfully teamed together. In a production environment, it will depend on which adapters will be used for the NIC Team. In the lab environment, the first NIC is used for a remote connection and will cause the device to lose connectivity. Hence, it cannot be used for NIC teaming.*

Step 7

In the **Taskbar,** right-click the Network icon and select **Open Network & Internet settings**.

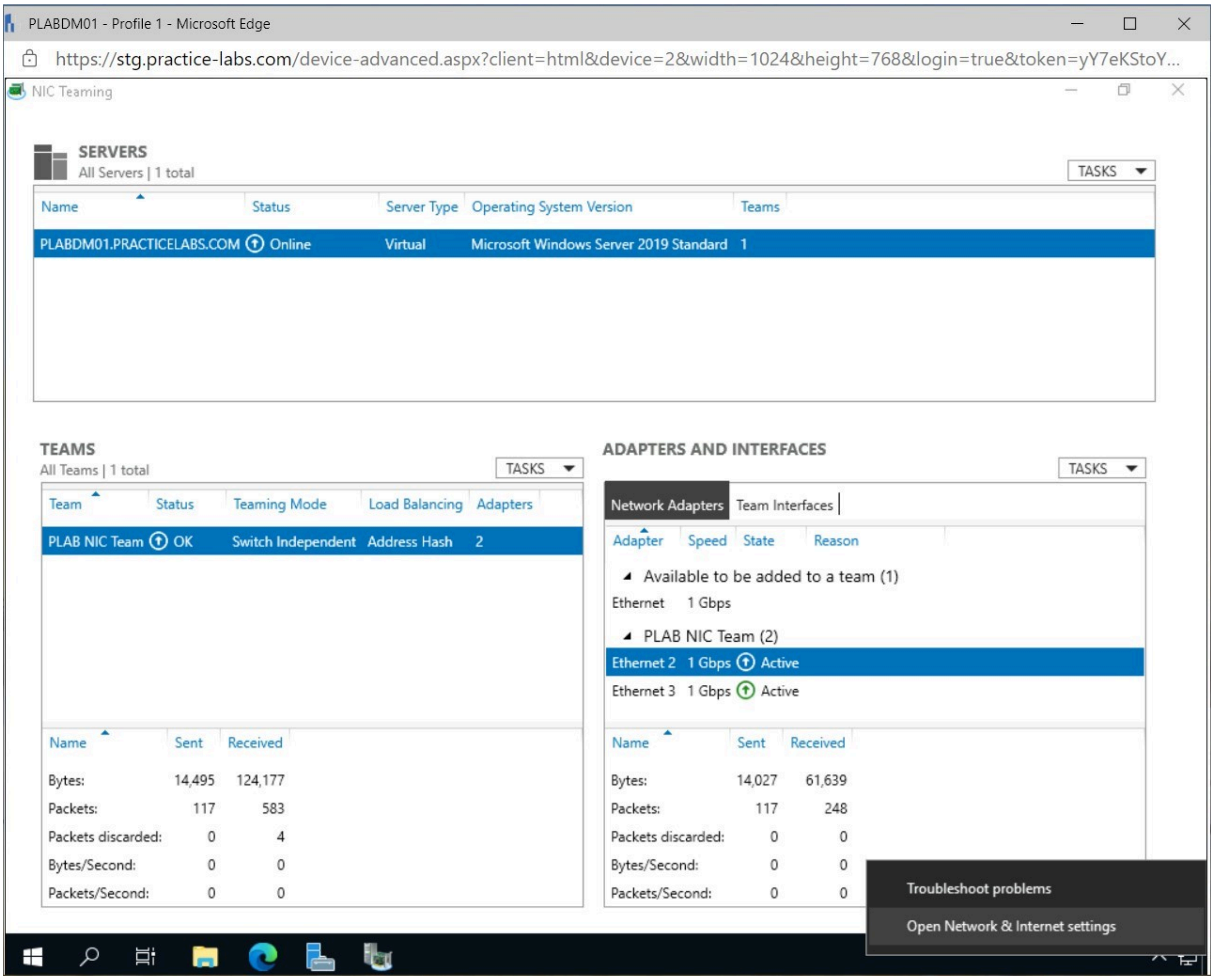

Figure 2.14 Screenshot of PLABDM01: Displaying opening Network & Internet settings from the Taskbar.

Step 8

On the **Settings - Network & Internet** window, click **Change adapter options** in the **Network status** pane.

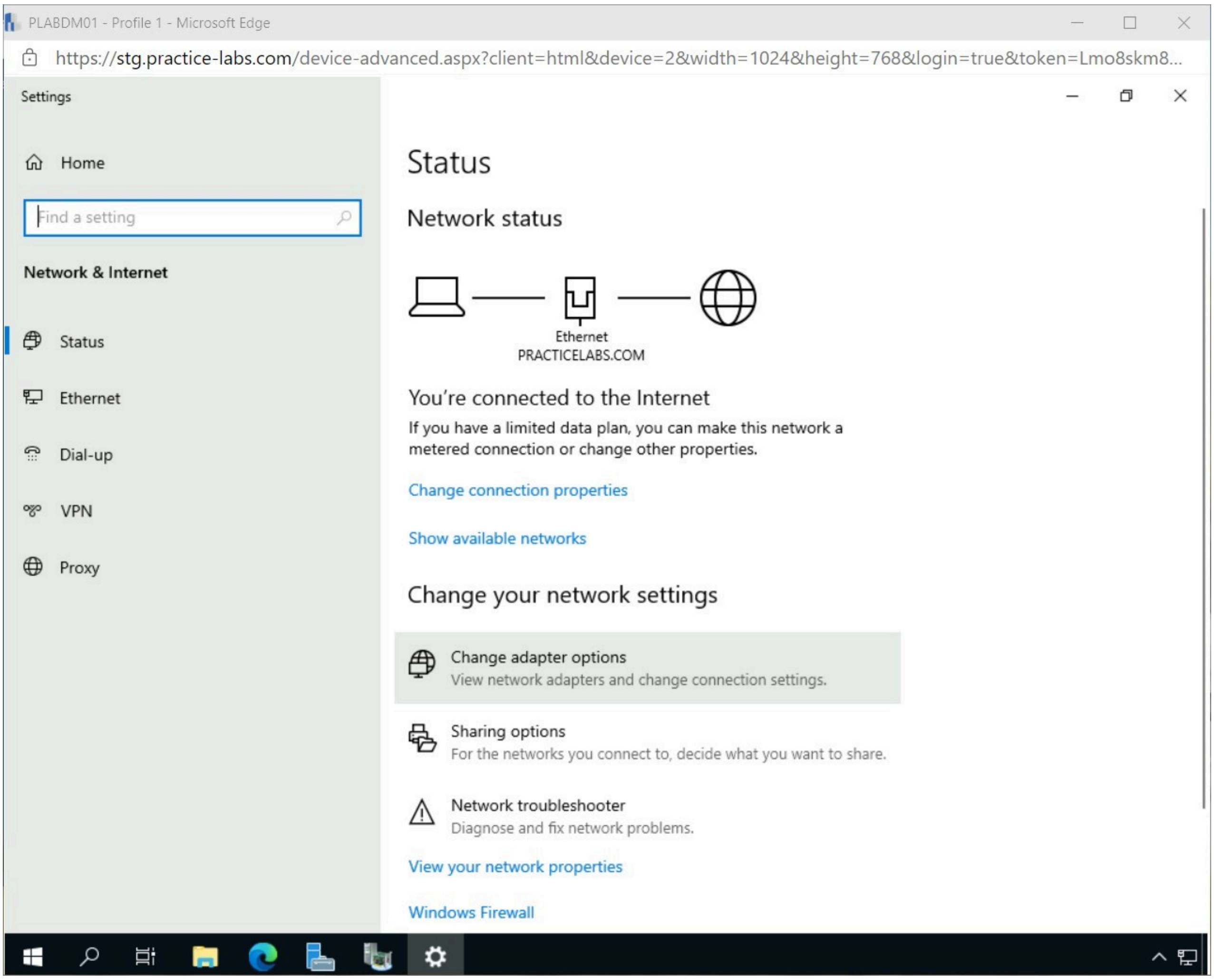

Figure 2.15 Screenshot of PLABDM01: Displaying selecting Change adapter options in the Network Status pane of the Network & Internet settings window.

Step 9

Right-click the **PLAB NIC Team** network interface in the **Network Connections** window and select **Properties**.

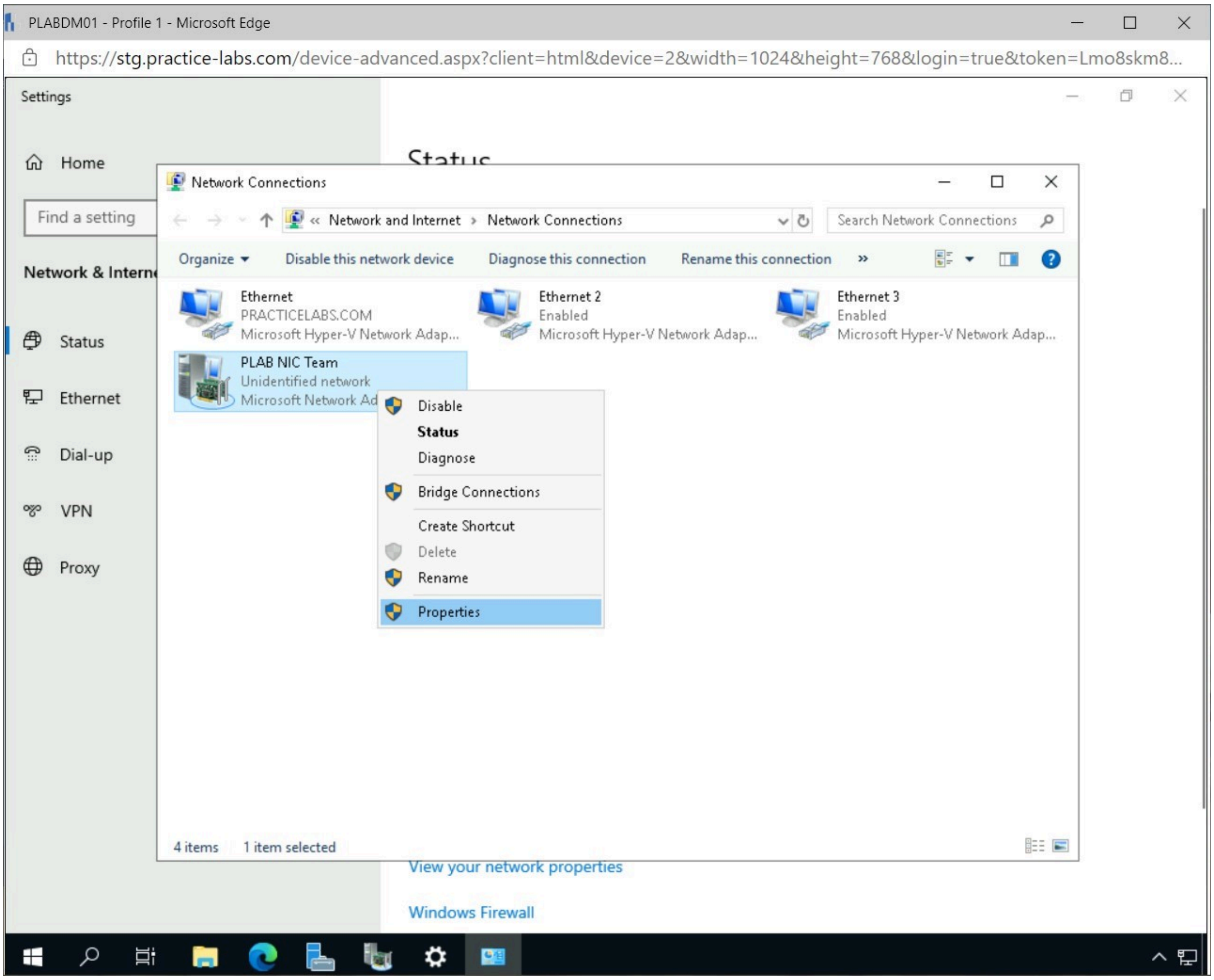

Figure 2.16 Screenshot of PLABDM01: Displaying opening the properties of the PLAB NIC Team network device.

Step 10

On the **PLAB NIC Team Properties** window, select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**.

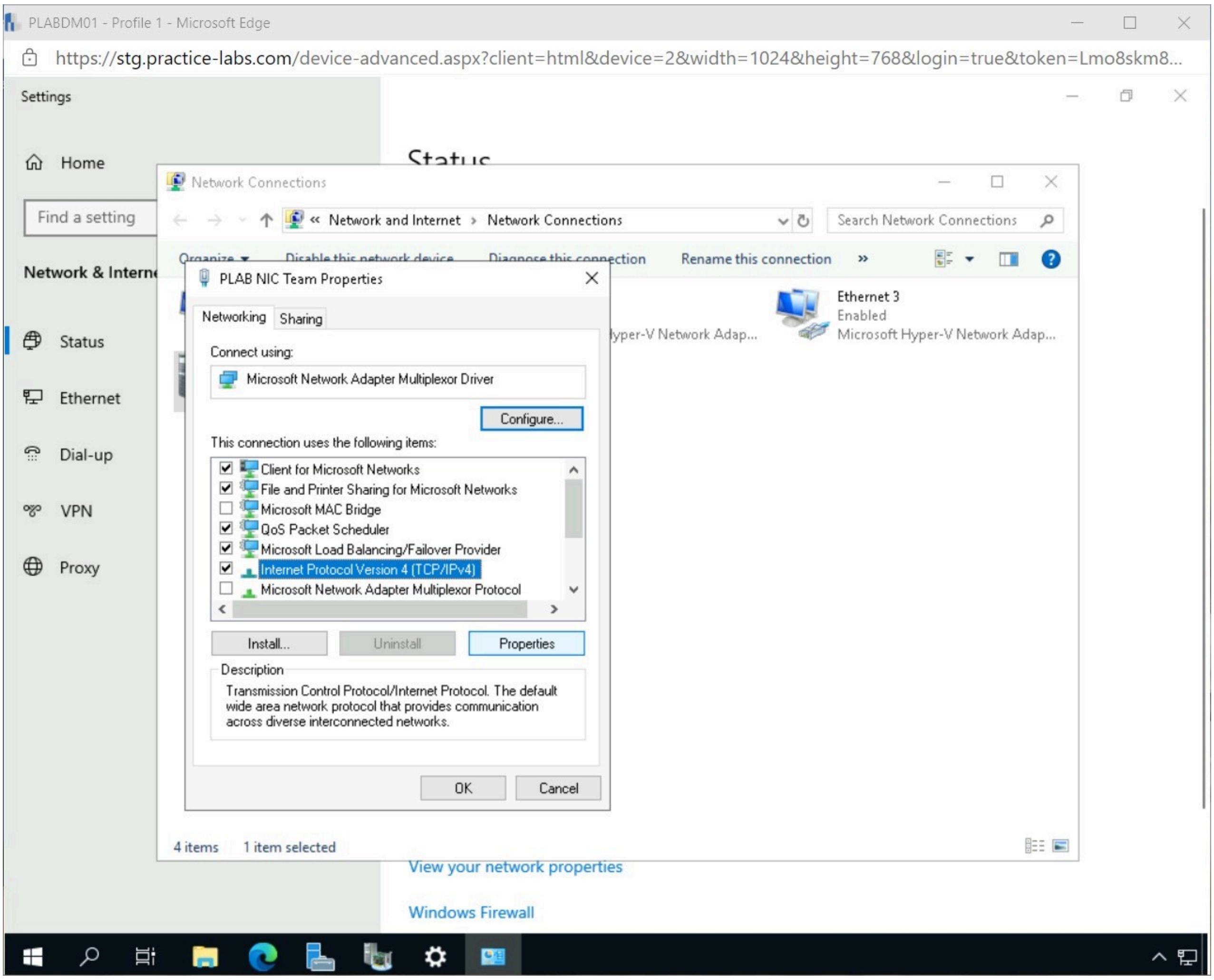

Figure 2.17 Screenshot of PLABDM01: Displaying configuring the PLAB NIC Team network device.

Step 11

Click **Use the following IP address** radio button in the **Internet Protocol Version 4 (TCP/IP) Properties** window.

Enter the following and click **OK**.

```
IP address: 192.168.0.9
Subnet mask: 255.255.255.0
Default gateway: 192.168.0.250
Preferred DNS server: 192.168.0.2
```

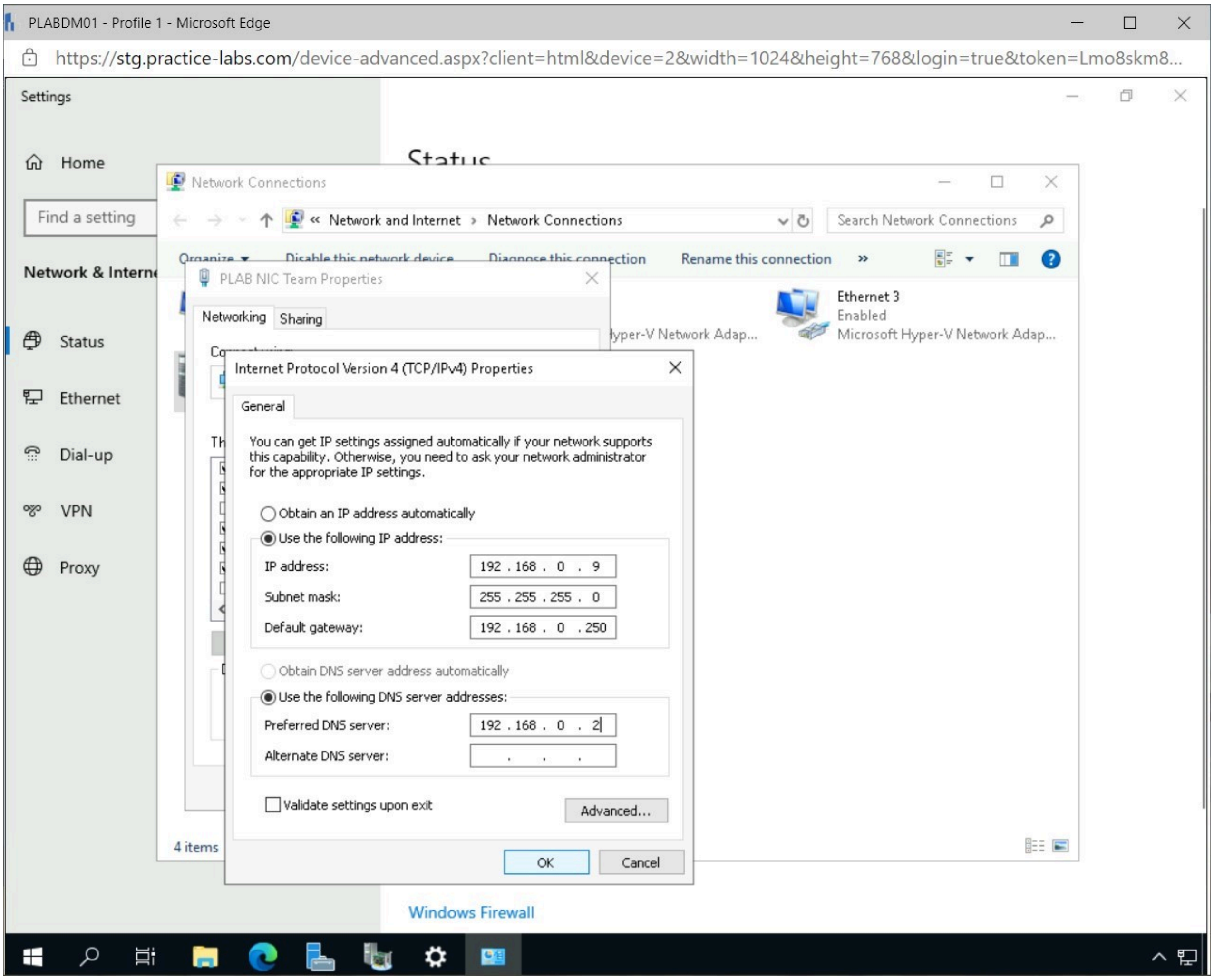

Figure 2.18 Screenshot of PLABDM01: Displaying configuring the PLAB NIC Team with an IP address.

Step 12

In the **Microsoft TCP/IP** pop-up window, click **Yes**.

Click **Close** to close the **PLAB NIC Team Properties** window.

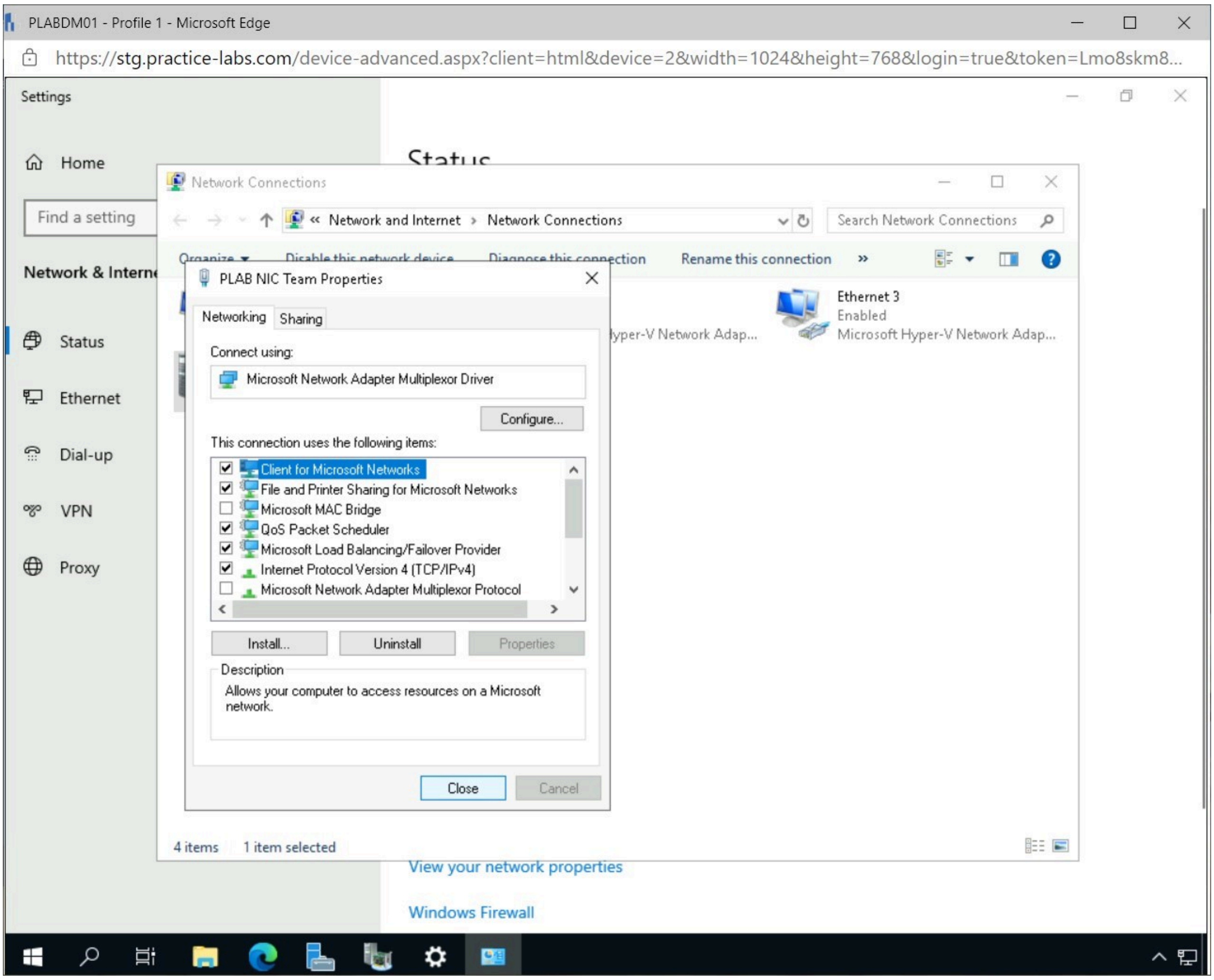

Figure 2.19 Screenshot of PLABDM01: Displaying closing the PLAB NIC Team Properties window.

Step 13

Double-click the **PLAB NIC Team** network device.

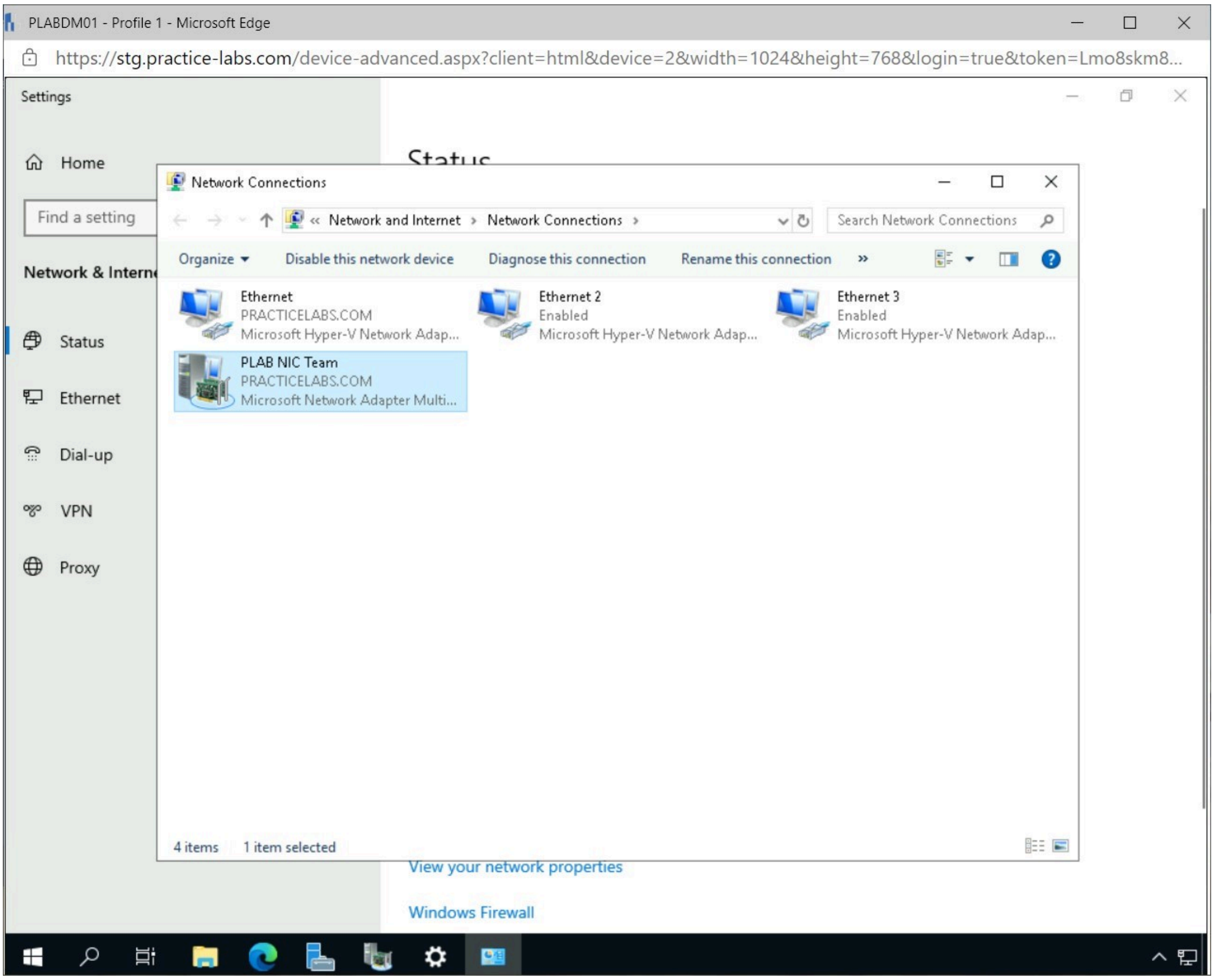

Figure 2.20 Screenshot of PLABDM01: Displaying opening the PLAB NIC Team device.

Step 14

Connect to **PLABDC01**.

Right-click **Start** and select **Windows Powershell**.

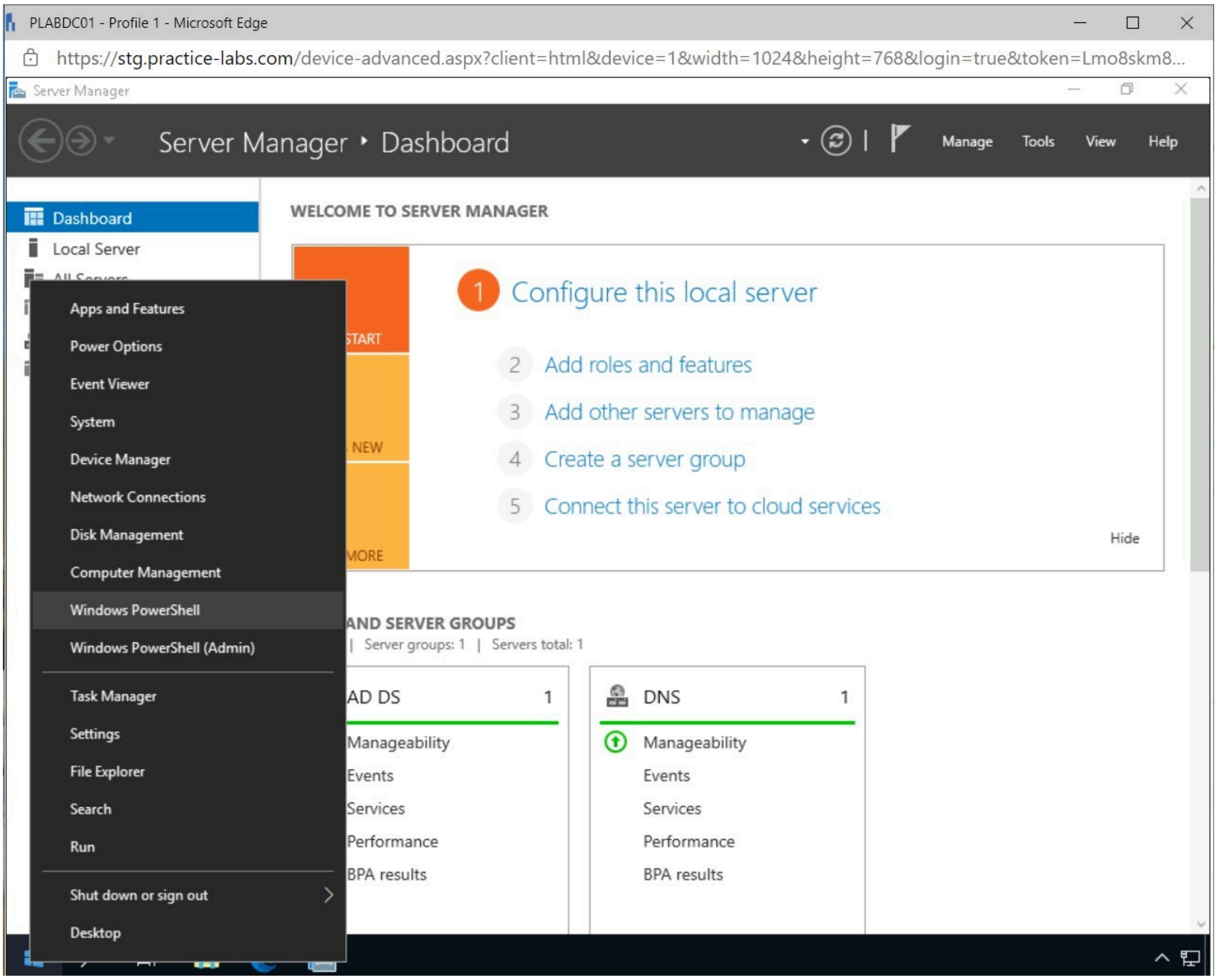

Figure 2.21 Screenshot of PLABDC01: Displaying opening Windows Powershell from the Start menu.

Step 15

In the **Windows Powershell** window, type the following and press **Enter**:

```
ping 192.168.0.9
```

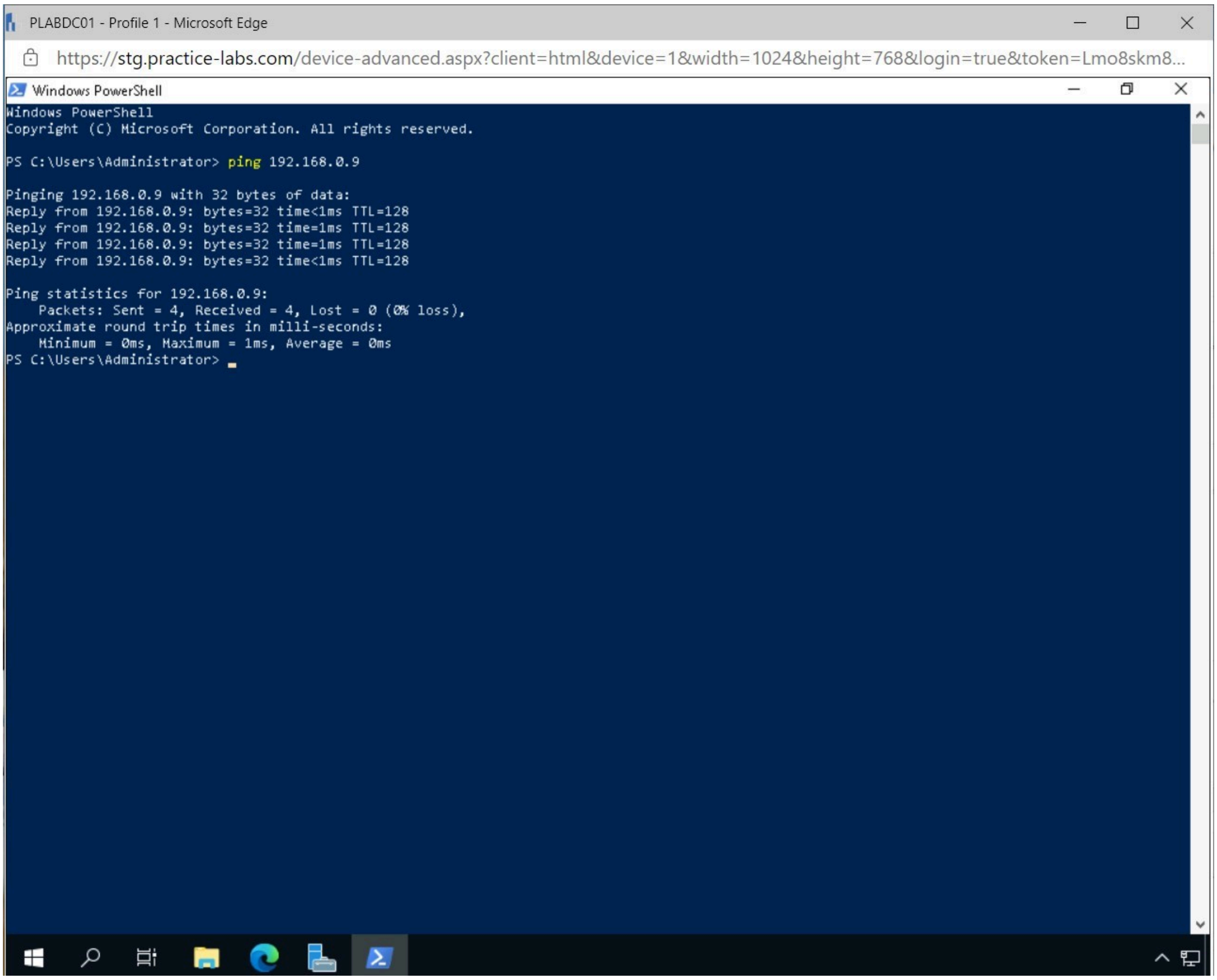

Figure 2.22 Screenshot of PLABDM01: Displaying executing the command in Windows Powershell.

Note: *The two Network Interface cards on the* ***PLABDM01*** *device have been teamed together and are accessible using the 192.168.0.9 IP address.*

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Configuring Server High Availability** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Configure Network Load Balancing
- Exercise 2 - Configuring NIC Teaming on a Server

You should now be able to:

- Install and Configure a Web Server
- Install the Network Load Balancing Feature
- Configure a Network Load Balancing Cluster
- Configure a Forward Lookup Zone for Network Load Balancing Cluster
- Verify the Created Network Load Balancing Cluster
- Configure a Secondary Network Interface Card (NIC)
- Configure NIC Teaming

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.